

Integrating Multi-Source Data with Sentiment Analysis and Language Models to Enhance Stock Market Decision Making

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I declare that this written submission represents my ideas in my own words and that, where the ideas or words of others have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in my submission. I understand that any violation of the above will be cause for disciplinary action by the University and may also evoke penal action from the sources which have thus not been properly cited, or from whom proper permission has not been taken when needed.

The work reported in this thesis has been carried out by me under the supervision of Dr. Sujata Deshmukh and has not been submitted, in whole or in part, for the award of any other degree or diploma of this or any other institution.

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Certificate

This is to certify that the thesis entitled “**Integrating Multi-Source Data with Sentiment Analysis and Language Models to Enhance Stock Market Decision Making**”, submitted by **Anandkumar Pardeshi** (Registration No. [**PhD registration number**]) to University of Mumbai for the award of the degree of **Doctor of Philosophy** in [**degree discipline, e.g. Computer Engineering**], is a record of bona fide research work carried out by him under my supervision and guidance at Fr. C. Rodrigues College of Engineering, Bandra, Mumbai.

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I acknowledge the reviewers and editors of the venues in which the constituent studies of this thesis were published or are under review. Their criticism materially improved the work; in one instance a reviewer's insistence on a robustness analysis exposed two implementation defects that had silently invalidated an entire results section, and correcting them changed the central claim of that study.

Finally, I thank my family for their patience across the years this research has taken.

Anandkumar Pardeshi

Abstract

Equity markets are driven simultaneously by structured price and volume records, by qualitative information carried in news and social media, and by slower macroeconomic conditions. Each stream is individually incomplete, and each becomes informative or uninformative at different times. Conventional forecasting systems compress this heterogeneity into a single numerical pipeline, report a statistical accuracy figure, and stop short of the decision the investor actually has to take. This thesis addresses that shortfall by developing, implementing and validating an integrated framework in which multi-source fusion, language-model-based sentiment extraction, transparent reasoning and decision-level allocation are treated as one connected engineering problem rather than as four separate ones.

The first objective develops a comprehensive model integrating financial metrics, news sentiment, social media sentiment and macroeconomic indicators. A hierarchical Multi-Source Fusion Network is designed in which convolutional encoders process indicator streams, transformer encoders process text, engineered temporal features carry macroeconomic conditions, and a cross-modal attention layer assigns time-varying weight to each source before a bidirectional recurrent encoder produces the forecast. Evaluated on ten years of National Stock Exchange data across ten sectoral indices, the fused model attains a normalised mean absolute error of 0.018, a coefficient of determination of 0.988 and a directional accuracy of 84.7 per cent at the index level, against 76.3 per cent for a simple averaging ensemble over identical inputs. Removing both textual sources costs 10.9 per cent of explained variance, exceeding the sum of their individual removals and confirming that sentiment carries information that price history does not. The fusion mechanism is then strengthened by a Bayesian formulation in which the weight assigned to a source is derived from its own recent predictive uncertainty through a softmax over negative error variances, so that a stream is discounted at the moment it becomes unreliable rather than at the end of a training cycle. On the Nifty 50 this dynamic rule improves the Sharpe ratio by 22.7 per cent and reduces maximum drawdown by 18.3 per cent against the strongest static-fusion baseline, while the same expert networks fused with fixed weights perform no better than that baseline, which locates the gain in the weighting mechanism rather than in the networks.

The second objective designs a hybrid framework coupling large language models with conventional financial indicators. A transformer sentiment encoder converts news headlines into a continuous polarity score, which is time-aligned to trading sessions and fused with fifteen technical indicators before gradient-boosted classification. On Indian equity data the sentiment score emerges as the highest-gain predictor, correlating 0.74 with momentum yet only 0.04 with price, and the sentiment-augmented model raises directional accuracy from a 46.43 per cent baseline average to 47.72 per cent while producing the highest prediction confidence of all models examined. The absolute level is reported without embellishment: at single-name daily horizons all models cluster near the random classifier, so language-model sentiment supplies a measurable but bounded increment rather than a decisive edge.

The third objective establishes transparency and examines sentiment noise and sentiment-to-price temporal dynamics. A Dynamic-Causal Hybrid Framework replaces correlation-based feature selection with a directed causal adjacency matrix recovered by conditional independence testing, encodes intraday, daily and weekly horizons through a multi-scale memory encoder, and allocates capital through a credibilistic optimiser. On 308 trading sessions of NIFTY50 data it recovers a causal graph of density 0.850 with an identifiable causal hub, and demonstrates that strong causal influence can coexist with weak linear correlation — the mechanistic justification for preferring causal to correlational feature selection. Sentiment quality is then treated as a measurable property rather than an assumption: a sentiment composite is validated against four independently published series, carries the expected sign against every one of them, and separates the identified market states at $p = 1.8 \times 10^{-8}$.

The fourth objective evaluates and validates the framework as a decision system. Latent market states are identified by an expanding-window Gaussian mixture model, the fused signal enters a constrained quadratic program with covariance shrinkage and a turnover penalty, and the resulting portfolio is scaled by a regime-dependent risk budget. Over eighteen years of ten liquid multi-asset instruments under a walk-forward protocol with transaction costs, the framework attains a Sharpe ratio of 0.878 against 0.628 for an equal-weighted benchmark and reduces maximum drawdown from -36.1 to -19.0 per cent. It outperforms in all four stress episodes, wins on drawdown in eighteen of nineteen calendar years and in all eighteen specifications examined, and on a 665-session holdout that postdates every design decision it records a higher Sharpe ratio and

a smaller drawdown than all three benchmarks. Ablation at matched average exposure isolates the source of the gain: timing risk by market state improves both the Sharpe ratio and the drawdown with no change in mean return, whereas the signal fusion layer contributes nothing detectable to cross-sectional selection. Cross-market transfer to a concentrated mega-capitalisation universe reproduces the defensive behaviour but loses on risk-adjusted return, which bounds applicability to universes with genuine dispersion in asset volatility.

Taken together, the six publications underlying this thesis demonstrate a coherent progression from information integration, through language-model-based signal extraction and transparent reasoning, to validated decision making. The component-level nulls are retained rather than removed, because they identify precisely which layers of a fusion architecture earn their place and which do not.

Keywords: multi-source data fusion; sentiment analysis; large language models; explainable artificial intelligence; causal discovery; regime detection; portfolio construction; Indian equity market.

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List of Abbreviations

ADX	Average Directional Index	MAE	Mean Absolute Error
ANOVA	Analysis of Variance	MDD	Maximum Drawdown
ARIMA	Autoregressive Integrated Moving Average	ML	Machine Learning
ATR	Average True Range	MLP	Multi-Layer Perceptron
AUC	Area Under the Curve	MOSPI	Ministry of Statistics and Programme Implementation
BERT	Bidirectional Encoder Representations from Transformers	MSE	Mean Squared Error
Bi-LSTM	Bidirectional Long Short-Term Memory	MSFN	Multi-Source Fusion Network
BSE	Bombay Stock Exchange	NLP	Natural Language Processing
CAGR	Compound Annual Growth Rate	NSE	National Stock Exchange
CIT	Conditional Independence Test	OBV	On-Balance Volume
CNN	Convolutional Neural Network	OHLCV	Open, High, Low, Close, Volume
CPI	Consumer Price Index	PC	Peter–Clark (causal discovery algorithm)
CVaR	Conditional Value at Risk	QP	Quadratic Program
DA	Directional Accuracy	RBI	Reserve Bank of India
DCHF	Dynamic-Causal Hybrid Framework	RDM-DEA	Range Directional Measure Data Envelopment Analysis
DL	Deep Learning	RMSE	Root Mean Square Error
DM	Diebold–Mariano (test)	ROC	Receiver Operating Characteristic
EMH	Efficient Market Hypothesis	RSI	Relative Strength Index
ETF	Exchange-Traded Fund	SEBI	Securities and Exchange Board of India
GARCH	Generalised Autoregressive Conditional Heteroskedasticity	SHAP	Shapley Additive Explanations
GMM	Gaussian Mixture Model	SMA	Simple Moving Average
GRU	Gated Recurrent Unit	SVM	Support Vector Machine
IC	Information Coefficient	VADER	Valence Aware Dictionary and sEntiment Reasoner
IIP	Index of Industrial Production	VIX	Volatility Index
LLM	Large Language Model	VWAP	Volume-Weighted Average Price
LSTM	Long Short-Term Memory	WPI	Wholesale Price Index
MACD	Moving Average Convergence Divergence	XAI	Explainable Artificial Intelligence

List of Symbols

$h_m^{(l)}$	representation of modality m at encoder layer l
α_m	cross-modal attention weight assigned to modality m
F_t	fused multi-source representation at session t
$\sigma_{i,t}^2$	predictive uncertainty of source i at session t
$w_{i,t}$	uncertainty-derived fusion weight of source i at session t
τ	trailing window length used to estimate predictive uncertainty
T	softmax temperature in the dynamic fusion rule
s_t	continuous news-sentiment polarity score, $s_t \in [-1, +1]$
$A(t)$	time-varying directed causal adjacency matrix
e_{ij}	directed causal edge from series i to series j
r_t	identified latent market state (regime) at session t
$T_{i,t}, M_{i,t}$	technical and sentiment composites for asset i at session t
$c_{i,t}$	composite-agreement confidence term
$z_{i,t}$	fused cross-sectional signal for asset i
$\alpha_{i,t}$	signal placed on a return scale by information-ratio scaling
Σ_t	shrinkage-estimated return covariance matrix
λ, κ	risk-aversion and turnover-penalty coefficients
$\tilde{w}_t, w_t^*$	risky-sleeve weights before and after the regime risk budget
k_{r_t}	regime risk-budget multiplier, $k \in \{1.00, 0.75, 0.50\}$
$\tilde{R}$	fuzzy (triangular) return variable in the credibilistic optimiser

Chapter 1

Introduction

1.1 Stock Market Decision Making as an Information Problem

The price of a traded security is the outcome of a continuous negotiation between participants who do not observe the same information at the same time. Part of that information is structured and numerically clean: the open, high, low, close and volume records of each session, and the technical indicators derived from them. A larger part is unstructured and arrives irregularly, in news articles, corporate disclosures, regulatory announcements, analyst commentary and social media discussion. A third part changes slowly and shapes the environment rather than the individual security, in the form of policy rates, inflation, industrial output, currency movements and commodity prices. Any forecasting system that reads only one of these streams is, by construction, reasoning from an incomplete description of the market it is trying to anticipate [1, 2, 4].

This is not a new observation. What is new is that acting on it has become technically feasible. Two developments are responsible. The first is the maturation of multi-source data fusion, which supplies a principled account of how heterogeneous streams may be combined so that their complementary content is preserved instead of being averaged away [1, 3, 30]. The second is the arrival of transformer-based large language models, which read financial text with a level of contextual sensitivity that lexicon and bag-of-words methods never achieved, and which can therefore convert narrative information into a numerically usable signal [61, 46, 37]. Together they make it reasonable to ask a question that was previously out of reach: whether a single system can absorb quantitative, textual and macroeconomic evidence and convert it into a decision that an investor could actually execute.

The framing matters. A very large share of the published literature in this area treats the problem as one of *prediction*: given a feature vector, forecast tomorrow's return or its sign, and report an accuracy figure. That framing is legitimate but incomplete, because an investor does not consume a prediction. An investor consumes a position — a vector of holdings, sized against a risk budget, rebalanced at a cost, and held through

drawdowns. The translation from the first object to the second is not a formality. It involves a covariance structure that must be estimated, position limits that bind, turnover that must be paid for, and a decision about how much total risk to carry that is logically separate from the decision about what to hold. Each of those steps can destroy the value of a forecast as easily as preserve it [108, 137, 104].

1.2 Three Obstacles Between a Fused Signal and a Defensible Decision

Three difficulties sit between the raw heterogeneous information and a defensible decision, and this thesis is directed at all three simultaneously rather than at any one of them in isolation.

1.2.1 The reliability of a source is not constant

Technical indicators are informative in quiet trending conditions and degrade when the market is disturbed; news and volatility signals behave in the opposite manner, carrying little information in placid stretches and a great deal during a shock. A fusion rule with fixed weights therefore misweights its inputs precisely when accurate weighting matters most [5, 11]. This is a structural problem rather than a tuning problem. Attention mechanisms, which are the standard remedy, learn weights over an entire training set; the resulting weight expresses the *average* relevance of a source across the training period and cannot express the proposition that a particular source has become unreliable in the last fortnight. Between retraining cycles, a stream that has degraded continues to influence the forecast at its historical weight.

What is missing is a measurement. If the reliability of a source can be estimated continuously from observable evidence — specifically, from the variance of that source’s own recent prediction errors — then the fusion weight becomes a function of that estimate and the architecture self-corrects without human intervention and without waiting for the next training run. Chapter 5 develops this idea formally and tests it.

1.2.2 A prediction is not an explanation

A forecast produced by a deep architecture is not, by itself, an account of why the forecast was made. In a domain where capital is committed on the basis of the output, and where

a regulator or an investment committee may reasonably demand a justification, opacity is a practical obstacle and not merely a philosophical one [81, 82].

The standard response is post-hoc attribution: methods such as Shapley additive explanations [82] or local surrogate models [83] report which inputs the fitted model relied upon. This is genuinely useful, and it is not what this thesis pursues as its primary transparency instrument. The reason is that post-hoc attribution explains a *model*; it does not establish that the relationships the model has learned are anything more than correlations that happened to hold in the training window. A feature that co-moves with the target for spurious reasons will be attributed importance exactly as a genuine driver would be. Chapter 7 therefore pursues transparency at the level of *structure*: the feature set entering the model is constrained, before any learning takes place, to variables with demonstrable directed influence, recovered by conditional independence testing over lagged series [98, 97, 87].

1.2.3 Statistical accuracy and economic usefulness are different properties

The third obstacle is the one most often elided. A model that is accurate is not thereby profitable, and a model that is profitable is not necessarily accurate. The two properties come apart for reasons that are well understood in principle and routinely ignored in practice: predictions are not equally costly to act on, a small edge can be consumed entirely by transaction costs, an unconditional accuracy figure says nothing about performance in the states where losses are concentrated, and — most importantly — a portfolio’s risk-adjusted return depends on the timing of risk as much as on the selection of assets.

This thesis treats the decision layer as a first-class object of study rather than as a postscript. Chapter 8 develops a decision-theoretic allocation framework, evaluates it over eighteen years with transaction costs, and — unusually — disassembles it to determine which of its layers actually produce the measured outcome. The answer, reported in full including the parts that do not favour the original design, is the central finding of the thesis.

1.3 Research Context and the Position of the Four Objectives

Figure 1.1 places the four research objectives on the path from heterogeneous raw information to an executable decision. Objective 1 governs the integration of sources. Objective 2 governs the extraction of signal from language and its combination with

conventional indicators. Objective 3 governs the transparency of the reasoning and the behaviour of sentiment as a noisy, time-lagged input. Objective 4 governs the conversion of a signal into a position and the validation of that conversion across markets.

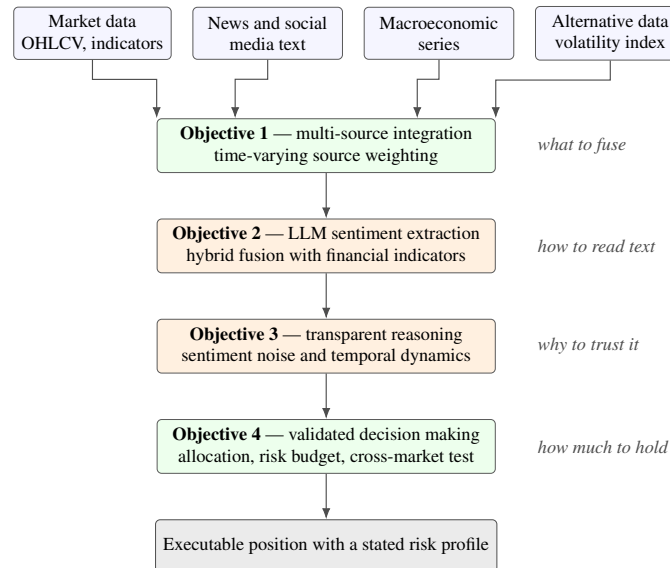


Figure 1.1: Research context: the path from heterogeneous market information to an executable decision, and the position of the four research objectives on that path.

The objectives are sequential in the sense that each consumes the output of the one before it, but they are not merely stages of a pipeline. Each raises a distinct research question, each is answered by distinct experimental evidence on a study universe chosen for that question, and — importantly for how this thesis should be read — the answer obtained at one stage redirected the design of the next. The framework reported in Chapter 8 is not the framework that was envisaged at the outset. It is the framework that survived the evidence.

1.4 The Indian Equity Market as the Primary Empirical Setting

The Indian equity market is the primary empirical setting of this research. The choice is deliberate rather than convenient, and it is worth stating the reasons explicitly because they also bound the generality of several results.

The National Stock Exchange combines four features that make it an unusually demanding test of a multi-source system. First, retail participation is high and has risen sharply, so sentiment expressed on social platforms is not merely commentary on the market but a partial cause of short-term price movement [35, 8]. Second, the market is strongly sensitive to domestic policy events — the annual Union Budget, Reserve Bank of India

monetary policy committee announcements, and regulatory decisions by the Securities and Exchange Board of India — which produce discrete structural breaks rather than smooth transitions. Third, the index carries pronounced sectoral concentration in financial services, so a shock in that sector propagates to the aggregate in a way it would not in a more evenly weighted index. Fourth, the market is simultaneously exposed to global factors, including crude oil prices, the rupee–dollar exchange rate and United States monetary policy, so domestic and international information can and frequently do disagree [12, 28].

That disagreement is the point. A fusion architecture is only interesting when the sources it fuses carry genuinely different information; in a market where every stream says the same thing, the fusion rule is irrelevant and a single source would do. The Indian market supplies sources that disagree, and the disagreement is informative rather than noise.

Two qualifications are recorded here and revisited where they bind. The first is that instrument-level, time-stamped news corpora of the depth available for United States large-capitalisation equities do not exist for broad Indian index instruments, which constrains what can be measured at the allocation layer and is the direct reason the sentiment construct used in Chapter 8 is market-derived rather than textual. The second is that the allocation study of Chapter 8 requires eighteen years of daily data on a universe with genuine dispersion in asset volatility, including two severe crises; the instruments that satisfy that requirement with clean, survivorship-free history are United States exchange-traded funds, and the study is therefore conducted there and treated as a cross-market test of the mechanism rather than as a claim about Indian allocation. Both decisions are stated rather than concealed, because they determine precisely which claims in this thesis are about the Indian market and which are about the mechanism.

1.5 Research Questions

The thesis addresses five research questions, each of which is answered by evidence reported in a specific chapter.

RQ1. Does fusing financial, textual and macroeconomic streams under a learned, time-varying weighting improve forecast quality over the best single-source and fixed-weight alternatives, and can the improvement be attributed to individual streams rather than merely asserted? (Chapter 4.)

- RQ2.** Can the weight assigned to a source be made a function of that source’s measured, instantaneous predictive reliability rather than its historical average relevance, and does doing so improve risk-adjusted outcomes? (Chapter 5.)
- RQ3.** How much of the improvement that large language models deliver in sentiment *measurement* survives into forecast quality once the sentiment score is temporally aligned, fused with technical indicators, and evaluated out of sample at the horizon and granularity at which decisions are actually taken? (Chapter 6.)
- RQ4.** Can transparency be grounded in directed causal structure rather than in post-hoc attribution, and can the noise carried by the sentiment channel be measured rather than assumed? (Chapter 7.)
- RQ5.** When the fused signal is converted into an executable portfolio under realistic constraints, which layers of the resulting framework actually produce the measured outcome, and under what conditions does the framework transfer to a different market? (Chapter 8.)

1.6 Contributions of the Thesis

The contributions claimed are stated here in summary and developed with their supporting evidence in the chapters indicated. They are stated as bounded claims; where the evidence supports only a qualified statement, the qualification is part of the contribution.

C1. An attributable four-stream fusion architecture for an emerging market.

A Multi-Source Fusion Network integrating technical indicators, news sentiment, social media sentiment and Indian macroeconomic series through modality-specific encoders and a cross-modal attention layer, evaluated on ten years of National Stock Exchange data across ten sectoral indices under a strictly chronological protocol with expanding-window walk-forward confirmation. The contribution is not that fusion improves accuracy, which was already known, but that the improvement is *attributable*: source ablation isolates the marginal value of each stream and demonstrates superadditivity between the two textual sources, and the learned attention distribution demonstrates that source importance is state-dependent. (Chapter 4.)

C2. A fusion rule conditioned on measured source reliability. A Bayesian dynamic uncertainty-weighted formulation in which each source is modelled by its own

expert network, its predictive uncertainty is estimated continuously from the variance of its recent errors, and the fusion weights are obtained by a softmax over the negative uncertainties. The contribution is isolated by construction: the same expert networks fused with fixed weights perform no better than the established static baselines, which locates the measured gain in the weighting mechanism rather than in the networks. (Chapter 5.)

C3. A measured account of what language-model sentiment contributes. A hybrid pipeline coupling a transformer sentiment encoder with fifteen technical indicators, evaluated against technical-only baselines under a like protocol. Three findings constitute the contribution: sentiment is the highest-gain feature in the fused space; it is orthogonal to price, correlating 0.04 with price level and 0.74 with momentum; and the incremental gain in raw directional accuracy is small, with all models close to the random classifier at a single-name daily horizon. The third finding is reported as prominently as the first two, and it is what redirects the thesis toward the decision layer. (Chapter 6.)

C4. Structural rather than post-hoc transparency, with sentiment noise quantified. A Dynamic-Causal Hybrid Framework that constrains the feature set by directed causal structure recovered through conditional independence testing, encodes intraday, daily and weekly horizons through a multi-scale memory encoder, and allocates through a credibilistic optimiser. It recovers an interpretable causal graph of density 0.850 with an identifiable causal hub, and demonstrates that strong directed influence can coexist with weak linear correlation — the mechanistic justification for causal rather than correlational feature selection. Separately, the sentiment construct is validated against four independently published series and shown to separate identified market states at $p = 1.8 \times 10^{-8}$, while its predictive content is measured independently and found not to be significant. Keeping construct validity and predictive usefulness apart is itself a methodological contribution. (Chapter 7.)

C5. A decision framework ablated at matched risk, with its applicability bounded by evidence. A regime-conditional allocation framework that separates relative composition from total risk, evaluated over eighteen years with transaction costs, a fully disclosed specification sweep, four pre-selected stress episodes, a 665-session holdout postdating every design decision, and a secondary market universe. The ablation is conducted *at matched average exposure*, which is what

makes it possible to distinguish the timing of risk from its level; it finds that regime-conditional risk timing improves both the Sharpe ratio and the drawdown with no significant change in mean return, and that the signal fusion layer contributes nothing detectable. (Chapter 8.)

C6. Two generalisable engineering failure modes. Both were found only by disassembling the framework and both are reported because they are silent: an absolute volatility target calibrated for a concentrated equity portfolio binds in one rebalance out of 224 on a diversified universe, leaving the regime layer present in the specification but inert; and a signal-to-allocation conversion in which mismatched units between the linear and quadratic terms of a mean-variance objective, rather than the information in the signal, determine the optimiser’s solution. Neither produces an error message. (Chapter 8.)

1.7 Scope and Delimitations

Six delimitations bound the work and are stated at the outset so that no claim in later chapters is read more widely than the evidence supports.

Horizon. All studies operate at a daily decision horizon. Intraday microstructure, order-book dynamics and execution algorithms are outside scope. The multi-granularity encoder of Chapter 7 uses intraday-derived features but produces daily decisions.

Direction of exposure. Every portfolio constructed in this thesis is long-only and unlevered. Risk budgeting is one-sided: the framework de-risks toward cash in disturbed states but never levers up. This is a deliberate restriction, because it guarantees that any measured performance difference is attributable to risk reduction rather than to leverage.

Language coverage. Sentiment extraction is English-only. A substantial share of Indian retail participants consume financial news in regional languages, and the sentiment channel measured here therefore observes only part of the information actually reaching the market. This is identified as future work in Chapter 9 rather than dismissed.

Transfer. The cross-market evidence in Chapter 8 comes from applying the framework to a second universe *without modification*. This is direct transfer, not transfer learning: no parameter was fine-tuned on the target market, no representation was pre-trained and re-used, and no domain-adaptation objective was optimised. The result is therefore a lower bound on cross-market performance, and it is labelled as such wherever it appears.

Explainability methods. Transparency in this thesis rests on directed causal structure and on the interpretability of the fusion weights. Attribution methods such as Shapley additive explanations and local surrogate models are reviewed and cited but are not run as experiments; where the thesis speaks of explainability it means structural transparency, and the distinction is maintained throughout.

Costs and frictions. Transaction costs are modelled as a one-way charge of ten basis points in the allocation study and are absent from the prediction studies of Chapters 4 to 6, where the evaluation is of forecast quality rather than of realised profit. Slippage, market impact and financing costs are not modelled. The consequence — that directional accuracy reported in Chapter 4 does not translate directly into trading profit — is stated in that chapter rather than left implicit.

1.8 Organisation of the Thesis

The thesis is organised into nine chapters. Chapter 2 presents the technical foundations required to read the later chapters without recourse to external material: technical indicator construction, the three generations of sentiment extraction, fusion strategies, regime modelling, portfolio construction, and the evaluation metrics and statistical procedures used throughout. Chapter 3 reviews the research literature along the four axes of the objectives and derives the five research gaps that define the scope of the work. Chapter 4 develops and evaluates the Multi-Source Fusion Network. Chapter 5 develops the Bayesian dynamic uncertainty-weighted formulation of the fusion rule. Chapter 6 develops the hybrid language-model and indicator framework. Chapter 7 develops the explainable and causal framework and the sentiment diagnostics. Chapter 8 develops the decision-theoretic allocation framework and reports the ablation, the holdout and the cross-market test. Chapter 9 consolidates the findings, states the limits of the evidence and sets out the future scope. A statement of the publications underlying the thesis and the consolidated reference list follow.

Chapter 2

Background and Preliminaries

This chapter assembles the technical material on which the rest of the thesis depends. It is deliberately self-contained: a reader who works through it should be able to read Chapters 4 to 8 without recourse to external references for notation, for the definition of a metric or for the justification of a statistical procedure. The chapter does not survey the research literature — that is the task of Chapter 3 — but establishes the objects the literature operates on.

2.1 The Market Environment and Its Information Streams

2.1.1 Market efficiency and the scale of the achievable edge

Any claim about forecastable structure in asset prices must be positioned against the efficient market hypothesis in the form given by Fama [131]: if prices reflect available information, systematic outperformance net of costs should not persist. The hypothesis is not falsified by the existence of a small measurable edge; it constrains its size. The benchmark machine-learning study in empirical asset pricing, by Gu, Kelly and Xiu [137], reports exactly this pattern — statistically detectable predictability whose magnitude is modest and whose economic value depends heavily on how it is harvested.

This framing is used throughout the thesis in a specific way. When Chapter 6 reports a directional accuracy of 47.72 per cent against a baseline mean of 46.43 per cent, the correct reaction is not disappointment but recognition that the measured increment is of the order the literature leads one to expect at a single-name daily horizon. Conversely, when Chapter 4 reports 84.7 per cent at the index level, the correct reaction is to ask what object is being predicted and over what horizon, because the two figures are not comparable. Section 9.3 of Chapter 9 treats that reconciliation as a substantive finding rather than as a caveat.

Two qualifications to the strict efficiency reading are relevant. The first is empirical: robust cross-sectional regularities exist, of which the size and value factors of Fama

and French [132] and the momentum effect of Jegadeesh and Titman [133] are the most durable, and their persistence establishes that predictable structure is a matter of degree rather than of existence. The second is behavioural. Prospect theory [134] and the behavioural-finance programme that followed it [135] describe systematic departures from the rationality that strict efficiency assumes; Lo's adaptive markets hypothesis [136] reconciles the two positions by treating efficiency as an evolving property of a population of strategies rather than a fixed state. On that reading, an edge can exist, be competed away and re-emerge — which is precisely the non-stationarity of source reliability that Chapter 5 is built to accommodate.

2.1.2 Structural characteristics of the Indian equity market

The National Stock Exchange of India and the Bombay Stock Exchange together constitute the primary venue for Indian equity trading. Three characteristics are relevant to the design decisions taken in this thesis.

Retail participation and sentiment transmission. Retail participation in Indian equities is high and has risen sharply over the study period. Sentiment expressed on social platforms is therefore not only a commentary on the market but a partial determinant of short-term price movement, and the relationship is asymmetric: Khan and Chahal [35] find that the influence of social media sentiment on Indian sectoral returns is strongest at market extremes. This motivates the inclusion of social media as a distinct stream in Chapter 4 rather than as an adjunct to news.

Policy sensitivity and discrete structural breaks. The annual Union Budget, monetary policy committee announcements by the Reserve Bank of India, and regulatory decisions of the Securities and Exchange Board of India produce discrete, scheduled events that reset relationships between instruments within hours. This is the reason a *static* causal graph is inadequate for the Indian market and a time-varying one is required, as Chapter 7 argues; it is also the reason the stationarity assumption embedded in standard debiasing layers fails at exactly the moments that matter.

Sectoral concentration and dual exposure. The index carries pronounced concentration in financial services, so shocks in that sector propagate to the aggregate disproportionately; and the market is simultaneously exposed to global factors, including crude oil prices, the rupee-dollar exchange rate and United States monetary policy. Domestic and international information can therefore disagree, and the disagreement is informative.

This is what makes the market a demanding test of a fusion architecture rather than a convenient one.

2.1.3 The four information streams

Table 2.1 summarises the four streams used in this thesis, their native sampling frequency, the transformation applied to reach a common daily representation, and the distortion that transformation introduces. The final column is the important one: every alignment step buys comparability at the cost of some fidelity, and stating the cost is part of the methodology rather than an afterthought.

Table 2.1: The four information streams, their native frequency, the transformation applied to reach a common daily representation, and the distortion that transformation introduces

Stream	Native frequency	Transformation to daily	Distortion introduced
Market data (OHLCV and derived indicators)	Per session, and intraday where available	Indicators computed on closing series; intraday series aggregated to session features	None at the daily horizon; intraday aggregation discards microstructure ordering
News text	Irregular, time-stamped	Scored per article, aggregated per session, with post-close publications attributed to the next session	Within-day ordering of articles is lost; relevance weighting is a modelling choice, not a measurement
Social media text	Irregular, high volume	Scored per post, aggregated into daily polarity, mention volume and dispersion	Bot and coordinated-posting contamination is not separated; dispersion partly compensates
Macroeconomic series	Monthly or quarterly	Interpolated to daily frequency	Announcement-day effects are discontinuous and interpolation understates them; the direction of the resulting bias is known and stated

2.2 Technical Indicators

Technical indicators compress the price and volume record into features intended to expose trend, momentum, volatility and participation. Fifteen indicators are used in Chapters 4 to 6; the families and the reasons for their inclusion are given here, with the canonical definitions, which follow the standard references [140, 138, 139]. No claim is made that any individual indicator carries independent predictive power; they are used as a feature basis that the learning architecture combines, and the question of which of them

matters is answered by the ablations of Chapter 4 rather than assumed here.

2.2.1 Trend and moving averages

The simple moving average over n sessions,

$$\text{SMA}_n(t) = \frac{1}{n} \sum_{k=0}^{n-1} P_{t-k}, \quad (2.1)$$

is the elementary trend statistic; the exponential moving average, $\text{EMA}_n(t) = \beta P_t + (1 - \beta) \text{EMA}_n(t-1)$ with $\beta = 2/(n+1)$, weights recent observations more heavily and responds faster to a turn. Windows of 5, 10 and 20 sessions are used, capturing weekly, fortnightly and monthly trend respectively. The slope of the exponential average supplies a directional feature that the level does not.

2.2.2 Momentum oscillators

The relative strength index of Wilder [138] normalises the ratio of average gains to average losses over n sessions onto a bounded scale,

$$\text{RSI}_n(t) = 100 - \frac{100}{1 + \text{RS}_n(t)}, \quad \text{RS}_n(t) = \frac{\overline{\text{gain}_n(t)}}{\overline{\text{loss}_n(t)}}, \quad (2.2)$$

and is conventionally read as indicating overbought conditions above 70 and oversold conditions below 30. The moving average convergence divergence statistic is the difference between two exponential averages, $\text{MACD}(t) = \text{EMA}_{12}(t) - \text{EMA}_{26}(t)$, with a signal line given by EMA_9 of the statistic itself; the crossing of the two is the conventional trigger. The Aroon indicators measure the number of sessions since the highest high and the lowest low within a window, and the average directional index summarises trend strength independently of direction.

2.2.3 Volatility and range

Bollinger bands place a k -standard-deviation envelope around a moving average, $\text{SMA}_n(t) \pm k \sigma_n(t)$, with $k = 2$ conventional; the band width and the position of the close within the band are both used as features. The average true range averages the maximum of the session range and the gaps to the previous close, and realised volatility over a window is computed as the standard deviation of log returns $r_t = \log(P_t/P_{t-1})$,

annualised by $\sqrt{252}$.

That volatility is itself predictable, clusters in time, and responds asymmetrically to negative returns is the standard description of financial return series since the autoregressive conditional heteroskedasticity model of Engle [142] and its generalisation by Bollerslev [143] — a description that the linear autoregressive-integrated-moving-average machinery of Box and Jenkins [141], which supplies the classical baseline in Chapter 4, cannot represent. This matters twice in the thesis: it is the reason a regime-conditional risk budget is worth constructing at all (Chapter 8), and it is the reason a debiasing layer that assumes stationary noise fails under stress (Chapter 7).

2.2.4 Participation and volume

On-balance volume accumulates signed volume, $OBV(t) = OBV(t-1) + \text{sgn}(P_t - P_{t-1})V_t$, and the volume-weighted average price, $VWAP(t) = \sum_k P_k V_k / \sum_k V_k$, supplies a participation-weighted reference level. Volume momentum, the ratio of current volume to its own moving average, distinguishes a price move made on conviction from one made on thin trading.

A caution applies to the unbounded accumulators. On-balance volume grows without bound and takes values of the order 10^8 to 10^9 on liquid instruments, while a risk term in a mean-variance objective is of the order 10^{-4} . Passing such a feature into an optimiser without rescaling lets the ratio of units, rather than the information in the signal, determine the solution. This is not a hypothetical concern: it is one of the two engineering failure modes diagnosed in Chapter 8.

2.3 Sentiment Extraction from Financial Text

Financial text analysis has passed through three generations, and the distinctions between them determine what a sentiment score means.

2.3.1 Lexicon methods

The first generation scores text against a predefined polarity dictionary. The decisive finding for finance is that of Loughran and McDonald [48]: general-purpose lexicons misclassify financial language badly, because words that are negative in ordinary usage (“liability”, “tax”, “cost”) are neutral descriptors in a filing. Domain-specific dictionaries

are therefore required. VADER [47] remains a widely used reference implementation for social text, augmenting a lexicon with rules for intensifiers, negation, capitalisation and punctuation. Lexicon methods are fast, fully interpretable and deterministic; they are insensitive to context, sarcasm and syntax.

2.3.2 Supervised machine learning

The second generation learns polarity from annotated corpora using conventional classifiers. Accuracy improves over lexicon baselines but at the cost of annotation, and generalisation across market conditions is weak: a classifier trained on one regime’s vocabulary degrades when the vocabulary shifts [23].

2.3.3 Transformer language models

The third generation replaced both. The transformer architecture [61] computes representations through scaled dot-product self-attention,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (2.3)$$

where Q , K and V are query, key and value projections of the input and d_k is the key dimension. The same mechanism, applied across modalities rather than across tokens, is the cross-modal attention used for fusion in Chapter 4; the notational continuity is deliberate.

The bidirectional pre-training objective of BERT [49] and the optimised training recipe of RoBERTa [50] established the encoder family on which financial adaptation is built, while the demonstration that sufficiently large autoregressive models perform new tasks from instructions alone [51] opened the generative route to sentiment extraction that Chapter 6 does not take but that Section 6.1 discusses.

Domain adaptation followed quickly. FinBERT [46] further-trains a bidirectional encoder on financial text; larger domain models [43, 44, 45] extend the approach to generative reasoning over financial documents, and surveys of their practical application record both the gains and the operational obstacles [40]. Jung and Jang [13] showed that masking numerically significant tokens during adaptation preserves quantitative context that generic pre-processing destroys — “GDP growth at 7.8 per cent” carries information that “GDP growth at [NUM] per cent” does not.

2.3.4 Continuous versus categorical scores

A design decision recurs throughout this thesis: sentiment is retained as a continuous polarity score $s_t \in [-1, +1]$ rather than as a categorical label. Categorical output discards intensity, and intensity is exactly what distinguishes a routine positive report from a significant one. Retaining the continuous score, together with the model’s own confidence, preserves that information for the fusion layer and makes it possible to examine the empirical distribution of sentiment as a diagnostic — a symmetric distribution about neutral with moderate tails indicates that the pipeline is measuring variation rather than producing a biased categorical output, and this check is reported in Chapter 6.

2.4 Multi-Source Data Fusion

2.4.1 The three canonical strategies

Given M heterogeneous sources with per-source representations $x_1, \dots, x_M$ and a target y , three fusion strategies dominate.

Early (feature-level) fusion concatenates representations before modelling, $\hat{y} = f([x_1; \dots; x_M])$. It is simple and preserves cross-feature interactions, but forces one architecture to learn interactions across domains with very different statistical properties, and is prone to modality dominance if scales are not equalised [6].

Late (decision-level) fusion trains source-specific models and combines their outputs, $\hat{y} = \sum_m w_m f_m(x_m)$. It preserves modularity and allows each architecture to match its source, but the weights w_m are typically fixed in advance or learned once and then frozen [5, 17].

Hybrid (model-level) fusion combines intermediate representations inside the network, usually through attention or gating [7, 9, 36]. This is the current state of the art and is a genuine advance, because the combination is learned rather than assumed.

2.4.2 What attention weights do and do not express

The limitation that motivates Chapter 5 can be stated precisely. A learned attention weight α_m is a point estimate obtained by minimising a loss over an entire training set. It expresses the *average* relevance of modality m across that set. It is not a statement about the reliability of modality m at a particular moment, and it carries no uncertainty. Conse-

quently, if a source degrades after training — because a news feed changes composition, because a social channel is contaminated, or because a regime shift alters which signals carry information — the architecture continues to weight it at its historical value until the next retraining cycle.

The remedy pursued in this thesis is to make the weight a function of a *measured* quantity. If $e_{i,t} = y_t - \hat{y}_{i,t}$ is the prediction error of source i , then

$$\sigma_{i,t}^2 = \frac{1}{\tau} \sum_{k=1}^{\tau} (e_{i,t-k} - \bar{e}_{i,t})^2 \quad (2.4)$$

is an empirical estimate of that source’s recent predictive volatility over a trailing window of τ sessions, and the fusion weights

$$w_{i,t} = \frac{\exp(-\sigma_{i,t}^2/T)}{\sum_j \exp(-\sigma_{j,t}^2/T)} \quad (2.5)$$

discount a source in proportion to how unreliable it has recently been, with temperature T controlling the sharpness of the discounting. The estimate in Equation (2.4) is *aleatoric*: it captures variation attributable to noise in the environment. It does not capture *epistemic* uncertainty — the model’s ignorance about regions of input space it has not seen — for which a Bayesian treatment of the network weights would be required [75, 76]. The distinction is maintained in Chapter 5 and the epistemic extension is identified as future work.

2.4.3 Distribution-aware normalisation

Heterogeneous sources have heterogeneous distributions, and a single scaler applied to all of them lets the heaviest-tailed input dominate the fused representation. Three scalers are used according to the shape of the input: standard scaling, $(x - \mu)/\sigma$, for approximately symmetric series such as log returns; min–max scaling, $(x - x_{\min})/(x_{\max} - x_{\min})$, for bounded series such as the sentiment score, which preserves the interpretable range; and robust scaling, $(x - \text{med})/\text{IQR}$, for heavy-tailed series such as a volatility index, where the mean and standard deviation are themselves unstable.

Every scaler in this thesis is fitted on an expanding window of strictly earlier observations. Fitting a scaler on the full sample is a subtle and common form of look-ahead: the test-period mean and variance leak into the training representation even though no test label

is used.

2.5 Temporal Alignment and Look-Ahead

The streams of Table 2.1 arrive on different clocks, and reconciling them is where look-ahead bias most often enters. Four rules are applied throughout this thesis.

Publication-time attribution. A news item or post is attributed to the first trading session that begins strictly after its time stamp. An article published after the close is attributed to the next session, never to the session that has already ended.

Holidays are gaps, not repeats. Market holidays are treated as genuine absences rather than forward-filled. Forward-filling a closed session injects artificial autocorrelation into the return series and inflates any statistic that depends on persistence.

Macroeconomic release lag. A macroeconomic series is available only after its release date, not after the period it describes. Interpolation to daily frequency is applied between releases, with the consequence — understatement of announcement-day discontinuities — stated explicitly.

Strictly chronological partitioning. Training, validation and test blocks are contiguous and ordered in time. Random or stratified splitting of a time series is a defect, not a variant, because it permits the model to learn from the future of the observation it is scored on.

2.6 Learning Architectures

2.6.1 Recurrent networks

Long short-term memory [63] maintains a cell state regulated by input, forget and output gates, which permits gradients to propagate across long sequences without vanishing — the failure mode that Bengio, Simard and Frasconi [64] showed to be intrinsic to simple recurrent networks trained by gradient descent. The bidirectional variant runs two such networks in opposite directions and concatenates their hidden states, $h_t = [\vec{h}_t; \overleftarrow{h}_t]$, so that the representation at each step reflects both past and future context *within the input window*. This is not look-ahead provided the input window closes at or before the decision time; the distinction is between context inside a supplied window and informa-

tion from beyond it. A temporal attention layer over the hidden sequence, in the form introduced for sequence alignment by Bahdanau, Cho and Bengio [62], allows the output to weight positions in that window unequally.

The gated recurrent unit [65] merges the forget and input gates into a single update gate and dispenses with the separate cell state. It has fewer parameters than long short-term memory and trains faster on comparable data, which is why it is the market-series expert in Chapter 5 while the full bidirectional variant is used for the fused representation in Chapter 4.

2.6.2 Convolutional encoders for indicator streams

One-dimensional convolution applied along the time axis of an indicator stream extracts local patterns — a crossover, a squeeze, a volume spike — with far fewer parameters than a dense layer, and with translation invariance that is appropriate because the same pattern carries the same meaning wherever it occurs. The encoder used in Chapter 4 is

$$h_m^{(l)} = \text{ReLU}\left(\text{Conv1D}\left(W_m^{(l)} * h_m^{(l-1)} + b_m^{(l)}\right)\right), \quad (2.6)$$

with $h_m^{(0)}$ the raw feature matrix of modality m , and max-pooling between layers.

2.6.3 Gradient-boosted trees

Gradient boosting, in the general formulation of Friedman [67], fits an additive ensemble of regression trees, each fitted to the gradient of the loss with respect to the current prediction. The implementation used here [66] adds second-order optimisation and explicit regularisation on tree complexity; histogram-based alternatives [68] trade a small amount of fidelity for speed and are not required at the sample sizes involved. Tree ensembles are used in Chapter 6 rather than a deep network for three reasons: they are invariant to monotone rescaling of features, which matters when combining indicators measured in incompatible units; they handle heterogeneous feature types without embedding layers; and their split-gain statistics supply a feature-importance measure that is intrinsic to the fitted model rather than a post-hoc approximation. Random forests [69] and support vector machines [70], in the standard implementations of Pedregosa et al. [74], serve as baselines.

2.6.4 Regularisation and optimisation

Dropout [71] randomly zeroes activations during training, which prevents co-adaptation and acts as an implicit ensemble; it is also the mechanism by which an approximate predictive distribution can be obtained at inference time, as Section 2.4.2 noted. Batch normalisation [72] stabilises the distribution of layer inputs and permits larger learning rates. Adaptive moment estimation [73] is used as the optimiser throughout, with early stopping on validation loss and a fixed patience. Random seeds are fixed for data shuffling and weight initialisation in every reported experiment, so that a difference between two configurations reflects the architecture rather than the initialisation.

2.7 Regime Modelling

2.7.1 Why regimes

Financial series are not homogeneous through time. The insight formalised by Hamilton [116] is that a series may be generated by a small number of distinct states with different means, variances and persistence, and that identifying the current state is a precondition for acting sensibly. Ang and Bekaert [117] demonstrated that accounting for regime shifts materially changes optimal allocations, and Kritzman et al. [118] made the practical case for regime-aware dynamic strategies.

2.7.2 Gaussian mixture identification

The regime model used in Chapter 8 is a K -component Gaussian mixture over a two-dimensional feature vector of rolling realised volatility and rolling momentum. The mixture density is

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \quad \sum_k \pi_k = 1, \quad (2.7)$$

fitted by expectation maximisation [144], and the state assigned to a session is the component with the highest posterior responsibility. Three components are used, interpreted after anchoring as calm, transitional and stress states; the number is fixed *a priori* on interpretability grounds rather than selected by an information criterion such as Schwarz's [145], because selecting K on the evaluation sample would be a further specification decision requiring disclosure, and three states map onto a distinction a practitioner

already makes.

Two implementation points are consequential and are stated here because they generalise.

Label anchoring is mandatory. The component indices returned by expectation maximisation are arbitrary and permute freely between refits. If the mixture is refitted at every rebalance — which it must be, to avoid look-ahead — then component 0 in one refit need not correspond to component 0 in the next. Components must therefore be sorted by a stable, interpretable statistic; ascending fitted volatility is used here. Omitting this step produces no error message, only a silently scrambled regime series, and the downstream results will look plausible while being meaningless.

Refitting must use only earlier data. The mixture is refitted at each rebalance on an expanding window of strictly earlier sessions. Fitting once on the full sample and then labelling the history is a severe and easily overlooked form of look-ahead, because the identification of “stress” is informed by knowing when stress occurred.

2.8 Causal Discovery

2.8.1 Predictive precedence

The oldest operational notion of directed influence in time series is that of Granger [97]: series X is said to Granger-cause series Y if past values of X improve the prediction of Y beyond what Y ’s own past achieves. Formally, comparing the residual variances of the restricted and unrestricted regressions

$$Y_t = \sum_{k=1}^p a_k Y_{t-k} + \varepsilon_t \quad \text{versus} \quad Y_t = \sum_{k=1}^p a_k Y_{t-k} + \sum_{k=1}^p b_k X_{t-k} + \eta_t, \quad (2.8)$$

the null $b_1 = \dots = b_p = 0$ is tested by an F statistic. The notion is predictive rather than interventional: it establishes precedence in an information set, not a manipulable cause, and the distinction between the two is the subject of the structural-causal literature [99, 100]. The thesis uses the term “causal” in this operational sense throughout and does not claim interventional identification. Two practical cautions from the time-series literature apply and are observed: inference in a vector autoregression with possibly integrated series requires the lag augmentation of Toda and Yamamoto [102] if the usual asymptotics are to hold, and the identification problems that Sims [103] raised for structural interpretation of such systems are not solved by the procedure used here.

Modern constraint-based methods designed for large nonlinear time-series datasets [101] address the scalability limitation discussed below.

2.8.2 Constraint-based structure learning

Constraint-based discovery procedures of the kind formalised by Spirtes, Glymour and Scheines [98] recover a directed structure by testing conditional independences. Starting from a complete undirected graph, an edge between X_i and X_j is removed whenever a conditioning set S is found such that $X_i \perp\!\!\!\perp X_j \mid S$; the surviving skeleton is then oriented using the conditioning sets that produced the removals. Applied to lagged financial series, the procedure yields a directed adjacency matrix $A(t)$ whose entry (i, j) records the strength of directed influence of series i on series j .

The computational cost is the binding practical constraint: the number of conditional independence tests grows combinatorially with the number of variables. This is why the study in Chapter 7 uses five instruments rather than fifty, and why scalable discovery is identified as a prerequisite for production-scale causal transparency.

2.8.3 Correlation is not causality, and the difference is exploitable

Two series can be strongly linearly correlated with neither influencing the other, and — more usefully for this thesis — one series can exert strong directed influence on another while their linear correlation is weak, if the influence operates with a lag or through a nonlinear channel. A correlation-based feature selector discards exactly those relationships. The empirical demonstration that such pairs exist in the studied universe is the mechanistic justification for the causal approach, and it is reported in Chapter 7.

2.9 Portfolio Construction

2.9.1 Mean-variance and its estimation problem

The mean-variance formulation of Markowitz [104] selects weights w maximising $\mu^\top w - \lambda w^\top \Sigma w$ for expected returns μ , covariance Σ and risk aversion λ . The formulation is correct and the estimation is fragile: sample estimates of μ and Σ carry large errors that the optimiser amplifies, because it allocates aggressively to whatever the estimates most favour — the “optimization enigma” described by Michaud [109], who characterised

mean-variance optimisers as error-maximisers. Four responses matter here. Ledoit and Wolf [107] shrink the sample covariance toward a structured target, $\hat{\Sigma} = (1 - \delta)S + \delta F$, producing a well-conditioned estimator at the cost of a small bias. Black and Litterman [110] anchor expected returns on equilibrium and blend views into them, which stabilises μ rather than Σ . Jagannathan and Ma [111] showed that a long-only constraint of the kind used in Chapter 8 is itself a form of shrinkage and improves out-of-sample performance for that reason. And risk-parity constructions [112] dispense with expected returns altogether, equalising risk contributions. DeMiguel, Garlappi and Uppal [108] demonstrated that naive equal weighting frequently outperforms optimisation that ignores estimation error — which is why equal weighting is used as the primary benchmark in Chapter 8 rather than as a straw man. Ang [115] provides the systematic treatment of the factor-based alternative.

2.9.2 Converting a signal into an expected return

A predictive signal is not an expected return and cannot be substituted for one. The standard conversion is information-ratio scaling in the form given by Grinold and Kahn [114]: a cross-sectionally standardised score $z_{i,t}$ becomes an expected excess return through

$$\alpha_{i,t} = \text{IC} \cdot \sigma_{i,t} \cdot z_{i,t}, \quad (2.9)$$

where IC is the assumed information coefficient of the signal and $\sigma_{i,t}$ the asset's forecast volatility. The scaling matters for a reason that is not cosmetic: the linear term $\alpha^\top w$ and the quadratic term $w^\top \Sigma w$ in the objective must be in compatible units, or their ratio — rather than the information in the signal — determines the solution. Chapter 8 reports what happens when this is neglected.

2.9.3 Position sizing under an estimated edge

The growth-optimal criterion of Kelly [105] gives the bet size maximising long-run log wealth; for continuous returns with signal strength s and forecast variance σ^2 , $f^* = \text{IC} \cdot s / \sigma^2$. Full Kelly is rarely appropriate in practice, because the criterion assumes IC and σ are known rather than estimated; a fractional multiple is standard, and the prototype system described in Chapter 8 uses one quarter of full Kelly with per-name and per-sector caps.

2.9.4 Risk budgeting: relative rather than absolute

Volatility-managed portfolios [120] scale total exposure by recent volatility. Chapter 8 adopts the idea but with one modification that proves decisive: the risk budget is expressed as a *fraction of the portfolio's own realised volatility* rather than as an absolute annualised target. An absolute target of, say, 15 per cent is not scale-free; it is calibrated for a concentrated equity sleeve and will essentially never bind on a diversified multi-asset sleeve whose own volatility is 6 per cent. The layer is then present in the specification, consumes computation, and does nothing. A relative budget is scale-free and remains active in any universe.

2.10 Evaluation Metrics

2.10.1 Forecast quality

For predicted values $\hat{y}_i$ and realised values y_i over n observations,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (2.10)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{DA} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{sgn}(\Delta y_i) = \text{sgn}(\Delta \hat{y}_i)]. \quad (2.11)$$

Where errors are reported as normalised, prices have been min–max scaled before the computation, so the reported magnitudes are dimensionless and comparable across instruments. Directional accuracy must always be read against the base rate of the majority class; on approximately balanced daily equity direction that base rate is close to one half, and this is the correct reference in Chapter 6.

2.10.2 Classification quality

Precision, recall and the F_1 score are reported alongside accuracy in Chapter 6 because accuracy alone conceals degenerate behaviour: a classifier that predicts the positive class unconditionally attains perfect recall and uninformative precision, and the reported results contain exactly such a case. Receiver operating characteristic analysis summarises the trade-off across thresholds; an area under the curve close to 0.5 indicates performance indistinguishable from random ranking.

2.10.3 Portfolio quality

For a return series r_t with risk-free rate r_f ,

$$\text{Sharpe} = \frac{\bar{r} - r_f}{\sigma_r} \sqrt{252}, \quad (2.12)$$

following Sharpe [106]. Maximum drawdown is the largest peak-to-trough decline of the cumulative wealth series, $\text{MDD} = \min_t (W_t / \max_{s \leq t} W_s - 1)$, and the Calmar ratio is the compound annual growth rate divided by the absolute maximum drawdown. Downside and upside capture measure the fraction of benchmark movement participated in when the benchmark falls and rises respectively; their ratio distinguishes genuine asymmetry from uniform de-risking, and Chapter 8 uses exactly this distinction.

Because the Sharpe ratio penalises upside and downside deviation alike, downside-sensitive alternatives exist — the Sortino ratio [127] divides excess return by downside deviation only — and the tail-risk measure used in the credibilistic optimiser of Chapter 7 is conditional value at risk in the form of Rockafellar and Uryasev [125], which unlike value at risk satisfies the coherence axioms of Artzner et al. [126] and is therefore well behaved under aggregation. The primary reporting in this thesis nonetheless uses Sharpe ratio and maximum drawdown, because they are what the comparison literature reports and substituting a more favourable measure would obscure rather than clarify.

2.10.4 Ablation at matched exposure

A methodological point is recorded here because it is used as an instrument rather than merely cited. Comparing a de-risked strategy against a fully invested benchmark confounds two effects: the strategy holds *less* risk on average, and it holds risk at *different times*. The first effect alone will improve a drawdown statistic and may improve a Sharpe ratio, without any skill in timing. The remedy is a matched-exposure control: a variant carrying the same average invested exposure as the full framework but allocating it uniformly rather than by regime. Any difference between the framework and that control is attributable to the timing of risk alone. This construction is possible only because the framework separates relative composition from total risk by design, and it is the central evaluation instrument of Chapter 8.

2.11 Statistical Procedures

2.11.1 Inference under serial dependence

Daily financial returns are serially dependent and heteroskedastic, so ordinary standard errors understate sampling variability. The Newey–West estimator [151] produces a heteroskedasticity- and autocorrelation-consistent covariance matrix and is used for every t statistic on a time series in this thesis. For statistics whose sampling distribution is not analytically tractable — a Sharpe ratio difference, for example — the stationary bootstrap [152] resamples blocks of random geometric length, preserving short-range dependence, and the reported interval is the empirical quantile range across resamples.

2.11.2 Comparing forecasts

The Diebold–Mariano test [153] compares the predictive accuracy of two forecasts by testing whether the mean of the loss differential $d_t = L(e_{1,t}) - L(e_{2,t})$ is zero, using a long-run variance estimate for the standard error. It is used in Chapter 6 to establish whether an observed accuracy difference between the sentiment-augmented model and its baselines is distinguishable from sampling noise.

2.11.3 Multiple testing and the credibility of a reported gain

The most important statistical point in this thesis is not a test but a disclosure discipline. Harvey and Liu [146] showed that conventional significance thresholds are inadequate once the number of strategies examined is taken into account, and Bailey and López de Prado [147] formalised the correction through the deflated Sharpe ratio, which adjusts for the number of trials, the non-normality of returns and the length of the sample. Formal multiple-comparison procedures for exactly this setting exist — White’s reality check [148] and Hansen’s test for superior predictive ability [149] both construct a null distribution over the full set of candidate strategies rather than over the selected one. The practical consequence adopted here is simpler and stricter: the *entire* specification sweep is reported rather than its best cell. Chapter 8 accordingly reports all eighteen cells of a three-seed by two-covariance-window by three-rebalance-interval grid, and states the range rather than the maximum.

2.11.4 Robustness and significance are different properties

A final distinction, applied consistently: a result is *robust* if it holds across specifications, sub-periods and universes; it is *significant* if the observed effect is unlikely under a null of no effect. These are different properties and neither implies the other. Eighteen years of daily data contain thousands of observations but only a handful of independent crisis episodes, so a drawdown claim can be highly robust and simultaneously fail a significance test, because the effective sample size for the events that matter is small. Chapter 8 reports both and does not conflate them.

2.12 Summary

This chapter has established the objects the thesis operates on: the four information streams and the cost of aligning them; the indicator families and the scale hazard that unbounded accumulators introduce; the three generations of sentiment extraction and the reason a continuous score is retained; the three fusion strategies and the precise limitation of learned attention that motivates uncertainty-weighted fusion; the learning architectures and the regularisation protocol; regime identification and the two implementation points that make it valid; causal discovery in the predictive-precedence sense and the reason correlational selection destroys usable information; portfolio construction, signal scaling and the case for a relative risk budget; the evaluation metrics, including the matched-exposure ablation that is this thesis's principal analytical instrument; and the statistical procedures, including the disclosure discipline that makes a reported gain credible. Chapter 3 now turns to what the research literature has established using these objects, and to what it has left unresolved.

Chapter 3

Literature Survey and Research Gaps

This chapter reviews the research directly relevant to the thesis. It is organised along the same four axes as the research objectives, so that the gap identified at the end of each section is the gap that the corresponding objective addresses. Emphasis is placed throughout on what each body of work establishes and, more importantly, on what it leaves unresolved. Two quantitative surveys conducted as part of this research — one of the data types used in recent multi-source studies and one of the fusion mechanisms they employ — are reported in Section 3.1, because the central gap of the thesis is visible in them directly.

3.1 Multi-Source Data Fusion in Financial Prediction

3.1.1 That fusion helps is established

The premise that heterogeneous sources outperform any single source is now well supported. Snášel et al. [1] formalised a generalised multi-source fusion framework for stock selection and demonstrated that complementary streams mitigate the individual weaknesses of each, establishing the conceptual basis for combining numerical market data with unstructured text. Long et al. [2] reached the same conclusion from a multi-view heterogeneous data perspective, showing that different views of the same security carry non-overlapping information. Nti, Adekoya and Weyori [4] reported accuracy gains of 15–20 per cent from a deep fusion of technical, fundamental and sentiment features, and — a point that recurs throughout this thesis — emphasised that sources sampled at different frequencies must be synchronised before they are combined, because misalignment can manufacture apparent predictability where none exists.

The pattern recurs across a wide range of designs. Mu et al. [8] combined investor sentiment with optimised deep learning and reported consistent gains over price-only baselines. Fu and Zhang [27] examined news and social sentiment jointly and found that the two behave differently: news sentiment supplies the more stable signal, social media the more frequent directional one — which is the empirical justification for treating

them as two streams rather than one. Shaban et al. [26] confirmed the pattern for trend forecasting; Dong and Hao [28] extended it to multi-dimensional, multi-level dynamic fusion of macroeconomic and market data; Chen et al. [29] combined multi-feature computation, penalised feature selection and a calendar-aware recurrent network to stabilise performance in unsettled periods; Zhang et al. [31] aligned text with price movements through multiple instance learning at appropriate time scales; Zhu [32] processed multi-source social media data with a long-sequence transformer to capture long-range patterns; Hong [33] predicted several correlated financial variables jointly through a multi-task structure; and Alsheebah and Al-Fuhaidi [34] reported the emerging-market case with endogenous and exogenous variables in a recurrent model.

Parallel work on single-source architectures establishes the ceiling that fusion must beat, and it is a moving ceiling. Fischer and Krauss [77] established long short-term memory as a strong baseline for financial market prediction on a large cross-section, and the subsequent surveys of deep learning in financial time series [78, 79, 80] document a decade of architectural refinement on price-derived inputs. Song [24] enriched the feature space for trend-reversal prediction and Javaid et al. [25] combined a peephole recurrent network with a time-attention layer; both improve on conventional deep baselines while remaining confined to price-derived inputs. A fusion architecture that beats an ordinary recurrent baseline has therefore demonstrated little; the relevant comparison is against the strongest available single-source model *and* against a fixed-weight combination of the same sources, and Chapter 4 is constructed to supply both.

3.1.2 A survey of data types and fusion mechanisms

To locate the gap precisely rather than by impression, thirty-six recent multi-source studies were surveyed and classified twice: by the primary data types they combine, and by the mechanism through which they combine them. Table 3.1 reports the first classification.

The distribution shows a field that has resolved the question of *what* to combine: a plurality of recent studies already fuse text with market data, and a further tranche incorporates alternative sources. Table 3.2 reports the second classification, and it shows that the question of *how* to combine has not been resolved in the same way.

The final row of Table 3.2 is the finding. Across the surveyed studies, the most sophisticated mechanisms in use learn a weighting, but they learn it once. Attention and

Table 3.1: Distribution of primary data types across thirty-six recent multi-source financial prediction studies surveyed as part of this research

Primary data category	Count	Share	Characteristic limitation
Technical / OHLCV only	9	25.0%	Cannot access narrative information by construction
Textual or sentiment combined with technical	16	44.4%	Combination mechanism is almost always static
Alternative data sources	4	11.1%	Feature extraction dominates; fusion is secondary
Reviews, multi-agent and other	7	19.4%	Do not report a directly comparable architecture
Total	36	100%	

Table 3.2: Taxonomy of fusion mechanisms in the surveyed multi-source studies, with the limitation each mechanism carries

Mechanism	Description	Key limitation	Representative work
Early fusion	Concatenation of feature vectors from all sources before modelling	Static integration; assumes fixed inter-source relationships; prone to modality dominance	[6, 8, 11, 30]
Late fusion	Averaging or stacking of predictions from source-specific models	Static, pre-determined weighting; ignores time-varying source reliability	[5, 17, 34]
Attention-based hybrid	Neural attention weights features or modalities inside the network	Weights are point estimates without uncertainty; can be unstable with noisy inputs	[7, 9, 21, 36]
Dynamic, uncertainty-aware	Weights derived from a time-varying estimate of each source’s predictive reliability	<i>Not found in the surveyed literature</i>	—

gating produce point estimates over an entire training set; normalising-flow formulations [11] model the joint distribution of sentiment and financial signals without an explicit per-source confidence; gated cross-attention [36] stabilises multimodal fusion without measuring which modality has recently degraded. Farimani et al. [5] describe their multimodal weighting as adaptive and move genuinely toward time variation, but the adaptation is learned rather than measured, and no explicit, probabilistic statement of source confidence enters the fusion. Bacco et al. [12] study prediction explicitly under high uncertainty yet still fuse sentiment and price with a static mechanism, which is precisely the configuration that should fail when uncertainty rises.

3.1.3 Emerging markets and the Indian setting

A further gap runs through the entire body of work reviewed above and is easiest to see when the studies are sorted by market. The architectures are overwhelmingly designed, tuned and validated on developed markets, and the calibrations they carry do not transfer cleanly.

Three instances make the point concrete. Yu and Liu’s causal graph formulation [89] is built on Fama–French factors [132], which are estimated on United States equities and encode a size–value–profitability structure that does not describe an index in which public-sector banks carry substantial weight, in which monsoon outcomes propagate through agricultural commodities into the stocks exposed to them, and in which regulatory decisions by a single domestic authority reset relationships across the board. Memory-augmented causal networks [90] store crisis patterns learned predominantly from United States and Chinese data; the Indian market has had crises of its own — the 2016 demonetisation shock and the 2024 budget-related volatility among them — whose dynamics do not resemble those patterns. Multi-granularity graph constructions [92] assume homogeneity across instruments, which fails where highly liquid constituents and thinly traded ones behave differently and where retail-driven noise is concentrated in the latter.

The specifically Indian evidence that does exist is narrow but pointed. Khan and Chahal [35] establish, across eleven sectoral indices, that the influence of social media sentiment on Indian sectoral returns is asymmetric and strongest at market extremes — a result that a fixed-weight fusion scheme cannot represent and that Chapter 4 recovers independently from its attention weights. Alsheebah and Al-Fuhaidi [34] demonstrate the value of exogenous variables in an emerging-market recurrent model. What is missing is a framework that jointly incorporates financial indicators, news sentiment, social media signals *and* Indian macroeconomic variables — policy rate, inflation, industrial production, currency — under one architecture and one strict temporal protocol, with the contribution of each stream measured rather than assumed. That absence is gap G2 of Section 3.6 and the subject of Chapter 4.

3.1.4 Fusion output rarely reaches a decision

Where the output of a fusion architecture is carried through to a portfolio at all, the conversion is typically heuristic. Sami et al. [20] construct holdings by statistical averaging of forecasts, without a covariance structure, position limits or an explicit risk objective. The gap between a good forecast and a good portfolio is treated as an implementation detail rather than as a modelling problem, and the consequence is that the reported economic performance of such systems is not attributable to any identifiable component.

3.2 Sentiment Analysis and Language Models in Finance

3.2.1 Three generations, and what each established

Before the technical evolution, the substantive question: does text carry market-relevant information at all? The affirmative evidence predates modern language models and is strong. Tetlock [52] showed that the pessimism of a widely read financial column forecasts downward pressure on market prices followed by reversion, and later established with co-authors [53] that the linguistic content of news stories predicts firm earnings and returns beyond the quantitative disclosure they accompany. Antweiler and Frank [55] found that the volume of internet message-board activity, not merely its polarity, predicts volatility. Bollen, Mao and Zeng [54] reported that aggregate mood extracted from social media carries predictive content for index movements. Da, Engelberg and Gao [56] demonstrated that retail attention itself, measured through search behaviour, moves prices, and Barber and Odean [57] documented the mechanism by which attention-grabbing news drives individual buying. This body of work establishes the premise on which the sentiment channel of this thesis rests; what it does not establish is how much of the information survives into an out-of-sample forecast, which is the question Chapter 6 measures.

The evolution from lexicon methods through supervised learning to transformer language models was described technically in Section 2.3. Its research significance is that each generation resolved the failure of the previous one. Loughran and McDonald [48] established that general-purpose lexicons misclassify financial language badly enough to require domain-specific alternatives, which is the finding that made financial sentiment analysis a distinct field rather than an application of general sentiment analysis; VADER [47] remains the reference rule-based implementation for social text. Super-

vised machine learning improved accuracy at the cost of annotation and generalised poorly across market conditions [23]. Transformer architectures [61] displaced both, and domain adaptation followed: FinBERT [46] adapted bidirectional encoders to financial language; BloombergGPT [43] and the FinGPT family [44, 45] extended the approach to generative reasoning over financial documents. Surveys [37, 38, 39] converge on the finding that contextual language models extract market-relevant meaning that lexicon methods miss.

3.2.2 Applications and their common evaluation weakness

Applications have followed quickly and cover a wide range of markets and instruments. Fatouros et al. [41] showed that conversational language models outperform established financial sentiment baselines. Wilksch and Abramova [42] built a dedicated model for sentiment in financial tweets. Haryono et al. [9] coupled a transformer news encoder with a gated recurrent unit for technical data and later extended the approach to generative aspect-based sentiment [10]. Ho and Huang [22] combined sentiment with candlestick chart representations. Jung and Jang [13] preserved quantitative context by masking numerically significant tokens. Ónózó et al. [15] applied language models to news analysis and macroeconomic nowcasting. Vargas et al. [17] fused sentiment with recurrent price models on Brazilian equities, Wang and Chen [18] built a sentiment-based recommendation system, Shang [19] combined high-dimensional indicators with social media dynamics, Elahi and Taghvaei [58] combined financial data with news articles using language models directly, Officioso et al. [60] coupled financial news sentiment with market data in a language-model-based predictor, and Mishra et al. [59] and Liu et al. [14] surveyed the resulting body of work.

What this literature establishes is that language models improve sentiment *measurement*. What it establishes far less clearly is how much of that improvement survives into *forecast quality*. Three evaluation weaknesses recur. First, the reported gain is frequently stated without the baseline against which it should be judged: an accuracy of 55 per cent is impressive against a balanced base rate and unremarkable against a class imbalance of the same magnitude. Second, temporal alignment is often described loosely, and the attribution of a post-close publication to the session that has already ended is a subtle form of look-ahead that inflates results without leaving a trace. Third, evaluation stops at the classification metric; the question of whether a measured increment in accuracy is

large enough to survive transaction costs is rarely posed.

3.2.3 Market-derived sentiment as a distinct construct

A second and older family of sentiment measures reads sentiment off the market rather than off text. Baker and Wurgler [128] construct investor sentiment from trading activity and dispersion, on the argument that participation reveals what surveys merely report. Singsiri and Yokrattanasak [129] build allocation signals from money-flow indicators on exactly this reasoning, and Naidoo et al. [130] model investor sentiment and market volatility as a joint object using a mixed-frequency conditional heteroskedasticity framework.

This family matters to the present thesis for a specific structural reason. Instrument-level, time-stamped news corpora of the depth required for a cross-sectional study do not exist for broad index instruments such as exchange-traded funds. A cross-sectional allocation study over such instruments therefore cannot use textual sentiment and must use a market-derived construct, which is what Chapter 8 does. The consequence — that a null result on the market-derived composite is a null about *that proxy*, not about textual sentiment in general — is recorded there and is one of the more important qualifications in the thesis.

3.3 Explainability, Causality and Uncertainty

3.3.1 Post-hoc attribution and its limit

Opacity is a recognised obstacle to the adoption of learned models in finance. Koa et al. [81] used self-reflective large language models to generate natural-language explanations alongside stock predictions, and the general-purpose attribution methods of Lundberg and Lee [82] and Ribeiro et al. [83] are now standard instruments for post-hoc interpretation of fitted models.

The limitation is structural rather than a matter of method quality. Post-hoc attribution explains a model: it reports which inputs the fitted function relied upon in reaching a particular output. It does not establish that the relationships the model has learned are anything more than correlations that happened to hold in the training window. A feature that co-moves with the target for spurious reasons receives an attribution exactly as a genuine driver would, and an audience shown that attribution has been given a faithful

account of the model and a potentially misleading account of the market.

This concern is not idiosyncratic to finance. The comprehensive survey of explainable artificial intelligence by Barredo Arrieta et al. [84] distinguishes transparency by design from post-hoc interpretability and notes that the two answer different questions; Doshi-Velez and Kim [85] argue that an interpretability claim is empty without an explicit statement of what the explanation is for and who must act on it; and Rudin [86] makes the stronger case that for high-stakes decisions an intrinsically interpretable model should be preferred to an explained black box. The position taken in this thesis is a version of Rudin's applied to the feature set rather than to the model class: constrain *what may enter* the model by a structural criterion, and the resulting system becomes auditable regardless of how the fitting is subsequently done.

3.3.2 Causal formulations

Causal formulations address this directly. Xu, Cheng and Lee [87] showed that causal feature selection is more robust out of sample than correlation-based feature engineering, separating genuine causal drivers from spurious associations. Their causal graph is static, however, and cannot represent the structural breaks produced by policy announcements — a limitation that is acute in the Indian setting, where budget and monetary policy events reset inter-instrument relationships within hours.

Liu et al. [88] introduced a dynamic-causal lens that encodes stock interactions as symbolic movement patterns and adapts to changing conditions. The formulation is the closest in the literature to what the Indian setting requires, and its cost is computational: the approach scales quadratically in the number of instruments, which is prohibitive for a fifty-constituent index. Yu and Liu [89] coupled factor investing with causal graph neural networks through a fuzzy cognitive map, demonstrating that multi-factor causal structure improves predictive power; their factors are Fama–French factors calibrated to United States equities, which do not reflect Indian idiosyncrasies such as the weight of public-sector banks or the seasonal influence of the monsoon on agricultural commodities and the stocks exposed to them. Luo et al. [90] added memory to a causal interaction network, with the attendant risk of over-fitting to historical crises whose character does not repeat; and Xing et al. [93] modelled the vagueness of inter-stock relations, recognising that inter-stock relationships lie on a continuum rather than being crisp, without supplying calibration guidance for an emerging-market setting.

3.3.3 Multi-granularity temporal structure

Alongside the causal line, multi-granularity architectures demonstrated that financial series carry structure at several time scales at once. Zhu et al. [92] processed minute-, day- and week-level data jointly and showed that cross-granularity information transfer improves trend prediction; their graph construction assumes homogeneity across instruments, which fails in a universe where highly liquid constituents behave differently from thinly traded ones. Wang et al. [91] added a debiasing layer to multi-scale temporal memory learning that reduces historical bias in trend forecasting.

The debiasing layer carries an assumption that deserves emphasis, because Chapter 7 reports it failing empirically. The correction assumes *stationary* noise properties, so that a bias estimated on past data remains valid. Indian market volatility is not stationary: it clusters, responds asymmetrically to negative returns, and changes character discretely at policy events. A debiasing layer built on a stationarity assumption is therefore least reliable at exactly the moments when reliability matters most.

3.3.4 Relational learning across the cross-section

Relational formulations extend the same idea to the cross-section. Patel, Jariwala and Chattopadhyay [94] treat securities as nodes in a learned relational graph and show that relational modelling improves directional accuracy over independent treatment; their model predicts without dynamic risk adjustment, which is a serious limitation for a market with intraday reversals. Liu et al. [95] model mutual interplay through dual-dynamic attention across cross-sectional and temporal dimensions, with scalability to large universes not fully addressed. Gao et al. [96] select stocks through probabilistic state-space learning, assuming linear Gaussian transitions that are strained by the nonlinear dynamics of election periods and policy announcements.

3.3.5 Uncertainty as a first-class quantity

A related and less examined issue is that a signal's reliability is itself uncertain. Most fusion architectures pass point predictions forward without any statement of confidence, so a source that has recently become noisy continues to be weighted as though it had not. In Bayesian machine learning a prediction is incomplete without an estimate of its uncertainty, and the distinction between *aleatoric* uncertainty — irreducible noise in the

environment — and *epistemic* uncertainty — the model’s ignorance about regions it has not observed — is well established [75, 76]. Khuwaja, Khowaja and Dev [16] address heterogeneity in financial data through adversarial learning without making per-source confidence explicit. Across the fusion literature surveyed in Section 3.1, explicit, time-varying uncertainty quantification per source is absent.

3.4 From Prediction to Decision: Allocation and Validation

3.4.1 Portfolio construction and the estimation problem

Portfolio construction has its own long literature. Markowitz [104] established the mean-variance formulation, whose extreme sensitivity to estimated inputs motivated both shrinkage estimators [107] and the demonstration by DeMiguel, Garlappi and Uppal [108] that naive equal weighting can outperform optimisation that ignores estimation error. That demonstration is the reason equal weighting is the primary benchmark in Chapter 8: a framework that cannot beat $1/N$ has not earned its complexity. Risk-based construction, of which the hierarchical approach of López de Prado [113] is representative, avoids the estimation of expected returns entirely by forgoing signal information — a defensible trade that nonetheless defines away the question this thesis asks.

3.4.2 Regime-aware and risk-managed allocation

Regime-switching models following Hamilton [116] showed that financial series are not homogeneous through time. Ang and Bekaert [117] demonstrated that accounting for regime shifts materially changes optimal allocations, and Kritzman, Page and Turkington [118] made the practical case for regime-aware dynamic strategies. Luo and Mulvey [119] extend the line with dual-regime signals and regime-dependent asset selection. Volatility-managed portfolios [120] scale total risk by recent volatility and improve risk-adjusted returns, but say nothing about composition: the mechanism is orthogonal to which assets are held.

Kumar and Yadav [121] proposed credibilistic portfolio selection under a range directional measure formulation, handling positive and negative returns without assuming normality, which is a genuine advance for return distributions that are neither symmetric nor thin-tailed. The formulation is not coupled to a dynamic signal, so rebalancing occurs at fixed intervals rather than in response to a change in the underlying relationships.

Gu, Du and Wang [122] generate risk-aware prediction intervals rather than point forecasts, requiring calibration that is not supplied for emerging-market instruments; Zhu et al. [123] build sparse portfolios driven by irregular news streams, using English-language sources only, which for an Indian deployment omits a substantial share of the information actually reaching retail participants.

3.4.3 Evaluation practice and the credibility of a reported gain

Evaluation practice is the final concern and, in the judgement of this thesis, the most consequential. Risk-adjusted performance is still summarised by the ratio introduced by Sharpe [106], and Gu, Kelly and Xiu [137] provide the benchmark machine-learning study in asset pricing, noting that measured predictability is small. Harvey and Liu [146] showed that conventional significance thresholds are inadequate once the number of strategies examined is taken into account, and Bailey and López de Prado [147] formalised the deflated Sharpe ratio, which adjusts for the number of trials, the non-normality of returns and the sample length. Valls and Steurer [150] document in detail how volatility-based strategies overfit and what the resulting reported performance is worth. Robust inference under serial dependence requires corrections [151], resampling schemes [152] and forecast-comparison tests [153].

The practical consequence adopted throughout this thesis is a disclosure discipline rather than a test: a reported gain is credible only when the specification search behind it is reported alongside it. Reporting the best cell of a grid and omitting the grid is not a presentational choice but a misstatement of the evidence.

3.5 Consolidated Review and Positioning

Table 3.3 consolidates the representative work reviewed above, stating for each the contribution that is relevant to this thesis and the limitation that this thesis must work around. The final column is not a criticism of the cited work, which is in each case addressing the question it set itself; it records the reason the work cannot be adopted unmodified here.

3.6 Research Gaps

The review supports five gaps. Figure 3.1 positions them against the state of the art along the three axes of fusion strategy, signal extraction and decision layer.

Table 3.3: Consolidated review of representative work, with the limitation relevant to this research

Reference	Technique	Contribution relevant here	Limitation to be addressed
Snášel et al. [1]	Generalised multi-source fusion for stock selection	Formal basis for combining numerical and textual streams	Fusion weights are not conditioned on the prevailing market state
Long et al. [2]; Nti et al. [4]	Multi-view and multi-source deep fusion	Views carry non-overlapping information; frequency alignment is essential	Developed and tested outside emerging-market conditions
Fu and Zhang [27]	Multi-source market sentiment with price data	News and social sentiment behave differently and should be separate streams	Combination weights are fixed across regimes
Haryono et al. [9]; Mu et al. [8]	Sentiment fused with technical indicators	Gated and optimised fusion of text and price streams	Gate learned globally; sentiment quality assumed rather than validated
Farimani et al. [5]	Adaptive multimodal weighting	Moves toward time-varying source importance	No explicit probabilistic confidence per source
Araci [46]; Wu et al. [43]	Domain-adapted language models	Contextual sentiment extraction from financial text	Gain in sentiment quality is not traced through to decision quality
Jung and Jang [13]	Numerical-masking adaptation	Preserves quantitative context that generic pre-processing destroys	Evaluated on measurement quality, not on forecast value
Koa et al. [81]; Lundberg and Lee [82]	LLM explanations; post-hoc attribution	Natural-language and feature-level justification	Explains the model, not the causal structure
Xu et al. [87]	Causal perspective on stock prediction	Causal features are more robust out of sample	Static graph cannot absorb policy shocks
Liu et al. [88]	Dynamic-causal graph from symbolic movement patterns	Structure adapts to changing conditions	Scales quadratically; no sector-specific adaptation
Zhu et al. [92]	Multi-granularity graph augmented learning	Several time scales processed jointly	Assumes homogeneity across instruments
Wang et al. [91]	Multi-scale temporal memory with debiasing	Multiple horizons processed jointly	Debiasing assumes stationary noise
Kumar and Yadav [121]	RDM-DEA credibilistic portfolio selection	Optimisation without a normality assumption	Not coupled to a dynamic signal for rebalancing
Moreira and Muir [120]	Volatility-managed portfolios	Total risk scaled by recent volatility	Does not address portfolio composition
Zhu et al. [123]	Sparse portfolios from irregular news	Sparse allocation responsive to news flow	English-only sources; omits regional-language sentiment
DeMiguel et al. [108]; Harvey and Liu [146]	Naive diversification; backtesting under multiple testing	Estimation error can defeat optimisation; gains must account for specification search	Sets the benchmark any framework must beat and requires every specification to be disclosed

G1. Static fusion of dynamically reliable sources. Existing frameworks fuse financial, textual and macroeconomic streams using fixed weights, globally learned attention or simple averaging. As Table 3.2 records, none of the surveyed studies conditions the contribution of a source on a measured, time-varying estimate of that source’s own predictive reliability. A stream that has become noisy therefore continues to influence the forecast at its historical weight until the next retraining cycle.

G2. Incomplete coverage of the Indian information environment. Most multi-source frameworks are designed and validated on developed markets. Frameworks that jointly incorporate financial indicators, news sentiment, social media signals and Indian macroeconomic variables — policy rate, inflation, industrial production, currency — under a single architecture and a strict temporal protocol are

rare, and the calibrations imported from developed markets (Fama–French factors, crisis memory, homogeneity assumptions) do not transfer cleanly.

G3. Unverified translation from language-model sentiment to decision quality.

Language models demonstrably improve sentiment measurement. Whether that improvement survives temporal alignment, fusion with technical indicators and honest out-of-sample evaluation at a single-name horizon has not been established, and reported gains are seldom accompanied by the baseline against which they should be judged or by an assessment of whether the increment is large enough to act on.

G4. Correlational rather than mechanistic transparency. Post-hoc attribution explains a fitted model. It does not distinguish a genuine driver from a spurious association, and it does not describe how a sentiment shock propagates into a price movement over time. A transparency mechanism grounded in directed causal structure and in multi-scale temporal dynamics is required, together with a direct measurement of how much noise the sentiment channel actually carries — as distinct from an assumption that it carries little.

G5. Prediction reported in place of decision. Statistical accuracy remains the dominant success criterion, and economic value is reported inconsistently. Where allocation is addressed it is often heuristic, and the contribution of individual architectural layers is almost never isolated, so architectures accumulate components whose value has never been tested. Cross-market transfer is asserted more often than it is demonstrated, and the conditions under which a framework should be expected to transfer are rarely stated in advance.

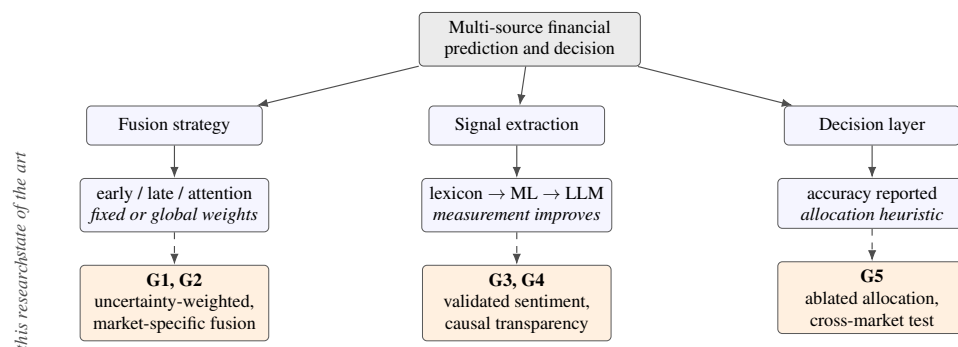


Figure 3.1: Taxonomy of the reviewed field and the location of the five research gaps addressed by this work.

3.7 Problem Statement and Research Objectives

3.7.1 Problem statement

To design, develop and validate an integrated decision-support framework that fuses heterogeneous financial information — market metrics, news and social media sentiment extracted using large language models, and macroeconomic indicators — into interpretable and reliability-aware predictions, and that converts those predictions into executable, risk-controlled stock market decisions whose performance is demonstrated across market states and across markets. The framework must address the limitations of existing approaches, namely static source weighting under non-stationary reliability, opaque and correlational reasoning, unquantified sentiment noise, and the routine substitution of statistical accuracy for economic value.

3.7.2 Research objectives

- RO1.** To investigate and develop a comprehensive model that integrates diverse data sources — such as financial metrics, news articles, social media sentiment and macroeconomic factors — aiming to enhance the accuracy of stock market predictions by providing a holistic analysis.
- RO2.** To design a hybrid framework that combines large language models, sentiment analysis techniques and conventional financial indicators, focusing on optimising deep learning architectures for real-time stock market predictions.
- RO3.** To establish and evaluate an explainable artificial intelligence framework capable of delivering transparent and interpretable predictions in stock selection, analysing the impact of sentiment data noise and exploring the temporal dynamics between sentiment shifts and stock price movements.
- RO4.** To evaluate and validate the result and practical application of the proposed model across different stock markets, exploring transfer learning techniques for cross-market applications and validating the model against benchmark datasets and real-world market data.

3.7.3 Alignment of gaps, objectives and contributions

Figure 3.2 traces each gap to the objective that addresses it and to the contribution reported in the corresponding chapter. The mapping is many-to-many by design: G1 and G2 are jointly answered by the fusion work of RO1 across Chapters 4 and 5, while G5 requires evidence from both the decision layer of RO4 and the transparency instruments of RO3.

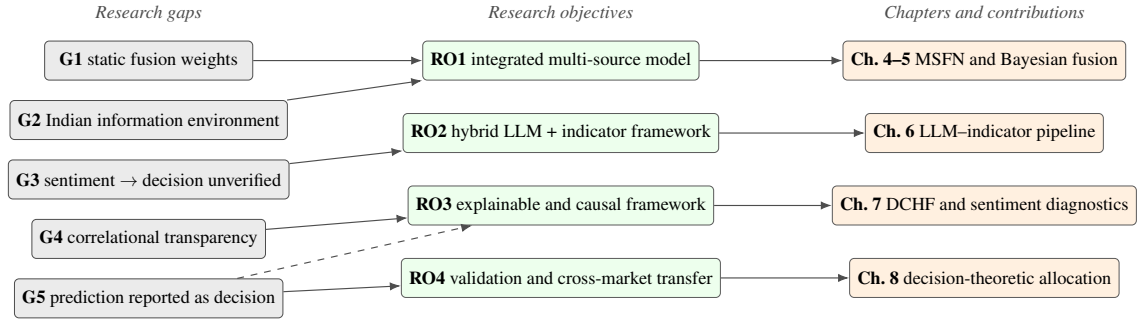


Figure 3.2: Alignment of research gaps, research objectives and the chapters in which the corresponding contributions are developed. Dashed links denote partial rather than primary coverage.

3.8 Summary

The literature establishes that multi-source fusion improves financial prediction, that language models improve sentiment measurement, that causal feature selection is more robust out of sample than correlational selection, and that regime awareness changes optimal allocations. It leaves four things unresolved that this thesis addresses: fusion weights are learned once rather than measured continuously; the Indian information environment is under-represented and the calibrations imported from developed markets do not transfer; the translation from improved sentiment measurement to improved decisions is asserted rather than demonstrated; and the decision layer is evaluated as an aggregate, so that the contribution of individual architectural components is unknown. The next five chapters address these in turn.

Chapter 4

Multi-Source Data Integration Framework

Research Objective 1. *To investigate and develop a comprehensive model that integrates diverse data sources — such as financial metrics, news articles, social media sentiment and macroeconomic factors — aiming to enhance the accuracy of stock market predictions by providing a holistic analysis.*

This chapter develops the first component of the thesis: a hierarchical architecture that integrates four heterogeneous streams of Indian market information and produces a forecast whose improvement over simpler alternatives can be *attributed* to identifiable causes rather than merely observed. Section 4.2 specifies the architecture, Section 4.3 the data and the alignment protocol, Section 4.4 the experimental design and its assumptions, and Sections 4.5 to 4.9 the evidence. Section 4.12 states what the chapter does not establish, which is what motivates Chapter 5.

The work reported in this chapter appears in publication [P1].

4.1 Design Rationale

Three design decisions distinguish the Multi-Source Fusion Network from the fusion architectures reviewed in Section 3.1, and each is a response to a limitation identified there.

Modality-specific encoding before cross-modal combination. Early fusion concatenates raw features from every source and asks a single architecture to learn both within-domain and cross-domain structure. That is inefficient and prone to modality dominance: a stream with many features or large numerical magnitudes crowds out one with few. The architecture developed here learns a representation *within* each modality first, using an encoder matched to that modality’s structure, and combines only the learned representations. Each stream is therefore encoded in the form appropriate to it — convolution over indicator sequences, transformer encoding over text, engineered temporal features over slow macroeconomic series — before any cross-modal interaction occurs.

Time-varying rather than fixed source weighting. The combination is performed by cross-modal attention, with relevance weights recomputed at every time step rather than fixed by the architecture. This is what allows the model to express the proposition that news mattered more than price history on a particular day. Section 4.7 examines the learned distribution and shows that it is genuinely state-dependent rather than approximately constant, which is the empirical justification for the mechanism.

Attributability as a design requirement. The architecture is constructed so that each stream can be removed and the model retrained without changing anything else. This is not free: it constrains the encoders to be modular and prevents certain forms of parameter sharing. The return on that constraint is Section 4.6, which reports the marginal value of each stream rather than asserting that all of them help.

4.2 Architecture of the Multi-Source Fusion Network

The architecture is organised as five layers, shown in Figure 4.1: acquisition, preprocessing, feature extraction, fusion and prediction.

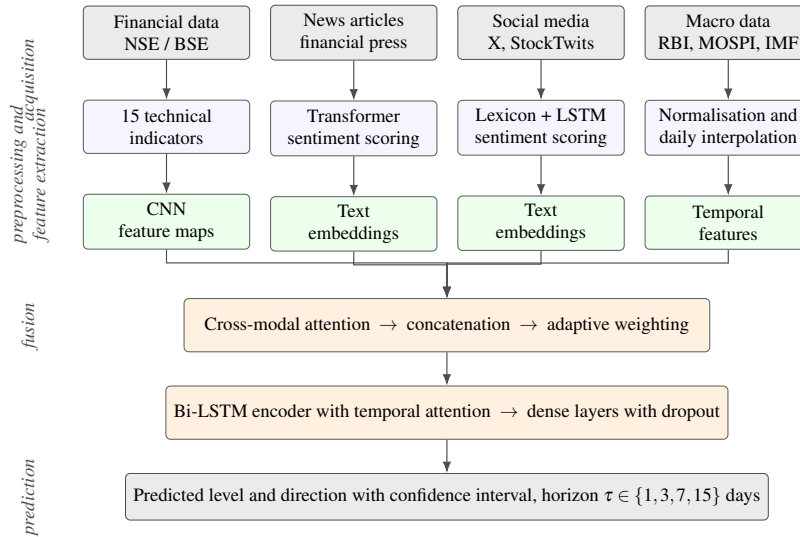


Figure 4.1: Architecture of the Multi-Source Fusion Network, showing the flow from heterogeneous acquisition through modality-specific preprocessing and feature extraction to cross-modal fusion and prediction.

4.2.1 Acquisition and preprocessing

The acquisition layer collects raw inputs from four interfaces: exchange price and volume records, financial press feeds, social media platforms, and official macroeconomic publications. The preprocessing layer standardises them. Technical indicators are com-

puted from closing series; news and social text are scored; macroeconomic series are normalised and aligned to a daily frequency. Section 4.3 gives the detail and states the cost of each alignment decision.

4.2.2 Modality-specific encoders

Each preprocessed modality m is passed through a stack of one-dimensional convolutional layers,

$$h_m^{(l)} = \text{ReLU}\left(\text{Conv1D}\left(W_m^{(l)} * h_m^{(l-1)} + b_m^{(l)}\right)\right), \quad (4.1)$$

where $h_m^{(0)}$ is the raw feature matrix of modality m , $W_m^{(l)}$ and $b_m^{(l)}$ are the learnable convolution kernel and bias at layer l , $*$ denotes convolution along the time axis, and max-pooling follows each convolutional block. Convolution is appropriate for indicator streams because the patterns of interest — a crossover, a band squeeze, a volume spike — are local and translation-invariant in time. For the two textual modalities the same notation applies to the sequence of daily aggregated scores; the contextual reading of individual documents has already been performed by the sentiment encoders described in Section 4.3.

4.2.3 Cross-modal attention

The fusion layer computes a relevance weight for each modality by scaled dot-product attention over the modality representations,

$$\alpha_m = \text{softmax}\left(\frac{q \cdot k_m}{\sqrt{d_k}}\right), \quad (4.2)$$

where q is a query vector learned during training, k_m is the key vector of modality m and d_k is the key dimension. The fused representation at session t is the attention-weighted combination of the modality representations, $F_t = \sum_m \alpha_m^{(t)} h_m^{(L)}(t)$, followed by concatenation with the unweighted representations so that no modality can be suppressed entirely by a degenerate attention distribution.

The essential property is that α_m is recomputed at every time step. A fixed-weight scheme — whether a hand-set constant or a single set of learned coefficients — asserts that the relative informativeness of price history and news is the same in a calm trending market as during an earnings announcement. Equation (4.2) does not make that assertion, and Section 4.7 shows that the data do not support it.

4.2.4 Temporal modelling and output

The fused representation is passed through two bidirectional recurrent layers,

$$\vec{h}_t = \text{LSTM}(F_t, \vec{h}_{t-1}), \quad \overleftarrow{h}_t = \text{LSTM}(F_t, \overleftarrow{h}_{t+1}), \quad h_t = [\vec{h}_t; \overleftarrow{h}_t], \quad (4.3)$$

with a temporal attention layer over the hidden sequence, and the forecast is produced by dense layers with dropout,

$$\hat{y}_{t+\tau} = W_o \cdot \text{Dropout}(h_t) + b_o, \quad \tau \in \{1, 3, 7, 15\} \text{ days}. \quad (4.4)$$

The backward pass in Equation (4.3) reads the input window in reverse; this is context *within* the supplied window, which closes at the decision time, and is therefore not look-ahead. Confidence intervals are produced from the dispersion of the dense-layer output under repeated dropout sampling.

4.3 Data, Sources and Temporal Alignment

4.3.1 Study universe and period

The study covers January 2015 to December 2024 — ten years, spanning demonetisation, the introduction of the goods and services tax, the COVID-19 crash and recovery, and the inflation-driven repricing of 2022 — on the NIFTY 50 index and ten sectoral indices: financials, information technology, pharmaceuticals, energy, fast-moving consumer goods, automobiles, metals, real estate, consumer durables and healthcare. Index-level data are used rather than single names, which removes survivorship bias by construction, although composition drift through constituent revisions remains.

Table 4.1 records the four streams, their sources, frequencies and extracted attributes.

4.3.2 Feature engineering by modality

Financial metrics. Fifteen technical indicators are derived from the price and volume record, following the definitions in Section 2.2: moving averages over 5, 10 and 20 sessions and the slope of the exponential average for trend; the relative strength index and the moving average convergence divergence with its signal line for momentum; Bollinger band width and position, average true range and realised volatility for volatility; and on-

Table 4.1: Multi-source data characteristics for the Indian market study, January 2015 to December 2024

Category	Source	Frequency	Attributes extracted
Financial metrics	NSE interfaces, Yahoo Finance	Daily	Open, high, low, close, volume; fifteen derived technical indicators covering trend, momentum, volatility and participation
News articles	Economic Times, Moneycontrol, Bloomberg Quint	Real time	Headline and body text, publication time stamp, source authority weight, computed polarity
Social media	X (formerly Twitter), StockTwits	Real time	Post text, engagement counts, author following; daily aggregate polarity, mention volume, sentiment dispersion
Macroeconomic	RBI Bulletin, MOSPI, IMF	Monthly / quarterly	GDP growth, CPI and WPI inflation, repo rate, index of industrial production, crude oil price

balance volume, volume-weighted average price derivatives and volume momentum for participation.

News sentiment. Financial news is scored by a domain-adapted transformer encoder of the FinBERT family [46]. For each trading session the scores of all articles referencing the target index or sector are aggregated, weighted by relevance to the instrument and by an authority weight attached to the source. Domain adaptation matters here for the reason Loughran and McDonald [48] identified: a general-purpose model reads “provision” and “liability” as negative in text where they are neutral accounting vocabulary.

Social media sentiment. Posts from X and StockTwits are scored by a hybrid model combining the VADER lexicon [47], which handles the intensifiers, negations and orthographic emphasis characteristic of social text, with a recurrent classifier fine-tuned on financial social media. Three daily aggregates are retained: mean polarity, mention volume and sentiment dispersion. Dispersion is included deliberately: a day on which opinion is uniformly positive carries different information from one on which the mean is the same but opinion is split, and it also serves as a partial diagnostic for coordinated posting.

Macroeconomic indicators. Six series are used: GDP growth, consumer price and wholesale price inflation, the RBI repo rate, the index of industrial production, and crude oil prices, obtained from RBI bulletins and official statistics. The series are normalised and interpolated to daily frequency.

4.3.3 Temporal alignment and its costs

Every stream is aligned to the NSE trading calendar under the four rules of Section 2.5. Two alignment decisions introduce distortions that are stated here rather than left implicit, because they bound the interpretation of the results.

The first is macroeconomic interpolation. Monthly and quarterly series are interpolated to daily frequency, which assumes that information flows smoothly between releases. It does not: macroeconomic effects are concentrated on announcement days. The direction of the resulting bias is known — interpolation *understates* short-horizon announcement-day effects — so the macroeconomic contribution measured in Section 4.6 should be read as a lower bound on the contribution an event-aware treatment would find.

The second is daily aggregation of text. Aggregating news and social posts to a daily score discards the ordering of items within the day, trading microstructure detail for tractability. At a daily decision horizon this is an acceptable trade, but it means the framework cannot express the difference between a positive headline that preceded a negative one and the reverse.

4.4 Experimental Setup

4.4.1 Partitioning and leakage control

The data are partitioned chronologically into three contiguous blocks: training from January 2015 to December 2021 (70 per cent), validation from January 2022 to June 2023 (15 per cent), and test from July 2023 to December 2024 (15 per cent). All features are standardised to zero mean and unit variance using *training-set statistics only*. Hyperparameters are selected on the validation block and the test block is examined once, after the configuration is fixed.

4.4.2 Hyperparameter selection

Hyperparameters were tuned by grid search on the validation block over learning rates $\{10^{-2}, 10^{-3}, 10^{-4}\}$, recurrent hidden sizes $\{64, 128, 256\}$, convolutional filter counts $\{32, 64, 128\}$, dropout rates $\{0.2, 0.3, 0.5\}$ and batch sizes $\{32, 64, 128\}$. The selected configuration is reported in Table 4.2.

Random seeds for data shuffling and weight initialisation were fixed across all runs, and

Table 4.2: Selected architecture and training configuration for the Multi-Source Fusion Network

Component	Setting	Selection basis
Convolutional encoder	64 filters, kernel size 3, max-pooling	Validation loss; larger filter counts did not improve validation error
Recurrent encoder	Two bidirectional layers, 128 hidden units	Validation loss; 256 units over-fitted within the patience window
Dropout	0.3 on dense layers	Validation loss
Optimiser	Adam, learning rate 10^{-3} [73]	Grid search
Batch size	64	Grid search
Early stopping	Patience 10 epochs on validation loss	Fixed <i>a priori</i>
Trainable parameters	≈ 2.1 million, majority in the recurrent layers	—
Training time	Under two hours on a single consumer-grade accelerator	—
Inference latency	Milliseconds per forward pass	—

an identical preprocessing pipeline was applied to every model variant, so that reported comparisons reflect architectural differences alone rather than initialisation noise.

4.4.3 Baselines

Six baselines are used, chosen so that the comparison isolates specific claims rather than merely demonstrating that the proposed model wins.

- **ARIMA**, the classical statistical reference, establishing what linear time-series structure alone achieves.
- **LSTM (price only)** and **Transformer (price only)**, establishing the ceiling for a single-source deep model.
- **LSTM + sentiment** and **LSTM + macro**, establishing the incremental value of each added stream under a simple architecture.
- **Simple-average ensemble**, combining the single-source models with fixed equal weights. This is the critical baseline: it consumes *exactly the same four sources* as the proposed model, so any difference between the two isolates the fusion mechanism from the data.

4.4.4 Experimental assumptions

Four assumptions are embedded in the evaluation and are recorded with their consequences. First, macroeconomic interpolation understates announcement-day effects, as discussed above. Second, daily sentiment aggregation discards within-day ordering. Third, the evaluation assumes frictionless markets: transaction costs, slippage and market impact are not modelled, so directional accuracy does not translate directly into trading profit — a point taken up seriously in Chapter 8, where costs are charged. Fourth, relationships learned in the training window are assumed to hold in the test window; structural breaks would violate this, which is precisely why the walk-forward analysis of Section 4.9 was conducted.

4.5 Results: Prediction Performance

Table 4.3 reports one-day-ahead prediction on the NIFTY 50 index against the six baselines, together with the source ablation that is discussed in Section 4.6. The two blocks are presented as a single table because the two questions they answer — does fusion help, and which source is responsible — are answered by the same experiment.

Table 4.3: One-day-ahead prediction on the NIFTY 50 index: comparison against baselines (upper block) and source ablation of the proposed model (lower block)

Configuration	Data sources	MAE	RMSE	R^2	DA (%)
<i>Baseline comparison</i>					
ARIMA	price only	0.124	0.167	0.721	52.3
LSTM	price only	0.087	0.112	0.845	61.7
Transformer	price only	0.079	0.103	0.869	64.2
LSTM + sentiment	price, sentiment	0.052	0.071	0.921	72.8
LSTM + macro	price, macro	0.061	0.082	0.903	68.5
Ensemble (simple average)	all, fixed weights	0.043	0.059	0.942	76.3
MSFN (proposed)	all, adaptive fusion	0.018	0.024	0.988	84.7
<i>Source ablation of the proposed model</i>					
– financial metrics	news, social, macro	0.041	0.055	0.946	73.2
– news sentiment	price, social, macro	0.027	0.036	0.971	79.1
– social media	price, news, macro	0.025	0.033	0.976	80.8
– macroeconomic	price, news, social	0.031	0.040	0.965	76.5
– both sentiment sources	price, macro	0.038	0.049	0.952	74.0
price and technical only	price only	0.079	0.103	0.869	64.2

Three readings follow from the upper block.

Deep architectures beat linear structure, but not decisively. The price-only recurrent

model improves on ARIMA from a mean absolute error of 0.124 to 0.087 and from 52.3 to 61.7 per cent directional accuracy; the price-only transformer improves further to 0.079 and 64.2 per cent. The gain is real and it is bounded: a better architecture on the same information buys roughly twelve percentage points of directional accuracy over a linear model and then plateaus.

Additional sources beat better architectures on the same source. Adding sentiment to a plain recurrent model raises directional accuracy to 72.8 per cent — a larger gain than replacing the recurrent model with a transformer on price alone. The ordering is informative: information is scarcer than modelling capacity in this problem.

The fusion mechanism, not the data, produces the headline result. The decisive comparison is between the proposed model and the simple-average ensemble, because the two consume identical inputs. The ensemble attains a mean absolute error of 0.043 and 76.3 per cent directional accuracy; the proposed model attains 0.018 and 84.7 per cent. Since the data are held constant, the difference is attributable to how the sources are combined. This is the central claim of the chapter, and it is the reason the ensemble baseline was constructed.

4.6 Source Contribution: Ablation

The lower block of Table 4.3 removes one stream at a time and retrains. Figure 4.2 presents the degradations directly.

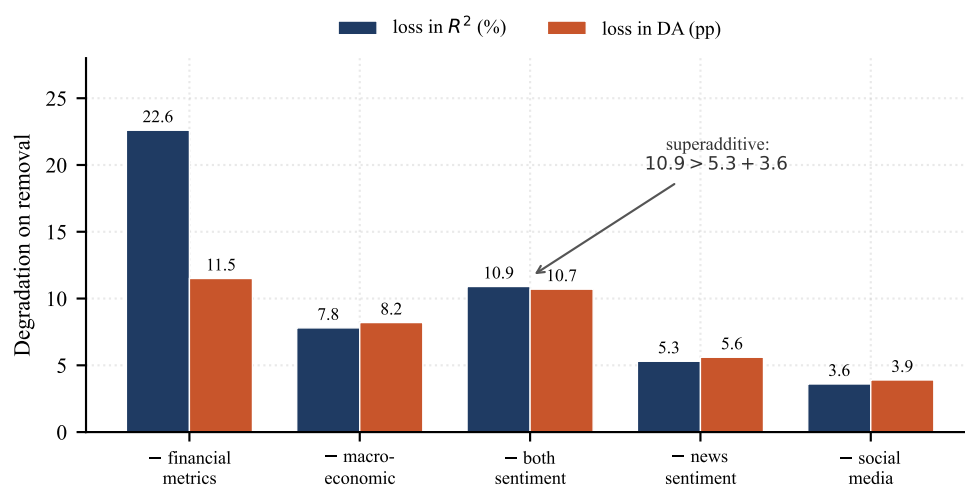


Figure 4.2: Marginal cost of removing each source from the fused model, measured as the loss in explained variance and in directional accuracy. The joint removal of both textual sources costs more than the sum of their individual removals.

Financial metrics are foundational but not sufficient. Their removal is the single most damaging ablation, costing 22.6 per cent of explained variance. This is the expected result and would be worrying if it were otherwise. The complementary observation is more interesting: restricting the model to price and technical features alone costs 32.0 percentage points of directional accuracy relative to the full model, so the numerical stream is necessary and a long way from sufficient.

Macroeconomic indicators carry substantial weight. Their removal costs 7.8 per cent of explained variance. Given that interpolation understates announcement-day effects, this is a lower bound. It is consistent with the structural observation of Section 2.1.2 that Indian equity valuations are strongly sensitive to policy rates, inflation and industrial output.

The two textual sources are complementary, not redundant. This is the most informative result in the table. Removing news sentiment alone costs 5.3 per cent of explained variance; removing social media alone costs 3.6 per cent; removing both costs 10.9 per cent. Since $10.9 > 5.3 + 3.6 = 8.9$, the joint effect exceeds the sum of the individual effects. That superadditivity is the quantitative signature of complementary information: each textual source partially covers for the other's absence, so removing one is cheap and removing both is expensive. It is also the empirical justification for a fusion architecture rather than a larger single-source model — a model with more capacity on price data alone cannot recover information that is not in the price data.

The finding is consistent with Fu and Zhang [27], who report that news sentiment supplies the more stable signal and social media the more frequent directional one. Two streams that differ in that way should be expected to substitute imperfectly for one another, which is exactly what the superadditivity measures.

4.7 Where the Gain Comes From: Attention Analysis

The ablation establishes that all four sources contribute. The attention distribution establishes *when* each contributes. Figure 4.3 reports the learned weights under three market conditions.

Averaged over the whole sample, financial metrics carry the highest weight at 35 per cent, macroeconomic indicators 25 per cent, news sentiment 22 per cent and social media 18 per cent — so the two textual sources jointly account for 40 per cent of the attention

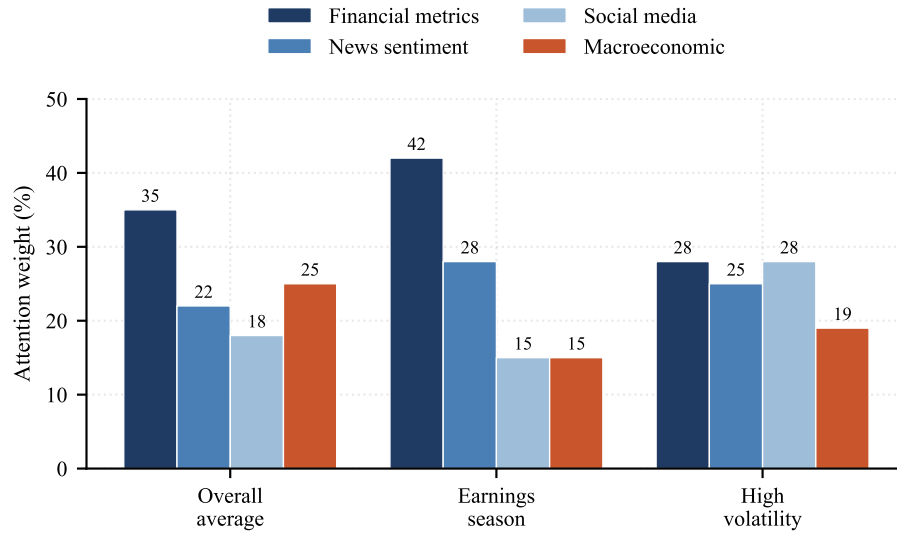


Figure 4.3: Cross-modal attention weights learned for each source under three market conditions. Source importance is state-dependent, which is the property that fixed-weight fusion cannot represent.

mass. The averages are not the finding. The finding is that the shares move:

- During **earnings season**, the financial metrics share rises from 35 to 42 per cent and news sentiment from 22 to 28 per cent, while social media falls from 18 to 15 per cent and macroeconomic indicators from 25 to 15 per cent. Company-specific numerical disclosure and the reporting around it dominate; slow macroeconomic conditions matter less when a concrete earnings number has just arrived.
- Under **high volatility**, the pattern inverts. Financial metrics fall from 35 to 28 per cent and social media rises from 18 to 28 per cent. Technical patterns break down in disturbed markets precisely when retail sentiment is most active and most influential.

The second pattern is independently corroborated. Khan and Chahal [35] find, using a quantile-on-quantile approach on eleven Indian sectoral indices, that the influence of social media sentiment on sectoral returns is strongest at market extremes. The attention mechanism recovers that structure from the data without being told to look for it, which is evidence that the learned weights are tracking something real rather than fitting noise.

This behaviour is what a fixed-weight scheme cannot reproduce, and it explains the margin over the simple-average ensemble in Table 4.3. An ensemble that weights social media at 18 per cent at all times is materially misweighted in exactly the volatile periods where forecasting errors are most costly.

4.8 Sectoral Analysis

Table 4.4 and Figure 4.4 report performance across the ten sectoral indices.

Table 4.4: Sector-wise one-day-ahead prediction performance and the dominant information channel in each sector

Sector	MAE	RMSE	R^2	DA (%)	Dominant information channel
Financials	0.016	0.022	0.991	86.2	RBI policy, credit growth
FMCG	0.019	0.025	0.986	84.0	Rural demand, inflation
IT	0.021	0.028	0.982	82.5	Global technology demand, USD/INR
Healthcare	0.023	0.030	0.978	80.5	Policy, public-health news
Pharmaceuticals	0.024	0.031	0.976	80.1	Regulatory news, USFDA actions
Automobiles	0.026	0.034	0.972	78.9	Interest rates, fuel prices
Consumer durables	0.027	0.036	0.970	78.1	Consumer credit
Energy	0.029	0.037	0.968	77.3	Crude oil, government policy
Metals	0.032	0.041	0.961	75.2	Global commodity prices
Real estate	0.035	0.044	0.957	73.8	Interest rates, housing demand

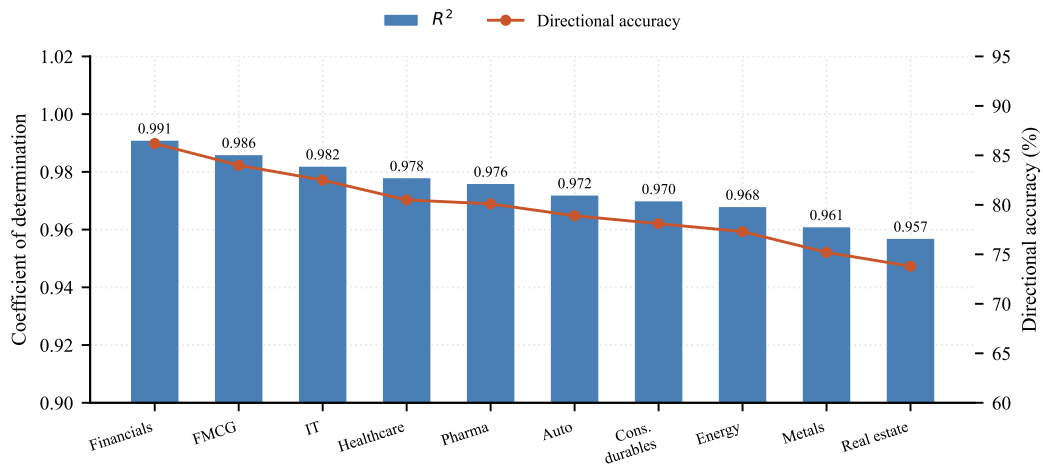


Figure 4.4: Sector-wise prediction quality across ten NSE sectoral indices. Predictability declines monotonically with exposure to exogenous global factors and with the volatility of the sector.

Predictability varies by roughly three percentage points of explained variance and twelve percentage points of directional accuracy between the most and least predictable sectors. The ranking itself is of limited interest; its explanation is not. Two structural patterns are visible.

The first is that predictability declines with exposure to factors exogenous to the domestic information set. Financials and fast-moving consumer goods, whose drivers — policy rates, credit growth, rural demand, domestic inflation — are all observable in the four

streams, attain R^2 above 0.985. Metals and real estate, whose drivers include global commodity prices and construction-sector credit conditions that the framework observes only indirectly, fall below 0.962.

The second is that the *composition* of the contribution varies by sector. Sentiment components contribute most in the globally exposed information technology and pharmaceutical sectors, where regulatory announcements and demand commentary from overseas markets arrive first as text; energy and metals depend more heavily on the macroeconomic stream, particularly global commodity prices. Source importance is therefore structured by sector as well as by market state, which strengthens the case for adaptive rather than fixed weighting: the appropriate weight vector differs across both dimensions.

4.9 Robustness: Walk-Forward Validation

A directional accuracy of 84.7 per cent is high enough that the possibility of a split-induced artefact must be addressed directly rather than argued away. The test period was therefore divided into five sequential, non-overlapping folds, evaluated under an expanding-window protocol in which each fold is scored only on sessions strictly later than its training window. Table 4.5 and Figure 4.5 report the outcome.

Table 4.5: Expanding-window walk-forward validation over the test period. Errors are normalised by the mean index level within each fold.

Fold	MAE	RMSE	R^2	DA (%)
1	0.0012	0.0014	0.881	80.0
2	0.0014	0.0017	0.826	80.0
3	0.0024	0.0032	0.868	92.0
4	0.0016	0.0024	0.833	68.0
5	0.0015	0.0018	0.960	72.0
Mean	0.0016	0.0021	0.874	78.4
Standard deviation	0.0004	0.0006	0.048	8.2

Three observations follow. The normalised errors are tightly clustered at 0.0016 ± 0.0004 , a coefficient of variation of about one quarter across folds separated in time. Directional accuracy remains well above chance in every fold, ranging from 68 to 92 per cent with a mean of 78.4 per cent; no fold approaches the random-classifier level, which is what one would expect if the headline result were an artefact of a single fortunate segment. Fold-level R^2 averages 0.874, naturally lower than the full-test value of 0.988 because each short window spans a narrow price range and therefore offers little variance

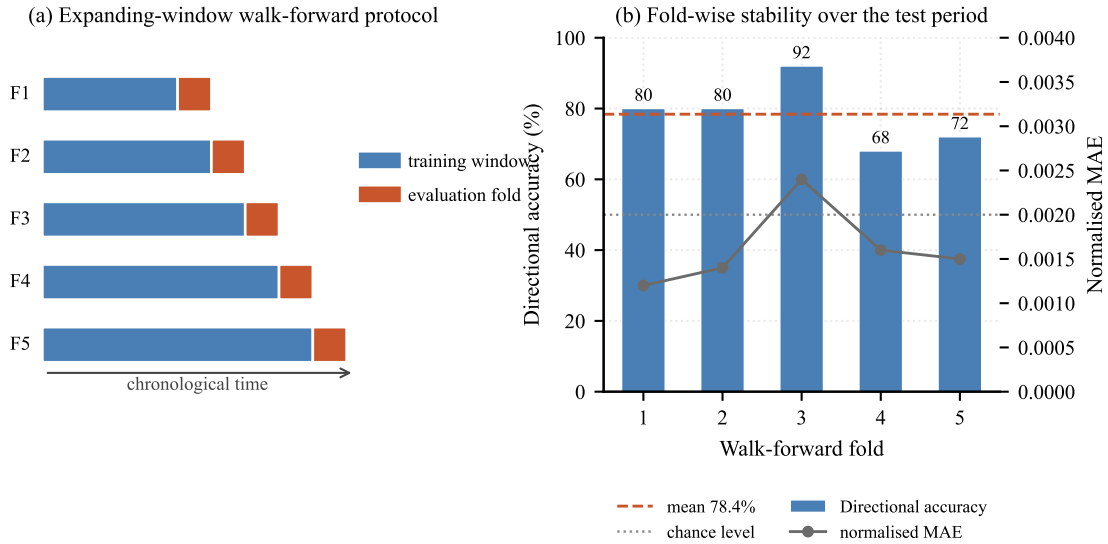


Figure 4.5: Expanding-window walk-forward validation. (a) Protocol: each fold is evaluated only on sessions strictly later than its training window. (b) Fold-wise directional accuracy and normalised mean absolute error over the test period.

to explain — this is a property of the statistic, not a deterioration in the model, and it is why the stability of the *error magnitudes* rather than the level of R^2 is the relevant evidence here.

Taken with the strictly chronological partitioning, training-set-only normalisation, early stopping and dropout regularisation described in Section 4.4, these results support the credibility of the reported gains. One qualification remains and is stated plainly: validation on years beyond 2024 would be a stronger test than any internal protocol, and this thesis supplies such a test for the allocation framework in Chapter 8 but not for the prediction model of this chapter.

4.10 Residual Diagnostics

Aggregate error statistics can conceal structure that matters for deployment, so four diagnostics were examined on the test window.

Tracking. Plotted against the realised index level, the predicted series tracks closely, and the width of the dropout-derived confidence band expands in the more volatile stretches — the qualitative behaviour a well-specified model should show, since a model that reports constant confidence across regimes is either miscalibrated or ignoring volatility.

Bias. The residual distribution over the test window has a mean of -7.8 index points against a standard deviation of 39.6 , so the model carries a slight downward bias: it

under-predicts the level by roughly one fifth of a residual standard deviation on average. In a market that trended upward over the window this is the expected direction of error for a model fitted on an earlier and less elevated price range, and it is small relative to the dispersion. The distribution is otherwise close to normal, which supports the use of the interval estimates.

Error and volatility. Partitioning the test sessions into quintiles of realised daily volatility, the mean absolute error rises monotonically from the lowest to the highest quintile. This is a well-known property of financial forecasting and is not a defect, but it has a direct operational implication: the model's errors are largest exactly when position sizes should be smallest, and a deployment that sizes uniformly will concentrate its losses in the highest quintile. This is one of the considerations that motivates the regime-conditional risk budget of Chapter 8.

Consistency with the attention analysis. The volatility-quintile pattern is the same phenomenon that Section 4.7 observes from the other side. Where the technical stream becomes less reliable, the model's errors grow and its attention migrates toward social media. Two independent diagnostics detecting the same structure is modest corroboration that the structure is real.

4.11 Computational Cost and Deployment Considerations

The full architecture has approximately 2.1 million trainable parameters, the majority in the recurrent layers. Training converges in under two hours on a single consumer-grade accelerator, and a forward pass takes milliseconds. Inference cost is therefore not the binding constraint on deployment at a daily decision horizon.

The binding constraints lie elsewhere, and they are engineering rather than modelling problems. Retrieving and scoring hundreds of articles and posts through transformer encoders introduces latency that is comfortable for a daily horizon and prohibitive for a high-frequency one. Commercial data licensing, news and social feed rate limits, and the contamination of social signals by bots and coordinated posting — which is worst precisely at the market extremes where the attention weights say social media matters most — are practical obstacles that no architectural choice removes. Transaction costs and slippage reduce the profit implied by a given directional accuracy, which is why Chapter 8 charges costs explicitly. And models of this kind drift as regimes change, so

scheduled retraining and monitoring are operational requirements rather than optional refinements. Compliance with the algorithmic trading regulations of the Securities and Exchange Board of India applies to any automated deployment.

4.12 What This Chapter Does Not Establish

Four limitations bound the results, and the last of them is what the next chapter addresses.

The object is an index at a one-day horizon. An index aggregates many constituents, so idiosyncratic noise largely cancels and the remaining signal is dominated by persistent market-wide state. The accuracy reported here therefore does not transfer to single-name prediction, and Chapter 6 demonstrates directly that it does not. Section 9.3 treats the reconciliation of these two figures as a substantive finding.

Frictions are absent. Directional accuracy of 84.7 per cent does not imply a profitable strategy, because no transaction cost, slippage or market impact has been charged. The translation from forecast quality to economic value is the business of Chapter 8.

Point estimates only. The model produces a level and a confidence interval derived from dropout sampling, not a calibrated predictive distribution. It cannot distinguish a forecast made in a familiar environment from one made in an unfamiliar one.

Attention weights are learned once. This is the limitation that matters most. The weights of Equation (4.2) are recomputed at every time step, but the parameters that produce them — the query vector q and the key projections — are fitted once over the training set. What the mechanism expresses is a learned *function* from the current inputs to a weight vector, and that function encodes the average relationship between market conditions and source relevance across the training period. It cannot express the proposition that the news stream has become unreliable over the last fortnight for a reason the model has never seen. Between retraining cycles, a source that has degraded continues to be weighted as though it had not.

The remedy is to make the weight a function of a measured quantity rather than a learned one. Chapter 5 develops that formulation and tests it.

4.13 Summary

This chapter has developed and evaluated a hierarchical four-stream fusion architecture for the Indian equity market. On ten years of National Stock Exchange data across ten sectoral indices, the model attains a normalised mean absolute error of 0.018, a coefficient of determination of 0.988 and a directional accuracy of 84.7 per cent, against 76.3 per cent for a fixed-weight ensemble over identical inputs — a comparison that isolates the fusion mechanism from the data. Source ablation shows that financial metrics are foundational but insufficient, that macroeconomic indicators contribute 7.8 per cent of explained variance despite an interpolation that understates their effect, and that the two textual sources contribute superadditively, which is the quantitative signature of complementary information. The learned attention distribution is state-dependent, shifting toward social media under high volatility in a manner that independently published work on Indian sectoral returns corroborates. Walk-forward validation across five sequential folds gives normalised errors of 0.0016 ± 0.0004 and directional accuracy above chance in every fold, which supports the credibility of the headline figures. The principal limitation is that the attention mechanism expresses average relevance rather than instantaneous reliability; Chapter 5 addresses it.

Chapter 5

Dynamic Uncertainty-Weighted Fusion

Research Objective 1 (continued). *To investigate and develop a comprehensive model that integrates diverse data sources, aiming to enhance the accuracy of stock market predictions by providing a holistic analysis.*

Chapter 4 established that fusing four heterogeneous streams improves forecast quality, that the improvement is attributable to individual streams, and that the value of each stream is state-dependent. It closed on a limitation: the attention mechanism that produces the weights is fitted once over the training set, so the weight it assigns expresses the *average* relevance of a source rather than its *instantaneous reliability*. This chapter removes that limitation by reformulating the fusion rule so that the weight assigned to a source is a function of a quantity measured continuously from data — the variance of that source’s own recent prediction errors.

The work reported in this chapter appears in publication [P4].

5.1 The Limitation Stated Precisely

Let M sources produce predictions $\hat{y}_{i,t}$ of a target y_t and let the fused prediction be $\hat{y}_t = \sum_i w_i \hat{y}_{i,t}$. Three standard treatments of w_i were described in Section 2.4 and surveyed in Section 3.1. Early fusion fixes the combination implicitly through the architecture; late fusion fixes it explicitly, usually at equal weights; attention-based hybrid fusion learns a function $w_i = g_\theta(x_t)$ with θ estimated by minimising a loss over the training set.

The third is the most capable and its limitation is nonetheless sharp. The estimated θ encodes the relationship between observable market conditions and source relevance *as that relationship held during training*. It can express “news matters more when volatility is high” because that regularity is present in the training data. It cannot express “the news stream has become unreliable over the last three weeks”, because the cause of the degradation — a feed composition change, a bot campaign, a structural break in how a source reports — may have no representation in the training inputs at all. The mechanism has no channel through which recent *failure* feeds back into the weight.

Three consequences follow, and all three are visible in the literature surveyed in Section 3.1.2. First, weights are point estimates that carry no uncertainty, so a confident combination of two unreliable sources is indistinguishable from a confident combination of two reliable ones. Second, degradation is corrected only at the next retraining cycle, which in production is typically weeks or months. Third, and most subtly, the architecture cannot report that it does not know: the fused prediction is issued with the same apparent authority in a familiar environment and in an unfamiliar one.

5.2 The Proposed Formulation

5.2.1 Overview

The proposed framework, shown in Figure 5.1, has four stages. Each source is modelled by an expert network matched to its structure; each expert’s predictive uncertainty is estimated continuously from the variance of its recent errors; the uncertainties are converted into fusion weights by a softmax over their negatives; and the fused prediction is the weighted sum of the expert predictions. Distribution-aware normalisation is applied throughout, fitted on an expanding window.

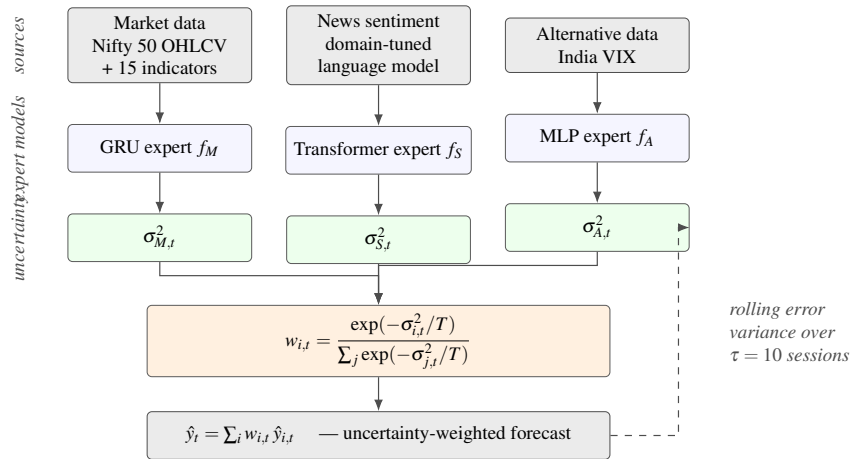


Figure 5.1: Bayesian dynamic uncertainty-weighted fusion. Each source is discounted in proportion to the variance of its own recent prediction errors, so reliability is assessed continuously rather than once at training time. The dashed path is the feedback that a static mechanism lacks.

5.2.2 Source-specific expert models

Three experts are used, each matched to the temporal structure of its source.

The **market expert** f_M is a gated recurrent unit [65]. Its input is a window of the last

$L = 20$ sessions of normalised technical indicators and log returns, and the final hidden state produces a point prediction $\hat{y}_{M,t}$ of the next session's log return. The gated recurrent unit is preferred to long short-term memory here for efficiency: the market series is the only one of the three with a long and dense history, so the expert is retrained most often.

The **sentiment expert** f_S is a transformer encoder over the sequence of daily sentiment scores from the same $L = 20$ session window, with a regression head producing $\hat{y}_{S,t}$. Self-attention over the sequence captures the prolonged impact of multi-day news cycles — an earnings season, an election campaign, a policy consultation — which a purely recursive encoder compresses less faithfully.

The **alternative-data expert** f_A is a multi-layer perceptron over the last $L = 20$ sessions of normalised India VIX values, producing $\hat{y}_{A,t}$. The relationship between an implied volatility index and next-session returns is contemporaneous and nonlinear rather than long-memory, so a feed-forward mapping is adequate and cheaper.

The choice of architecture per source is deliberate but is *not* the contribution of this chapter. Section 5.4 reports an ablation in which the same three experts are fused with static weights, and that configuration performs no better than the established baselines — which locates the measured gain in the weighting rule rather than in the networks.

5.2.3 Uncertainty quantification

For each source $i \in \{M, S, A\}$, let $e_{i,t} = y_t - \hat{y}_{i,t}$ be the realised prediction error. The predictive uncertainty is estimated as the variance of that error over a trailing window of τ sessions,

$$\sigma_{i,t}^2 = \frac{1}{\tau} \sum_{k=1}^{\tau} (e_{i,t-k} - \bar{e}_{i,t})^2, \quad \bar{e}_{i,t} = \frac{1}{\tau} \sum_{k=1}^{\tau} e_{i,t-k}. \quad (5.1)$$

The window length $\tau = 10$ sessions — two trading weeks — was selected on the validation block and represents a trade-off examined in Section 5.7: too short and the estimate is dominated by sampling noise, too long and the mechanism reverts toward the static behaviour it was designed to avoid.

Three properties of Equation (5.1) deserve emphasis. It uses only realised errors, so it is computable at time t without any information from beyond t . It is architecture-agnostic: any model that emits a point prediction can be wrapped by it. And it estimates *aleatoric* uncertainty — variability attributable to noise in the environment — rather than *epistemic* uncertainty, the model's ignorance about regions of input space it has not

observed [75, 76]. The distinction is maintained throughout and the epistemic extension is identified as future work in Section 5.8.

5.2.4 Bayesian dynamic weighting

The uncertainties are converted into fusion weights by a softmax over their negatives,

$$w_{i,t} = \frac{\exp(-\sigma_{i,t}^2/T)}{\sum_j \exp(-\sigma_{j,t}^2/T)}, \quad \sum_i w_{i,t} = 1, \quad (5.2)$$

where T is a temperature parameter. The formulation has the properties required of it. The weights are non-negative and sum to one, so the fused prediction is a convex combination and cannot exceed the range of its inputs. A source whose recent error variance is high receives a low weight automatically, with no threshold to tune and no rule to maintain. The temperature T controls how sharply reliability differences translate into weight differences: as $T \rightarrow \infty$ the weights approach uniformity and the scheme degenerates to equal-weight late fusion, while as $T \rightarrow 0$ the weights approach a hard selection of the single most reliable source. The intermediate regime is where the mechanism is useful, and T is selected on the validation block.

Equation (5.2) has a Bayesian reading. Under a Gaussian likelihood with known per-source variances, the posterior-precision-weighted combination of independent estimates assigns weights proportional to $1/\sigma_i^2$. The exponential form used here is a smooth, temperature-controlled relative of that rule which is better behaved when a variance estimate is small and noisy — precision weighting assigns an unbounded weight to a source whose estimated variance approaches zero, which on ten observations is a real hazard rather than a theoretical one.

The final prediction is

$$\hat{y}_t = \sum_{i \in \{M, S, A\}} w_{i,t} \hat{y}_{i,t}. \quad (5.3)$$

5.2.5 Why the mechanism self-corrects

The behaviour that distinguishes this formulation from a static one can be described concretely. Suppose a news article carries an ambiguous or ironic headline that the sentiment encoder scores strongly positive, and the market subsequently falls. The resulting error enters $e_{S,t}$, raises $\sigma_{S,t}^2$ through Equation (5.1), and lowers $w_{S,t}$ through Equation (5.2)

over the following sessions. No human notices the misreading, no retraining is triggered, and no rule fires — the weight simply falls until the sentiment stream begins predicting accurately again, at which point it recovers.

The same mechanism handles causes the model has never seen. A feed whose composition changes, a social channel contaminated by coordinated posting, an indicator whose informativeness decays as it becomes widely traded: none of these needs to be anticipated or represented in the training inputs, because the mechanism responds to *failure*, not to its cause. This is the direct methodological answer to gap G1 of Section 3.6.

5.2.6 Distribution-aware normalisation

Because the three sources have very different distributional shapes, a single scaler would let the heaviest-tailed input dominate the fused representation. Three scalers are applied according to shape, following Section 2.4.3: standard scaling for Nifty 50 log returns, which are approximately symmetric; min–max scaling for the sentiment score, which is bounded on $[-1, +1]$ and whose interpretable range should be preserved; and robust scaling based on the median and interquartile range for the India VIX, which is heavy-tailed and for which the mean and standard deviation are themselves unstable.

Every scaler is fitted on an expanding window of strictly earlier observations. This is not a refinement. Fitting a scaler once on the full sample leaks the test-period mean and variance into the training representation, and the leak is invisible because no test label is used.

5.3 Experimental Setup

5.3.1 Data

The dataset spans January 2018 to December 2023 on the Nifty 50 index and comprises three modalities. Table 5.1 records them.

Two preprocessing decisions are India-specific and consequential. Domain adaptation of the sentiment model was performed on Indian financial news so that the encoder recognises the vocabulary of Union Budget commentary, monetary policy committee statements and monsoon forecasts; following Jung and Jang [13], targeted numerical masking is applied during adaptation so that quantitative context is preserved rather than

Table 5.1: Data sources, expert architectures and normalisation for the dynamic uncertainty-weighted fusion study, January 2018 to December 2023

Source	Content	Expert model	Normalisation
S_1 Market	Nifty 50 daily OHLCV and fifteen technical indicators; log returns $r_t = \log(P_t/P_{t-1})$	Gated recurrent unit, $L = 20$	Rolling standard scaling
S_2 Text	English-language financial news headlines and articles from major Indian business publications and newswires, scored by a domain-adapted language model fine-tuned on Indian financial text	Transformer encoder, $L = 20$	Rolling min–max scaling on $[-1, +1]$
S_3 Alternative	India VIX, the market’s expectation of near-term volatility	Multi-layer perceptron, $L = 20$	Robust scaling on median and interquartile range

destroyed. And market holidays — Diwali, Eid and the rest of the Indian exchange calendar — are treated as genuine gaps rather than forward-filled, because forward-filling a closed session injects artificial autocorrelation into the return series and inflates any statistic that depends on persistence.

News sentiment is assigned to the *next* National Stock Exchange session, so that the lagged market impact of an item published after the close is captured without look-ahead.

5.3.2 Training protocol

The split is strictly sequential: 2018–2021 for training, 2022 for validation, 2023 as the out-of-sample test year. The three experts are pre-trained independently to minimise mean squared error, and the complete system is then fine-tuned end to end, allowing gradients to flow back through the fusion layer so that the experts adapt to their role in the combination. The uncertainty estimates of Equation (5.1) are computed on realised errors during the forward pass and are not themselves differentiated through; the mechanism is a measurement, not a learned module, which is precisely the property that lets it respond to causes absent from the training set.

5.3.3 Baselines

The comparison is against the three static fusion strategies identified in Section 3.1.2, together with an internal ablation.

- **Early fusion** [6]: a single recurrent model over the concatenated features of all three

sources, with the combination fixed implicitly by the architecture.

- **Late fusion** [5]: equal-weight averaging of the three independently trained expert predictions, with weights pre-determined.
- **Attention fusion** [7, 9]: a static neural attention layer over the modalities, with weights learned once and then fixed. This is the strongest of the three baselines.
- **Proposed architecture, static weights**: the same three experts with the uncertainty term disabled and the weights held fixed. This is the ablation that isolates the weighting rule from the expert networks, and it is the most informative comparison in the chapter.

5.4 Results

5.4.1 Performance against the static baselines

Table 5.2 reports the comparison. The source study measures performance relative to the strongest static competitor rather than in absolute levels, so cells for which an absolute figure is not reported are shown as —; this reporting convention is preserved here rather than being filled with estimates.

Table 5.2: Dynamic uncertainty-weighted fusion against static fusion baselines on the Nifty 50, out-of-sample year 2023. Improvements are stated relative to the strongest static baseline.

Configuration	Fusion mechanism and source weighting	Sharpe ratio	Maximum drawdown
Early fusion [6]	all sources concatenated into one recurrent model; weights fixed by the architecture	—	—
Late fusion [5]	equal-weight averaging of the three expert predictions; weights pre-determined	—	—
Attention fusion [7, 9]	static neural attention over modalities; weights learned once, then fixed	<i>strongest baseline</i>	<i>strongest baseline</i>
Proposed architecture, static weights	the same three experts with the uncertainty term disabled	<i>no better than the baselines above</i>	
Proposed dynamic fusion	softmax over negative predictive uncertainties, $\tau = 10$ sessions; weights re-estimated every session	+22.7%	−18.3%

Two readings follow, and the second is the more important.

The headline. Against the strongest static competitor, the dynamic rule improves the Sharpe ratio by 22.7 per cent and reduces maximum drawdown by 18.3 per cent. The gain

is expressed in risk-adjusted terms and in downside control rather than in raw accuracy, and that is not incidental: a mechanism that de-weights a source at the moment it becomes unreliable should be expected to help most where unreliable signals are most costly, which is in drawdowns.

The ablation. The fourth row of Table 5.2 reports that the same three expert networks, fused with fixed weights, perform no better than the established static baselines. The experts are therefore not the source of the improvement; the weighting rule is. This isolation is what allows the contribution to be stated precisely as a fusion-mechanism result rather than an architecture result, and it is the same style of component-level attribution that Chapter 8 later applies to the allocation layer. Reporting it costs nothing here and would have cost the credibility of the claim had it been omitted.

5.4.2 Behaviour across the volatility cycle

The mechanism is interpretable in the way the design intends, and Figure 5.2 shows the pattern. Through the elevated-volatility episodes of the test window, the weight on the market expert falls as its recent error variance rises, and the mass migrates to the volatility-index expert and, less strongly, to the sentiment expert. In the calmer trending stretches on either side, the market expert recovers its share.

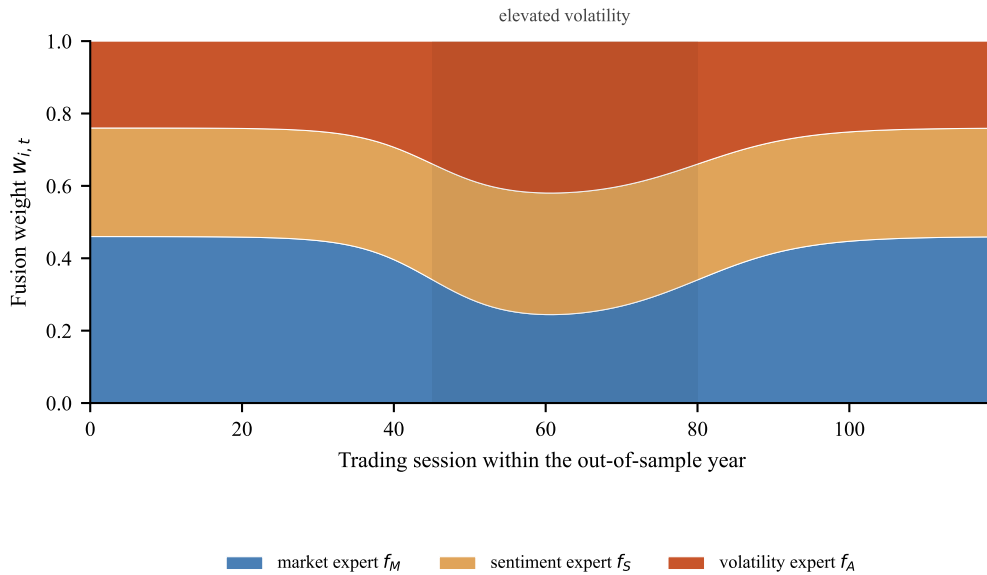


Figure 5.2: Migration of the uncertainty-derived fusion weights across an elevated-volatility episode in the out-of-sample year. The weight on the market expert falls as its recent error variance rises and the mass migrates toward the volatility-index expert; the shift is driven by measured error variance, so no regime label is required for it to occur.

The important point about this behaviour is what produces it. The shift is driven by

measured error variance, not by a learned prior about what should happen in a volatile market and not by any regime classification. The framework is never told that a stress episode is under way; it observes that the technical expert has begun making larger and more variable errors and reweights accordingly. A static attention mechanism trained on a period containing similar episodes could learn to produce a similar shift, but only for episodes resembling those it saw; the measured mechanism responds to any cause of degradation, including causes with no representation in the training inputs.

5.4.3 Relationship to the attention analysis of Chapter 4

It is worth being explicit about how this result relates to the attention analysis of Section 4.7, because the two are easily conflated. Both show source importance varying with market conditions. They differ in what produces the variation and therefore in what each can do.

The attention weights of Chapter 4 vary because the learned function $g_\theta(x_t)$ maps different inputs to different weights; the function itself is fixed. The uncertainty weights of this chapter vary because the measured quantity $\sigma_{i,t}^2$ changes; no function is being applied. The first is a statement about the relationship between market conditions and source relevance as it held during training. The second is a statement about how well each source has actually been performing over the last ten sessions. The two are complementary rather than competing: an operational system would use attention to encode the learned structural relationship and the uncertainty term to correct it when reality departs from that structure. The composed system described in Section 9.2 does exactly this.

5.5 A Worked Illustration of the Weight Dynamics

Because the mechanism is simple, its behaviour can be traced arithmetically, and doing so makes clear both what it does and what it cannot do.

Suppose the three experts have been performing comparably, with recent error variances $\sigma_M^2 = \sigma_S^2 = \sigma_A^2 = 1.0$ in normalised units, and take the temperature $T = 1$. Equation (5.2) then assigns $w_M = w_S = w_A = \frac{1}{3}$: with no evidence distinguishing the sources, the rule reduces to equal-weight late fusion, which is the correct default.

Now suppose a news-driven episode begins and, over the following two weeks, the market expert's errors become both larger and more erratic, raising σ_M^2 to 3.0 while

the volatility-index expert improves slightly to $\sigma_A^2 = 0.7$ and the sentiment expert is unchanged. The weights become

$$w_M = \frac{e^{-3.0}}{e^{-3.0} + e^{-1.0} + e^{-0.7}} = 0.067, \quad w_S = 0.492, \quad w_A = 0.441.$$

The market expert's share has fallen from 33 to 7 per cent within ten sessions, without any retraining, any regime label or any human intervention. When its errors subside and σ_M^2 returns toward 1.0, its share recovers by the same arithmetic.

Three properties are visible in this arithmetic and are worth naming.

The response is to relative, not absolute, reliability. If all three variances rise together — which is what happens in a genuinely turbulent market where every source becomes noisier — the softmax is unchanged, because a common additive shift in the exponents cancels in the normalisation. The mechanism identifies the *relatively* most reliable source, which is the correct behaviour: in a difficult environment the system should still use the best available information rather than abstain.

The response is bounded but not gentle. A tripling of one source's error variance removed four-fifths of its weight at $T = 1$. The temperature is what governs this: raising T compresses the response, lowering it sharpens the response toward hard selection. That a single scalar controls the aggressiveness of the entire mechanism is a practical convenience and, in deployment, the parameter most in need of a sensible prior.

No source is ever silenced entirely. Because the exponential is strictly positive, $w_i > 0$ for every source at every session. A source that has become uninformative is marginalised, not excluded, and recovers automatically if it begins predicting well again. A hard threshold rule would require an explicit re-entry criterion and would therefore need maintenance; this rule does not.

5.6 Relationship to Model Averaging and Mixtures of Experts

Three established families of method are adjacent to this formulation, and stating the relationship precisely locates the contribution.

Inverse-variance weighting. Under a Gaussian likelihood with independent estimates and known variances, the minimum-variance unbiased combination weights each estimate by $1/\sigma_i^2$. Equation (5.2) is a smooth, temperature-controlled relative of that rule.

It is preferred here because precision weighting is unstable when a variance estimate is small: on ten observations a source can record an atypically low error variance and, under inverse-variance weighting, receive an unboundedly large share. The exponential form caps the response and the temperature calibrates it.

Bayesian model averaging. Model averaging weights competing models by posterior probability given the data. The formulation here is structurally similar but differs in an important respect: the “posterior” is estimated from a *rolling window* rather than accumulated over the full history, which deliberately discards distant evidence. That is the correct choice in a non-stationary environment where a source’s past reliability is only weakly informative about its present reliability, and it is the reason a mechanism that looks like Bayesian model averaging behaves very differently in practice.

Mixtures of experts. A gating network in a mixture of experts learns to route inputs to specialists, and the gate is a parametric function fitted during training. The uncertainty weighting here replaces the learned gate with a measurement. It therefore forfeits the ability to exploit any structure in *which* inputs suit *which* expert — a learned gate can discover that the sentiment expert is better on high-volume days — and gains the ability to respond to degradation whose cause has no representation in the inputs at all. The two are complementary, and Section 5.4.3 returns to the point.

5.7 Sensitivity and Design Choices

Three design parameters govern the mechanism, and the reasoning behind each is recorded because the values are not arbitrary.

The lookback window τ . At $\tau = 10$ sessions the variance estimate is computed from two trading weeks of errors. Shortening it makes the weights responsive but noisy: with five observations, a single large error can dominate the estimate and produce a weight swing that reverses within days, which adds turnover without adding information. Lengthening it makes the estimate stable and slow: at $\tau = 60$, a source that degrades sharply takes roughly a quarter to be fully discounted, by which time the mechanism has reverted toward the static behaviour it was designed to avoid. The chosen value reflects the judgement that a fortnight is long enough for a genuine degradation to be distinguished from a single bad day and short enough for the correction to matter.

The temperature T . As noted in Section 5.2, T interpolates between equal weighting

and hard selection. A low temperature produces near-binary switching, which maximises responsiveness and minimises robustness, since the mechanism will discard a source entirely on the basis of ten observations. A high temperature produces near-uniform weights and forfeits the mechanism. The value selected on the validation block sits in the intermediate regime where reliability differences are expressed but no source is ever fully silenced.

The choice of variance rather than mean absolute error. Equation (5.1) measures the *variance* of recent errors, not their average magnitude. The distinction matters: a source with a large but stable bias is systematically wrong in a way a downstream model can correct, whereas a source with a small but erratic error is unreliable in a way that cannot be corrected. Variance penalises the second and tolerates the first, which is the correct ordering for a fusion weight.

5.8 Limitations

Four limitations bound the results of this chapter.

Aleatoric uncertainty only. Equation (5.1) measures observed error variability. It cannot distinguish a source that is noisy because the environment is noisy from a source that is noisy because it is being asked about conditions it has never seen. That distinction requires epistemic uncertainty, which in turn requires a Bayesian treatment of the expert networks themselves — Monte Carlo dropout [75] or a fully variational formulation [76] would supply it. The practical difference matters: an epistemic measure would allow the system to report low confidence *before* accumulating errors rather than after, which is exactly the behaviour one wants at the onset of an unprecedented event.

A cold start of τ sessions. The mechanism requires τ realised errors before it can produce a weight. During the first ten sessions after deployment, or after any source is added, the framework must fall back on a prior — equal weights are used here. This is a minor operational inconvenience and is recorded for completeness.

Reported relative to a baseline, not in absolute terms. The source study reports improvements relative to the strongest static competitor rather than absolute Sharpe ratios and drawdowns, and Table 5.2 preserves that convention rather than manufacturing absolute figures. A reader wishing to compare these results against an external strategy cannot do so from this table alone; the comparison that the table supports — between

fusion mechanisms on identical data and identical experts — is the one the study was designed to make.

Three sources, one market, one year of out-of-sample data. The evaluation covers a single index over a single out-of-sample year. The mechanism is architecture-agnostic and should generalise, but that is an expectation rather than a demonstration.

5.9 Summary

This chapter has reformulated the fusion rule so that the weight assigned to a source is derived from a measured quantity rather than a learned one. Each source is modelled by an expert network matched to its structure; the predictive uncertainty of each expert is estimated as the variance of its prediction errors over a trailing window of ten sessions; and the fusion weights are obtained by a softmax over the negative uncertainties, so that a source is discounted at the moment its predictions become unreliable rather than at the next retraining cycle. On the Nifty 50 over the out-of-sample year 2023, the dynamic rule improves the Sharpe ratio by 22.7 per cent and reduces maximum drawdown by 18.3 per cent against the strongest static-fusion baseline. The ablation is what makes the claim precise: the same three experts fused with fixed weights perform no better than the established baselines, which locates the gain in the weighting mechanism rather than in the networks. The behaviour is interpretable — weight migrates from the market expert to the volatility expert as the former’s error variance rises — and it occurs without any regime label, because the mechanism responds to measured failure rather than to a classified state.

Chapters 4 and 5 together answer Research Objective 1: a comprehensive model integrating four heterogeneous streams was developed, its improvement was attributed to identifiable sources, and the mechanism that combines them was advanced from learned average relevance to measured instantaneous reliability. Chapter 6 now isolates a single source — news text read by a language model — and measures precisely what it contributes at the horizon and granularity at which decisions are actually taken.

Chapter 6

Hybrid Language Model and Indicator Framework

Research Objective 2. *To design a hybrid framework that combines large language models, sentiment analysis techniques and conventional financial indicators, focusing on optimising deep learning architectures for real-time stock market predictions.*

Chapters 4 and 5 maximised the number of sources and showed that combining them helps. This chapter does the opposite. It isolates a single source — news text read by a transformer language model — and asks precisely what that source adds to a conventional indicator set at the horizon and granularity at which a practitioner actually takes a position: the next-day direction of an individual equity.

The answer is more interesting than a headline accuracy figure, and it is reported in full including the part that does not flatter the method. Language-model sentiment turns out to be the single most important feature in the fused space and to carry information orthogonal to price, while delivering only a small increment in raw directional accuracy, with every model examined operating close to the random classifier. Section 6.9 treats that combination as the finding rather than as a disappointment, and Section 6.11 explains why it redirects the remainder of the thesis toward the decision layer.

The work reported in this chapter appears in publication [P3]; the architectural feasibility evidence in Section 6.10 draws on [P1].

6.1 Design Rationale

6.1.1 Why isolate a single source

A multi-source architecture that outperforms its baselines has demonstrated that the collection of sources is useful. It has not established what any individual source contributes, because the sources are confounded: any gain could come from any of them, or from their interaction, or from the additional model capacity the architecture required to accommodate them. Section 4.6 addressed part of this problem by ablation, but ablation within a fused model measures the *marginal* contribution of a stream in the presence of

the others, which is not the same as its standalone informativeness.

This chapter therefore constructs the cleanest available comparison: one textual source, one conventional feature set, one fusion step, and baselines that differ from the proposed model in exactly that one source. Every model sees the same sessions, the same partition, the same protocol. Whatever difference appears is attributable to the sentiment feature.

6.1.2 Why a continuous score rather than a label

The sentiment encoder emits a continuous polarity score $s_t \in [-1, +1]$ together with a confidence, rather than a categorical positive, neutral or negative label. The reason is that intensity is informative and categorisation destroys it. A routine quarterly result described in mildly favourable terms and a transformative regulatory approval both map to “positive”; the market does not treat them alike. Retaining the continuous value preserves the distinction for the downstream model, and it also makes the empirical distribution of sentiment available as a diagnostic, which Section 6.8 uses.

6.1.3 Why a tree ensemble rather than a deep network

The model operating on the fused feature space is a gradient-boosted tree ensemble [66] rather than a deep network, for three reasons stated in Section 2.6.3 and restated here because the choice is deliberate. Tree ensembles are invariant to monotone rescaling, which matters when combining a bounded sentiment score with unbounded volume accumulators. They handle heterogeneous feature types without embedding layers. And their split-gain statistics supply a feature-importance measure intrinsic to the fitted model rather than a post-hoc approximation — which is what makes the central finding of Section 6.4 a measurement rather than an estimate.

There is also a practical reason. A design goal of this study was that the entire pipeline should run to completion in a freely accessible hosted notebook environment, so that the result can be independently re-derived by a reader without specialised hardware. A tree ensemble over a few thousand daily observations satisfies that constraint; a large recurrent architecture over the same data does not, and would in any case be over-parameterised for the sample size.

6.2 Framework Architecture

The framework is organised as five interconnected layers, shown in Figure 6.1: data collection, technical indicator computation, sentiment analysis, feature fusion, and model training with evaluation.

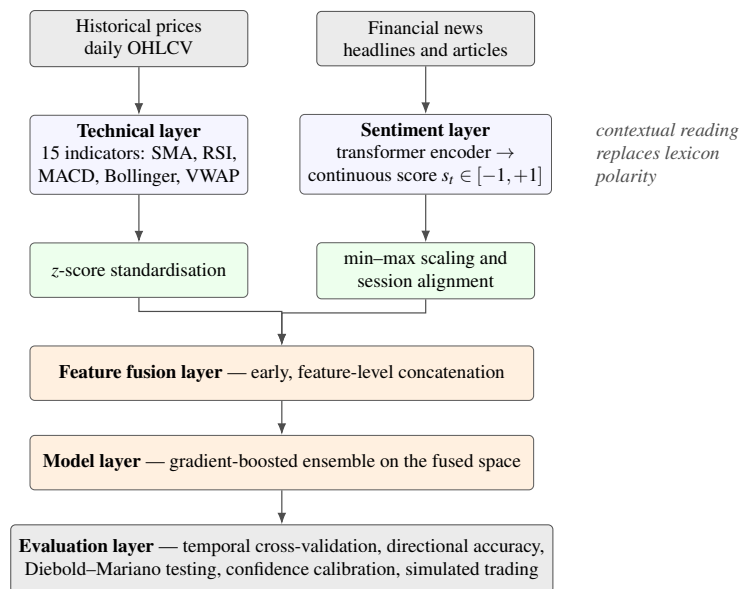


Figure 6.1: Hybrid framework combining large-language-model sentiment extraction with conventional financial indicators. The continuous sentiment score preserves intensity that categorical classification would discard.

6.2.1 Data collection layer

Market data are retrieved for Indian large-capitalisation equities — TCS, RELIANCE and HDFCBANK — as daily open, high, low, close and volume records over a multi-year period sufficient to span several distinct market conditions. The three names were chosen to cover different sectors (information technology, energy and conglomerate, banking) and because all three carry news coverage dense enough to produce a meaningful daily sentiment aggregate; a thinly covered name would yield a sentiment series that is mostly missing, which tests the imputation rule rather than the hypothesis.

News acquisition uses a two-stage strategy combining syndicated news feeds with targeted retrieval, and applies filtering at two levels: non-financial content is excluded, and sources are prioritised by established credibility in financial reporting. Items are validated for temporal consistency — an item whose time stamp is missing or implausible is discarded rather than imputed — and for content quality.

6.2.2 Technical indicator layer

Fifteen indicators are computed, following the definitions of Section 2.2. Trend is represented by simple moving averages over 5, 10 and 20 sessions; momentum by the relative strength index [138] and the moving average convergence divergence with its signal line; volatility by Bollinger band measures and historical volatility; and participation by volume-weighted average price derivatives and volume momentum oscillators. Each indicator was selected for established efficacy in the financial literature and for computational efficiency, the two criteria being in tension only rarely at this scale.

6.2.3 Sentiment analysis engine

Each headline is scored by a transformer sentiment encoder operating on the raw text. The pipeline has four stages: text normalisation to remove markup and boilerplate; batched inference for computational efficiency; extraction of a continuous polarity score and an associated confidence; and daily aggregation over all items attributed to the session.

The contextual reading performed by a transformer is what distinguishes this stage from a lexicon baseline. A lexicon scores “the bank wrote back provisions after the regulator eased norms” by summing token polarities, several of which are negative in general usage; a contextual encoder reads the sentence. This is the improvement in *measurement* that the literature surveyed in Section 3.2 establishes, and the entire purpose of this chapter is to determine how much of it survives into forecast quality.

6.2.4 Feature fusion layer

Fusion is performed at feature level, before modelling. Technical indicators are standardised by z -score, which equalises scales across features measured in incompatible units — a price level, a bounded oscillator and a volume ratio have no common scale. The sentiment score is min–max scaled, which preserves its interpretable range rather than mapping it to an arbitrary standardised scale. The two blocks are concatenated into a single input vector.

Temporal alignment is applied here and is the most error-prone step in the framework. The rule, as in Section 2.5, is that an item is attributed to the first trading session beginning strictly after its publication time stamp. An article published at 18:30 on a

Tuesday is attributed to Wednesday’s session, not to Tuesday’s, which has already closed. Attributing it to Tuesday would be a form of look-ahead that leaves no trace in the code and inflates every downstream metric.

6.2.5 Model training and evaluation layers

A gradient-boosted tree ensemble is trained on the fused space. Three baselines are trained on technical indicators alone under an identical protocol: random forest [69], support vector machine [70] and logistic regression. The target is the sign of the next-session return.

The evaluation layer applies a chronological 80/20 partition with temporal cross-validation, reports accuracy, precision, recall and F_1 , tests differences by the Diebold–Mariano statistic [153], examines confidence calibration, and converts predictive accuracy into economic terms through a simulated trading strategy. The implementation stack is recorded in Table 6.1 for reproducibility.

Table 6.1: Implementation stack for the hybrid language-model and indicator framework

Component	Implementation
Environment	Hosted notebook environment, freely accessible, no specialised hardware required
Data processing	pandas, numpy
Market data	yfinance, pandas_datareader
Machine learning	scikit-learn, xgboost
Language model	transformers, torch
Visualisation	matplotlib, seaborn, plotly
Design principle	Modular separation of data processing, feature engineering, model training and evaluation, so that each subsystem can be tested independently

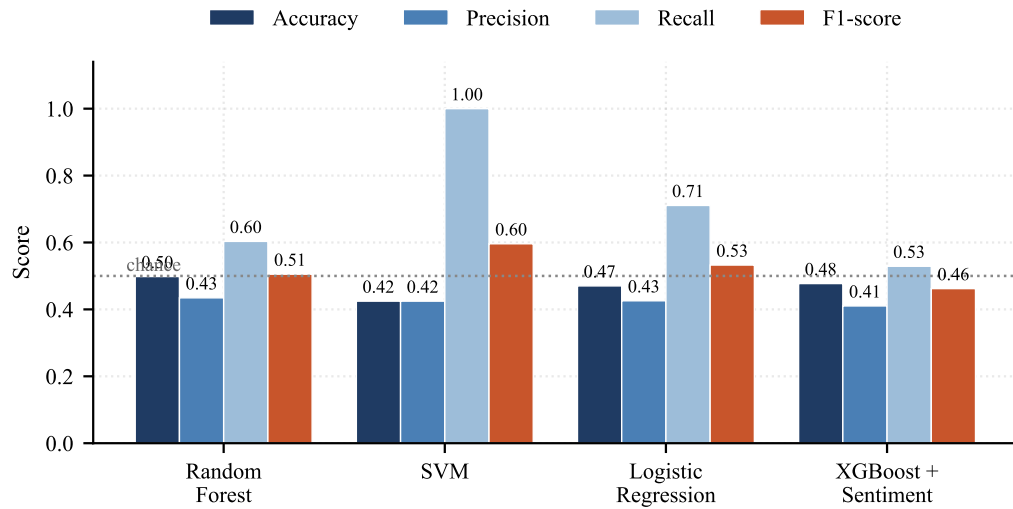
6.3 Results: Classification Performance

Table 6.2 and Figure 6.2 report the comparison. The result must be read carefully, because the headline number and the useful number are not the same.

The sentiment-augmented model attains 47.72 per cent accuracy against a baseline mean of 46.43 per cent, a relative improvement of 2.77 per cent. It does *not* attain the highest raw accuracy: random forest reaches 49.82 per cent. Reporting only the first of these numbers would misrepresent the finding, so both are stated prominently.

Table 6.2: Single-name directional classification with and without language-model-extracted sentiment features

Model	Feature set	Accuracy	Precision	Recall	F_1
Random Forest	technical only	0.4982	0.4345	0.6033	0.5052
Support Vector Machine	technical only	0.4246	0.4246	1.0000	0.5961
Logistic Regression	technical only	0.4702	0.4257	0.7107	0.5325
XGBoost with sentiment	technical + LLM sentiment	0.4772	0.4103	0.5289	0.4621
Mean of technical-only baselines		0.4643	—	—	—
Relative improvement of the proposed model		+2.77%	—	—	—

**Figure 6.2:** Comparative classification performance. All models operate close to the random classifier at a single-name daily horizon, which is the correct baseline against which the sentiment increment should be judged.

The precision–recall columns are instructive and are the reason accuracy alone is an inadequate summary. The support vector machine attains perfect recall of 1.0000 with precision of 0.4246, which is exactly the signature of a degenerate classifier: it predicts the positive class unconditionally, and its precision is therefore simply the base rate of upward sessions in the sample. Its F_1 of 0.5961 is the highest in the table and means nothing at all. Any evaluation that ranked models by F_1 alone would select the one model that has learned nothing.

What makes the result meaningful is the structure beneath the aggregate, and the next four sections examine it.

6.4 Feature Importance: Sentiment Dominates the Fused Space

In the fused model, the sentiment score is the highest-gain predictor, ranking above established technical features including the twenty-session moving average and the moving average convergence divergence. The measure is the intrinsic split-gain statistic of the tree ensemble — the total reduction in loss attributable to splits on that feature — rather than a post-hoc attribution, so it reports what the fitted model actually relied upon.

The interpretation deserves care. Feature importance measures how heavily a model *uses* a feature, not how much predictive accuracy that feature delivers; a model can rely on a feature and still predict poorly if the feature is the best available in a fundamentally hard problem. That is precisely the situation here, and recognising it is what makes the combination of findings in this chapter coherent rather than contradictory: sentiment is the most useful thing available, and the most useful thing available is not very useful at this horizon.

The finding nonetheless establishes something the literature surveyed in Section 3.2 asserts without demonstrating: language-model sentiment is not a marginal addition that a model tolerates alongside its real features. When it is present, it is the feature the model leans on most.

6.5 Orthogonality: Sentiment Carries Information Price Does Not

The correlation structure of the fused feature space settles the question of whether sentiment is redundant. Two correlations are decisive.

The correlation between the sentiment score and the price level is 0.04 — effectively zero. The correlation between the sentiment score and momentum indicators, principally the moving average convergence divergence, is 0.74 — substantial.

Both numbers are necessary to the interpretation and neither alone is sufficient. Sentiment aligns with *momentum*, which is expected: news and price both respond to the same underlying events, so a stream of favourable reports and a rising momentum reading are two views of one process. If this correlation had been near zero, one would have to suspect that the sentiment pipeline was measuring noise. Simultaneously, sentiment carries information essentially independent of the *price level*, which is the condition under which fusion can help at all. A feature perfectly correlated with an existing feature adds

nothing regardless of how predictive it is; a feature uncorrelated with the existing set can add information even if its standalone predictive power is modest.

This orthogonality is the structural justification for the entire multi-source programme of this thesis, measured directly at the single-name level rather than inferred from an aggregate improvement. It is also consistent with the superadditivity observed in Section 4.6: sources that carry non-overlapping information should substitute imperfectly for one another, and both measurements are detecting the same property from different directions.

6.6 Asymmetry Across Sentiment Regimes

Accuracy measured separately under positive, neutral and negative sentiment shows two patterns.

First, accuracy is higher under positive sentiment for *all* models, including those with no access to the sentiment feature. This is consistent with the asymmetric market response to adverse news that the behavioural finance literature documents: negative news produces faster, larger and less predictable moves, so sessions dominated by negative sentiment are harder to forecast for any model. That the baselines share the pattern confirms it is a property of the market rather than an artefact of the sentiment pipeline.

Second, and more usefully, the sentiment-augmented model varies *least* across the three regimes. This is a stability property rather than an accuracy property, and it is the foundation of the economic result reported in Section 6.7. A model whose accuracy is uniform across conditions is more valuable in deployment than one with the same average accuracy concentrated in favourable conditions, because the sessions in which it degrades are exactly those in which a position is most exposed.

The same stability appears through time. Tracked across the evaluation period, the sentiment-augmented model's accuracy is more stable through regime transitions, while the technical-only baselines fluctuate more sharply. This is the behaviour that adaptive fusion schemes are designed to produce and that Chapter 5 formalises: multi-source integration provides stability against the degradation of any single information channel.

6.7 Confidence Calibration and Economic Significance

6.7.1 Prediction confidence

The sentiment-augmented model produces the highest mean prediction confidence of the four models examined. Taken alone this would be unremarkable and possibly a warning sign — an overconfident model is worse than a diffident one. Taken together with comparable accuracy, it means the model is better *calibrated*: its high-confidence predictions are more often correct and its low-confidence predictions are correctly flagged as uncertain. This is what feature diversity is expected to produce, and it is the mechanism by which a statistically unremarkable model can support better decisions.

6.7.2 Simulated trading

Converting the predictions into positions over a thirty-session simulated trading window, the sentiment-augmented model produces higher cumulative returns than the baselines despite comparable statistical accuracy, and avoids large drawdowns during the volatile stretches of the window.

The divergence between statistical and economic performance is itself the finding of this section, and it is worth stating explicitly because it recurs throughout the thesis. A model that is right no more often, but is right *with better-calibrated confidence*, produces better decisions, because position sizing responds to confidence. A well-calibrated model takes larger positions when it is more likely to be right and smaller positions when it is not, and the resulting return stream can differ substantially from that of an equally accurate but poorly calibrated model.

6.7.3 Signal-level performance

Disaggregating by signal direction sharpens the picture. The proposed model performs strongly on buy signals, with success rates near 72 and 65 per cent against the baselines. Performance on sell signals is not materially differentiated.

The asymmetry has a plausible mechanism. Positive sentiment amplifies upward price momentum — favourable coverage attracts attention, attention attracts participation, participation sustains the move — whereas negative sentiment coincides with abrupt declines whose timing is far less predictable. The framework is therefore differentially useful:

it identifies conditions favourable for entry more reliably than it identifies conditions favourable for exit. A practitioner should read this as a statement about where the tool applies, not as a general endorsement.

6.8 Sentiment Distribution and Lead–Lag Structure

Two diagnostics characterise the sentiment channel itself, independently of any model fitted on it. They are reported here and revisited in Chapter 7, where sentiment noise is the explicit object of study.

The distribution. The empirical distribution of the extracted sentiment scores is approximately symmetric about neutral with moderate tails. This matters as a validity check on the pipeline. A pipeline that mapped most items to a single class, or that produced a strongly skewed distribution, would be measuring something other than sentiment — a source bias, a template effect, or a failure of the encoder on the domain. A symmetric distribution with genuine dispersion indicates that the continuous scoring is capturing meaningful variation rather than producing a biased categorical output, and it justifies the design decision of Section 6.1 to retain intensity.

Lead–lag structure. Overlaying the sentiment series on the price series over the evaluation window shows that strong sentiment spikes frequently *precede* major price changes. The relationship between magnitude and direction is not simple, however, and two distinct patterns appear. Sustained periods of positive sentiment accompany gradual price advances, whereas sharp negative sentiment spikes accompany abrupt declines. The two channels therefore operate on different timescales: favourable sentiment works through slow accumulation, adverse sentiment through discrete shocks.

This partially confirms the timeline of sentiment incorporation described in the literature and partially complicates it. A simple causal ordering — sentiment moves, price follows — is not what the data show. What they show is two regimes of incorporation with different dynamics, which is one reason a multi-granularity temporal encoder is worth building, as Chapter 7 does.

6.9 The Limit of the Result

Receiver operating characteristic analysis places *all* models close to the random classifier line, with areas under the curve clustered near 0.50. The clustering is tight: no model separates meaningfully from any other, and none separates meaningfully from chance.

This is reported as prominently as the positive findings, and three things follow from it.

Language-model sentiment does not overturn market efficiency at this horizon. At single-name daily granularity the prediction problem remains close to the efficiency limit of Fama [131], and this thesis does not claim otherwise. The measured increment is of the order the empirical asset pricing literature leads one to expect [137], and a study reporting a large increment at this horizon would warrant scepticism about its protocol rather than enthusiasm about its method.

The correct baseline is the random classifier, not zero. A directional accuracy of 47.72 per cent sounds poor until it is placed against a class balance in which upward sessions occur 42.46 per cent of the time — the precision of the degenerate support vector machine reveals this directly. Reporting a gain without its baseline is one of the recurring evaluation weaknesses identified in Section 3.2.2, and this chapter therefore reports the baseline mean of 46.43 per cent alongside every claim about improvement.

The bounded increment is informative, not merely disappointing. If the value of a language-model sentiment channel were large at a single-name daily horizon, the correct research programme would be to refine the channel. Since it is small, the correct programme is to ask whether value can be created elsewhere in the pipeline. Section 6.11 takes up that question and Chapter 8 answers it.

6.10 Architectural Feasibility at a Real-Time Horizon

The objective refers to optimising deep learning architectures for real-time prediction, and the feasibility evidence comes from the optimised architecture of Chapter 4. That model has approximately 2.1 million trainable parameters, converges in under two hours on a single consumer-grade accelerator, and executes a forward pass in milliseconds. Inference cost is therefore negligible relative to a daily decision horizon.

The binding constraint on “real time” in this domain is not inference but ingestion. Retrieving and scoring hundreds of news items and social posts through transformer

encoders introduces latency measured in seconds to minutes, depending on feed access and batching. For a daily decision this is comfortable. For an intraday or high-frequency decision it is not, and no architectural optimisation within the model changes that, because the cost lies in acquiring and reading the text rather than in predicting from it. The honest statement of scope is therefore that the framework is real-time at a daily horizon and is not a high-frequency system.

6.11 What This Chapter Establishes, and What It Changes

Four findings stand, and they must be read together rather than selectively.

1. Language-model sentiment is the **highest-gain feature** in the fused space, ahead of established technical indicators.
2. It is **orthogonal to price** — correlation 0.04 with price level, 0.74 with momentum — which is the condition under which fusion can add value.
3. It improves **confidence calibration and simulated economic performance**, with markedly stronger buy-side signal quality, and it stabilises accuracy across sentiment regimes.
4. It delivers only a **bounded increment in raw directional accuracy**, 47.72 per cent against a 46.43 per cent baseline mean, with every model near the random classifier.

The fourth finding is the one that changes the thesis. If the signal channel is weak at the granularity and horizon where decisions are taken, then refining the signal channel has limited expected return, and the value of the system must be sought elsewhere. Two candidate explanations for the weakness present themselves, and they have opposite implications. Either the *measurement* is inadequate — the sentiment construct is noisy, and a better extraction pipeline would recover a larger edge — or the measurement is sound and the *predictability* simply is not there at this horizon.

These are distinguishable empirically, and distinguishing them is the task of Chapter 7, which validates a sentiment construct against four independently published series and separately measures whether the validated construct forecasts anything. Chapter 8 then acts on the answer by moving the search for value from the prediction layer to the decision layer.

6.12 Summary

This chapter isolated a single information source — news text read by a transformer language model — and measured what it contributes to single-name daily directional prediction when fused with fifteen conventional technical indicators and evaluated against technical-only baselines under a like protocol.

The sentiment score is the highest-gain feature in the fused space, ranking above established technical indicators; it is orthogonal to price, correlating 0.04 with price level and 0.74 with momentum; it produces the highest prediction confidence of the models examined, the most stable accuracy across sentiment regimes, and the strongest buy-side signal quality, translating into higher simulated cumulative returns over a thirty-session window. The sentiment-augmented model attains 47.72 per cent directional accuracy against a baseline mean of 46.43 per cent, a relative improvement of 2.77 per cent, and does not attain the highest raw accuracy in the comparison. Receiver operating characteristic analysis places every model close to the random classifier.

The framework runs to completion in a freely accessible hosted environment, so the result is independently reproducible, and the architectural evidence of Chapter 4 establishes feasibility at a daily decision horizon with negligible inference cost. The bounded nature of the accuracy increment is reported rather than concealed, and it is what redirects the remainder of this thesis from the prediction layer to the decision layer.

Chapter 7

Explainable and Causal Framework

Research Objective 3. *To establish and evaluate an explainable artificial intelligence framework capable of delivering transparent and interpretable predictions in stock selection, analysing the impact of sentiment data noise and exploring the temporal dynamics between sentiment shifts and stock price movements.*

This objective has two halves that are usually pursued separately and are pursued together here. The first is transparency: making the reasoning behind a stock-selection decision inspectable. The second is a measurement problem: establishing how much noise the sentiment channel actually carries and how sentiment shifts relate in time to price movements. They belong together because an explanation that rests on a noisy input is not an explanation, and because Chapter 6 closed on precisely the question that the second half answers — whether the bounded value of the sentiment channel reflects inadequate measurement or genuine absence of predictability.

The transparency work is reported in Sections 7.2 to 7.5 and appears in publication [P2]; the sentiment diagnostics are reported in Sections 7.6 to 7.9 and draw on [P3] and [P5].

7.1 Transparency at the Level of Structure Rather Than Attribution

7.1.1 What post-hoc attribution can and cannot deliver

The standard response to model opacity is post-hoc attribution. Given a fitted model and an input, methods such as Shapley additive explanations [82] or local surrogate models [83] report how much each input contributed to the output; language models can then render that attribution in prose [81]. These methods are well founded and are genuinely useful for auditing a deployed system.

Their limitation is structural. Post-hoc attribution is faithful to the *model*: it reports what the fitted function did. It is silent on whether what the model did corresponds to anything in the market. A feature that co-moves with the target for entirely spurious reasons receives an attribution indistinguishable from that of a genuine driver, and an

audience shown the resulting explanation has been given an accurate account of the model and a potentially misleading account of the world. When the model is subsequently deployed and the spurious relationship breaks, the explanation will have provided no warning.

7.1.2 The alternative pursued here

This chapter therefore pursues transparency at the level of *structure* rather than of attribution. The feature set entering the model is constrained, *before any learning takes place*, to variables with demonstrable directed influence, recovered by conditional independence testing over lagged series. The explanation available afterwards is not “the model used feature X ” but “feature X was admitted because X improves the prediction of the target beyond the target’s own history, and here is the graph”. That statement is auditable by someone who does not trust the model, which is the property an investment committee or a regulator actually requires.

The trade is stated honestly. Structural transparency constrains the hypothesis space and may therefore cost accuracy relative to an unconstrained model. Section 7.4 reports the offsetting benefit: the causal constraint *retains* relationships that a correlational filter would have discarded, so in this universe the constraint is not merely a cost.

7.2 Design of the Dynamic-Causal Hybrid Framework

The framework operates in three sequential stages, shown in Figure 7.1.

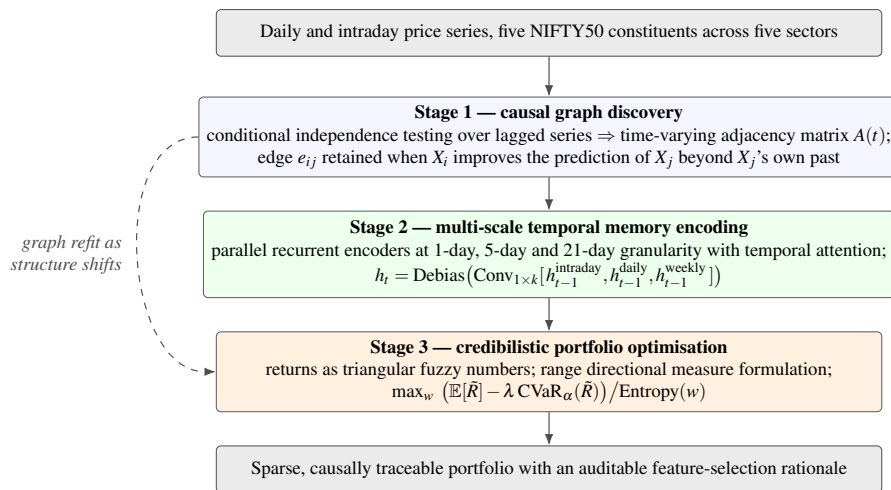


Figure 7.1: Three-stage Dynamic-Causal Hybrid Framework. Causal discovery constrains the feature set, multi-granularity encoding captures structure at several horizons, and credibilistic optimisation converts the signal into holdings under epistemic uncertainty.

7.2.1 Stage 1: causal graph discovery

Let $G_t = (V, E_t)$ be the causal graph at session t over N instruments. A directed edge $e_{ij} \in E_t$ is retained when series X_i improves the prediction of series X_j beyond what X_j 's own history achieves, tested by conditional independence over lagged closing-price series following the constraint-based procedure of Section 2.8.2 [98, 97]. The result is a time-varying adjacency matrix $A(t)$ whose entry (i, j) records the strength of the directed influence of instrument i on instrument j .

The graph is refitted as structure shifts rather than estimated once. This is the direct response to the limitation of static causal formulations identified in Section 3.3.2: a static graph cannot represent the structural breaks produced by Union Budget or monetary policy announcements, which in the Indian market reset inter-instrument relationships within hours.

7.2.2 Stage 2: multi-scale temporal memory encoding

Financial series carry structure at several time scales at once, a property consistent with the fractal description of market dynamics and with the multi-granularity architectures reviewed in Section 3.3.3. Three parallel recurrent encoders therefore process short (1-day), medium (5-day) and long (21-day) horizons, and their hidden states are combined through a convolutional temporal attention layer with a debiasing correction,

$$h_t = \text{Debias} \left(\text{Conv}_{1 \times k} \left[h_{t-1}^{\text{intraday}}, h_{t-1}^{\text{daily}}, h_{t-1}^{\text{weekly}} \right] \right), \quad (7.1)$$

where k is the kernel size. The purpose of the multi-scale construction is separation: volatility clustering operates at one scale and momentum at another, and an encoder that sees only one scale cannot distinguish a genuine trend from the low-frequency component of a volatility cycle.

The debiasing term carries an assumption that Section 7.5 reports failing empirically, and it is flagged here rather than after the fact. The correction assumes *stationary* noise properties, so that a bias estimated on earlier data remains valid later. Indian market volatility is not stationary.

7.2.3 Stage 3: credibilistic portfolio optimisation

The third stage converts the encoded signal into holdings. Rather than forcing a point estimate of expected return that the data do not support, returns are represented as triangular fuzzy numbers $[l, m, u]$, which admit vagueness in the market view explicitly — the epistemic uncertainty of Section 2.4.2, expressed in a possibilistic rather than probabilistic formalism. Weights are then obtained by

$$\max_w \frac{\mathbb{E}[\tilde{R}] - \lambda \text{CVaR}_\alpha(\tilde{R})}{\text{Entropy}(w)} \quad \text{subject to } w \in \text{RDM-DEA feasible set}, \quad (7.2)$$

following the range directional measure formulation of Kumar and Yadav [121], which handles positive and negative returns without assuming normality. The entropy denominator penalises concentration, and the conditional value at risk term penalises tail exposure rather than variance, which is the appropriate risk measure when the return distribution is asymmetric.

The coupling of Equation (7.2) to a *dynamic* causal graph is the extension this framework contributes: Kumar and Yadav’s formulation rebalances at fixed intervals, whereas here the feasible set is reconstructed when the causal structure shifts.

7.3 Experimental Setup

Data were obtained for five NIFTY50 constituents chosen to span five sectors: RELIANCE (energy and conglomerate), TCS (information technology), TITAN (consumer discretionary), NESTLEIND (fast-moving consumer goods) and HDFCBANK (financials). The period is 1 January 2024 to 31 March 2025, comprising 308 trading sessions.

The universe is deliberately small, and the reason is computational rather than methodological. Constraint-based causal discovery requires a number of conditional independence tests that grows combinatorially with the number of variables, so a five-instrument universe is tractable and a fifty-instrument universe is not with the same procedure. Section 7.11 treats scalable discovery as the principal open problem of this chapter. The five names were selected for sectoral spread and liquidity rather than by any performance criterion, which matters because selecting instruments on outcomes would invalidate the graph.

Evaluation is against a passive buy-and-hold benchmark over the same window, with the

causal structure, the recovered graph density and the behaviour around the 2024 Union Budget examined as separate diagnostics.

7.4 Results: Causal Structure as an Explanation Mechanism

7.4.1 The recovered graph

Table 7.1 reports the recovered adjacency matrix over the 308-session window.

Table 7.1: Dynamic causal adjacency matrix $A(t)$ over five NIFTY50 constituents, January 2024 to March 2025. Entry (i, j) is the strength of the directed influence of row i on column j .

Influencing ↓ / Influenced →	HDFCBANK	NESTLEIND	RELIANCE	TCS	TITAN
HDFCBANK	1.000	0.299	0.791	0.079	0.460
NESTLEIND	0.072	1.000	0.356	0.130	0.006
RELIANCE	0.913	0.324	1.000	0.650	0.849
TCS	0.110	0.670	0.516	1.000	0.436
TITAN	0.337	0.058	0.216	0.610	1.000

The graph has a density of 0.850 and a maximum causal score of 1.000, indicating strong directed interdependence among the constituents — unsurprising in an index whose members share macroeconomic exposure, but the *asymmetry* of that interdependence is informative.

7.4.2 A systematic causal hub

The most interpretable feature of Table 7.1 is the RELIANCE row. It exerts strong directed influence on HDFCBANK (0.913), TITAN (0.849) and TCS (0.650), while the corresponding column shows markedly weaker influence received in return: 0.791 from HDFCBANK, 0.216 from TITAN, 0.516 from TCS, 0.356 from NESTLEIND. Outgoing influence systematically exceeds incoming influence, which is the signature of a causal hub.

Figure 7.2 renders the structure above an edge threshold of 0.3.

The readability of Figure 7.2 is the point of the exercise. An analyst can inspect it, compare it against a prior belief about how the Indian market transmits shocks, and reject it if it conflicts. That is a different kind of object from a bar chart of attribution scores over a fitted model’s inputs, and it is what structural transparency delivers.

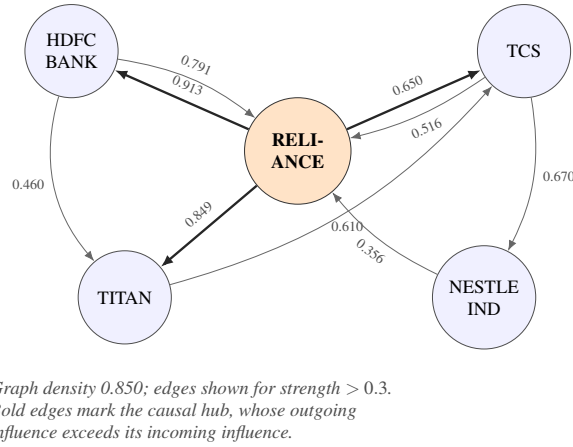


Figure 7.2: Dynamic causal graph recovered from NIFTY50 constituent series. The structure is readable, which is the property that makes the feature selection auditable.

7.4.3 Correlation and causality diverge, and the divergence is exploitable

The decisive evidence for the approach is the comparison between correlation and causality. Pairs exist in this universe — the influence of RELIANCE on TITAN being the clearest — for which the directed causal strength is high (0.849) while the linear correlation between the two series is low.

The mechanism is unsurprising once stated: influence that operates with a lag, or through a nonlinear channel such as an input-cost pass-through, does not produce contemporaneous linear co-movement. The consequence for modelling is decisive. A correlation-based feature selector applied to this universe would have discarded the RELIANCE–TITAN relationship as uninformative, and would have done so silently.

This is the mechanistic justification for the framework, and it settles a question that is usually treated as a trade-off. Transparency and accuracy are commonly presented as opposed: one accepts a simpler, more interpretable model and pays for it in performance. In this universe they are not opposed, because the causal constraint *retains* information that the correlational alternative destroys. The interpretable feature set is a superset of the correlational one along the dimension that matters.

7.5 Performance and the Failure of Stationary Debiasing

Performance over the 308-session window is reported in full, including the respects in which the framework underperforms.

Table 7.2: Performance of the Dynamic-Causal Hybrid Framework over 308 trading sessions, January 2024 to March 2025

Measure	Value	Reading
Causal graph density	0.850	Strong directed interdependence among constituents
Maximum causal score	1.000	At least one relationship of maximal measured strength
Sharpe ratio	0.377	Positive risk-adjusted return over the window
Win rate	48.8%	Below one half; combined with a positive Sharpe ratio this indicates asymmetric profit capture
Performance against buy-and-hold	−8.95%	Underperformance during a strong bull market
Peak volatility, 2024 Union Budget window	77.67%	Debiasing layer fails when the noise process changes

7.5.1 Asymmetric profit capture

A win rate below one half combined with a positive Sharpe ratio is not a contradiction; it is the signature of asymmetric profit capture, in which winning trades are larger than losing ones. The framework loses more often than it wins and still earns a positive risk-adjusted return, which means its position sizing is doing work that its directional accuracy is not. Transaction costs and signal lag account for the nominal shortfall against the passive benchmark.

The honest characterisation is therefore that this is a *capital-preservation* instrument rather than a return-maximising one, and it underperformed a rising market by 8.95 per cent over a window in which the market rose strongly. A framework that de-risks on structural signals will underperform during an uninterrupted advance, which is a property of the design rather than a defect in its implementation — but it is a property a practitioner must be told about in advance, and stating it is part of the contribution. The same characterisation recurs in Chapter 8 on a much longer sample, where the window includes the crises that this fifteen-month window does not.

7.5.2 The debiasing layer fails when the noise process changes

The most useful negative result in this chapter is the 77.67 per cent peak volatility recorded during the 2024 Union Budget announcement window. The debiasing correction in Equation (7.1) assumes stationary noise properties, and a budget announcement is

precisely an event at which the noise process changes character discretely: volatility clusters tighten, cross-instrument correlations jump, and the relationship between realised and expected dispersion breaks.

The result is diagnostic rather than fatal, and it is reported because it identifies a specific, addressable weakness rather than a general limitation. A debiasing layer whose noise model is itself state-dependent — estimating the correction separately within identified market states, or allowing its parameters to follow the volatility process — is the direct remedy, and it is listed as future work in Chapter 9. The general lesson generalises beyond this framework: any correction calibrated on historical noise will be least reliable at exactly the moments when the noise changes, which are the moments that determine a drawdown.

7.6 Sentiment Noise: Construct Validation Against External Evidence

The second half of Research Objective 3 concerns how noisy the sentiment channel is. This section and the two that follow treat that as a measurement problem with three instruments.

7.6.1 The claim to be tested

Naming a series a “sentiment measure” is a claim, and it is testable independently of whether the series makes money. The two questions are routinely conflated in the literature: a sentiment measure that fails to predict returns is described as a poor measure, and one that predicts returns is described as validated. Neither inference is sound. A measure can be an accurate gauge of market sentiment and still fail to forecast returns, if sentiment is priced contemporaneously; and a measure can forecast returns for reasons unrelated to sentiment.

The construct examined here is a market-derived composite in the tradition of Baker and Wurgler [128], built from exchange turnover and dispersion rather than from text. The reason it is market-derived rather than textual is structural and was stated in Section 3.2.3: instrument-level, time-stamped news corpora do not exist for the broad index instruments over which the cross-sectional study of Chapter 8 operates. The consequence for interpretation is important and is stated again in Section 7.8.

7.6.2 Validation against four independently published series

The composite was compared against four independently published series, *none* of which enters the framework at any stage: a household survey of consumer sentiment, an implied volatility index, a financial stress index and a financial conditions index. Two of these are survey-based or policy-institution constructions and share no inputs with the composite at all. Table 7.3 and Figure 7.3(a) report the outcome.

Table 7.3: Validation of the market-derived sentiment composite against four independently published series, in levels and in first differences, over 4,689 sessions from January 2008 to August 2026

External series	Expected sign	Freq.	r (levels)	p	r (diffs)	p
Household consumer sentiment survey	+	M	+0.262	0.056	+0.240	0.0004
Implied volatility index	—	D	−0.541	0.0003	−0.112	0.0025
Financial stress index	—	W	−0.606	0.0067	−0.254	0.0002
Financial conditions index	—	W	−0.513	0.0140	−0.123	0.0291

All four carry the expected sign; all four are significant in first differences at the five per cent level.

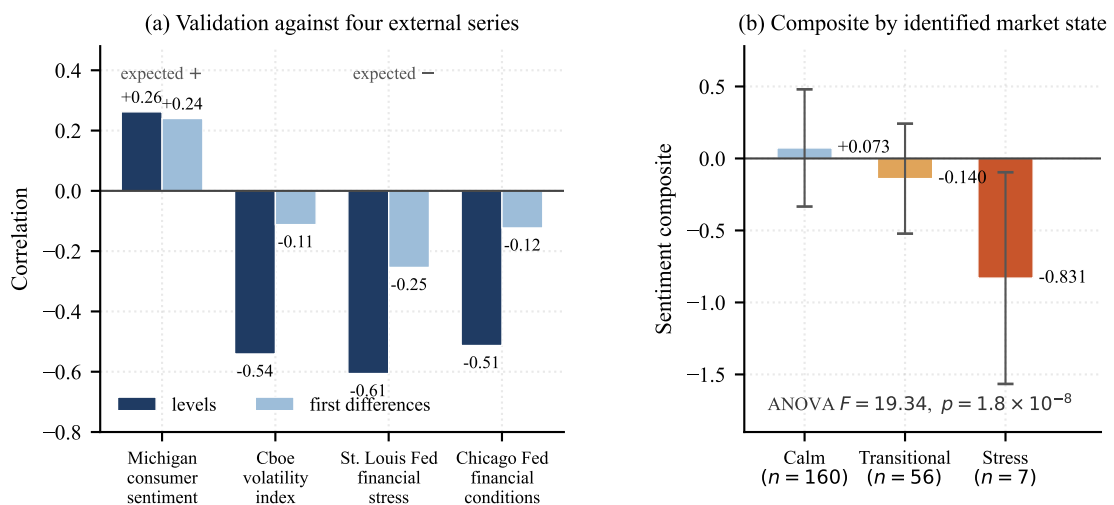


Figure 7.3: Sentiment diagnostics. (a) Correlation of the sentiment composite with four independently published series in levels and first differences. (b) The composite evaluated by identified market state; the states are labelled by fitted volatility and the composite plays no part in fitting them.

Every correlation carries the expected sign — positive against the consumer survey, negative against the three stress and volatility measures — and every one is significant in first differences at the five per cent level. Testing in first differences is the more demanding comparison, because two series that both trend will correlate in levels for reasons unrelated to their content.

The survey comparison is the informative one. A correlation of +0.240 between monthly

changes in a composite built from exchange turnover and dispersion, and monthly changes in what households report to an interviewer, cannot be a mechanical artefact of shared construction: the two objects have no common input. Either both are measuring a shared latent state, or the agreement is coincidence, and the significance level makes coincidence implausible.

7.6.3 Stability across sub-periods

A validation that holds only in aggregate may reflect a single episode. The comparison was therefore repeated within four sub-periods, giving sixteen series-by-sub-period cells. Figure 7.4 reports them.

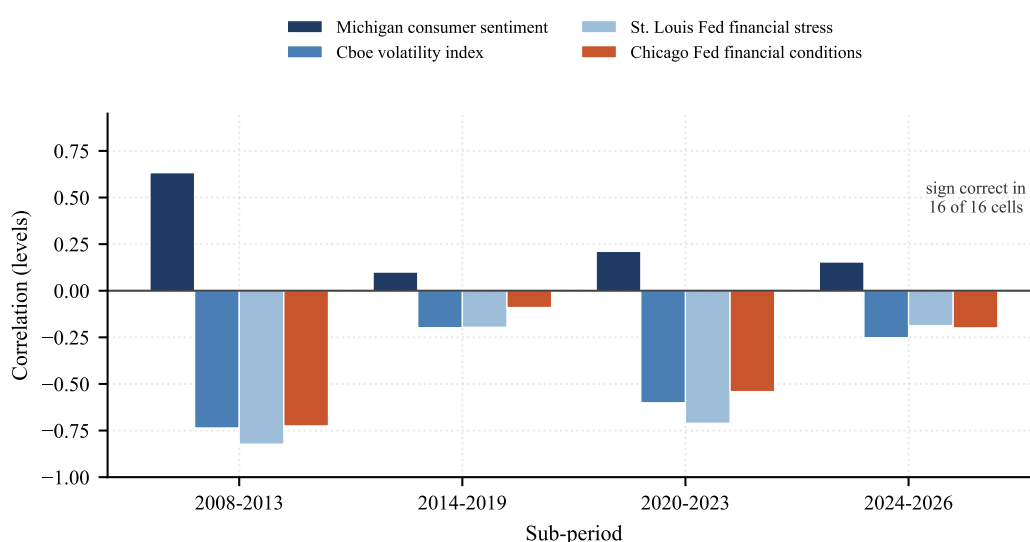


Figure 7.4: Correlation of the sentiment composite with each of the four external series within four sub-periods. The sign is correct in all sixteen cells, and magnitudes are largest in the crisis sub-periods, which is the expected pattern for a stress-sensitive measure.

The sign is correct in *all sixteen* cells. The magnitudes are far from constant — they are largest in the 2008–2013 and 2020–2023 sub-periods and weakest in the calm 2014–2019 stretch — but that variation is itself the expected pattern for a stress-sensitive measure: a gauge of market anxiety should co-move most strongly with other anxiety measures when there is anxiety to measure, and should be comparatively uninformative when there is not.

7.7 Separation Across Identified Market States

A second and independent test is available. If the latent states identified by the regime model of Chapter 8 are states of *sentiment* as well as of volatility, the composite should

differ systematically across them. The test has force because the mixture model is fitted on realised volatility and short-horizon momentum, and the sentiment composite is *not* among its inputs: any separation is separation on a withheld variable.

Table 7.4 and Figure 7.3(b) report the result.

Table 7.4: Mean sentiment composite by identified market state, with the state labels assigned by ascending fitted volatility. The composite takes no part in fitting the states.

Identified state	<i>n</i>	Mean	SD	Test statistic
Calm	160	+0.073	0.407	One-way ANOVA: $F = 19.34$, $p = 1.8 \times 10^{-8}$. Kruskal–Wallis: $H = 29.54$, $p = 3.9 \times 10^{-7}$
Transitional	56	−0.140	0.382	
Stress	7	−0.831	0.735	

The ordering is monotone and steep: +0.073 in the calm state, −0.140 in the transitional state and −0.831 in the stress state. One-way analysis of variance gives $F = 19.34$ with $p = 1.8 \times 10^{-8}$. Because the stress state contains only seven observations, the rank-based Kruskal–Wallis test was also computed as a distribution-free check and gives $H = 29.54$ with $p = 3.9 \times 10^{-7}$; the two agree, so the result does not depend on the normality assumption that the small stress sample would strain.

The conclusion supported is specific: the identified states may reasonably be described as *sentiment-and-volatility* states rather than volatility states alone. This matters for the allocation framework of Chapter 8, because a regime-conditional risk budget is more defensible if the regimes correspond to something an investor would recognise.

7.8 Predictive Content, Measured Separately

The construct is validated. The separate question is whether it forecasts returns, and the answer is that it does not.

Measured as a cross-sectional information coefficient against next-session returns, the predictive content of the validated composite over the full multi-asset sample is −0.026 with a Newey–West t statistic of −2.26: significantly *negative* rather than positive, and small. Chapter 8 reports the full set of these measurements, examines whether the negative relationship is stable enough to exploit by inversion, and finds that it is not.

Three points follow, and keeping them apart is the methodological contribution of this section.

A valid measure need not be a profitable one. A composite can track market senti-

ment accurately — as the sixteen-of-sixteen sign agreement and the $p = 1.8 \times 10^{-8}$ state separation establish — and still fail to forecast returns. If sentiment is incorporated into prices contemporaneously, an accurate contemporaneous gauge of sentiment will have no forward information by construction. The two findings are not in tension; they are answers to different questions.

Conflating the two is a recurring error. A study that reports only the predictive null invites the reading that the sentiment construct is poor. A study that reports only the validation invites the reading that sentiment is useful for forecasting. Reporting both is what allows the correct inference: the measurement is sound and the forward predictability is absent at this horizon on this universe.

The null is about this proxy, not about textual sentiment. The composite tested here is market-derived. The bounded but positive findings of Chapter 6 concern language-model-extracted *textual* sentiment on single names with dense news coverage. These are different constructs measured on different universes, and the null reported here does not transfer to the other. Section 7.10 returns to what this means for the question Chapter 6 left open.

7.9 Sentiment-to-Price Temporal Dynamics

The final component of Research Objective 3 concerns the temporal relationship between sentiment shifts and price movements. Three findings, drawn together from the diagnostics of Chapter 6 and the present chapter, characterise it.

Spikes precede moves, but the mapping is not simple. The temporal overlay of sentiment against price reported in Section 6.8 shows that strong sentiment spikes frequently precede major price changes. A simple causal ordering does not follow, because the relationship between magnitude and direction is not monotone.

Two distinct incorporation regimes. Sustained positive sentiment accompanies gradual price advances, whereas sharp negative sentiment spikes accompany abrupt declines. Favourable information is incorporated by accumulation; adverse information by discrete repricing. The two operate on different timescales, which is a direct argument for the multi-granularity encoder of Equation (7.1): a single-horizon model must choose which of the two dynamics to represent.

The channel is genuinely noisy, and the noise is measurable. The approximately

symmetric sentiment distribution with moderate tails reported in Section 6.8 establishes that the extraction pipeline captures real variation rather than producing a biased categorical output. The information-coefficient measurement of Section 7.8 establishes that the forward-looking content of the market-derived composite is not distinguishable from zero in a usable direction. Together these say that the sentiment channel is measurable, real, and at this horizon largely contemporaneous with price rather than leading it in a way that can be traded.

7.10 Answering the Question Chapter 6 Left Open

Chapter 6 closed by posing a fork. The bounded value of the sentiment channel at a single-name daily horizon admits two explanations: either the *measurement* is inadequate, in which case a better extraction pipeline should recover a larger edge; or the measurement is sound and the *predictability* is not there at that horizon.

The diagnostics of this chapter resolve the fork for the market-derived composite. The construct is demonstrably sound: correct sign against four independent external series, correct sign in all sixteen series-by-sub-period cells, and separation across withheld-variable market states at $p = 1.8 \times 10^{-8}$. Its forward predictive content is separately measured and is not usable. The limitation is therefore *not* one of measurement. Building a better gauge of the same construct would not produce a forecast.

The consequence for the thesis is the one stated at the end of Chapter 6 and acted upon in Chapter 8. If the signal channel is bounded and the boundedness does not stem from poor measurement, then the framework must earn its value somewhere other than the signal layer. The natural candidate is the decision layer, where the same information can be used to determine *how much risk to carry* rather than *which assets to hold*. That hypothesis is tested directly, with a matched-exposure ablation designed to distinguish the two, in the next chapter.

One qualification is restated so that it is not lost. This resolution applies to the market-derived composite on a broad-instrument universe. It does not close the question for textual sentiment on instruments with dense, time-stamped news coverage, where Chapter 6 found a small but real and orthogonal signal. Applying the fusion hypothesis where such a corpus exists would convert a proxy null into a genuine test, and this is identified as the first item of future work in Chapter 9.

7.11 Limitations

Four limitations bound this chapter.

Five instruments, not fifty. Constraint-based causal discovery scales combinatorially in the number of variables, so the recovered graph covers five constituents over fifteen months. Whether the hub structure identified here persists across the full index, and whether it is stable across longer periods, is not established. Scalable causal discovery is the principal open problem.

Predictive precedence, not intervention. The notion of causality used is the operational one of Section 2.8.1: X improves the prediction of Y beyond Y 's own past. This is not interventional identification, and no claim is made that acting on X would move Y . Confounding by an unobserved common driver — a macroeconomic shock that moves both instruments with different lags — is not excluded by the procedure.

Stationary debiasing. The failure documented in Section 7.5 is a limitation of the present implementation rather than of multi-scale encoding in general, and the remedy is identified.

No attribution experiments. Transparency here rests on structure. Shapley additive explanations and local surrogate models are reviewed and cited but are not run, so the thesis makes no comparative claim about structural versus attributional transparency; it argues for the former on the grounds given in Section 7.1 and demonstrates that in this universe it is not purchased at the cost of information.

7.12 Summary

This chapter pursued transparency at the level of structure rather than attribution, and treated sentiment noise as a measurement problem rather than an assumption.

The Dynamic-Causal Hybrid Framework constrains the feature set by directed causal structure recovered through conditional independence testing, encodes intraday, daily and weekly horizons through a multi-scale memory layer, and allocates through a credibilistic optimiser that admits vagueness in the market view. Over 308 NIFTY50 sessions it recovers a causal graph of density 0.850 with an identifiable causal hub whose outgoing influence systematically exceeds its incoming influence, and it demonstrates that strong directed causal influence can coexist with weak linear correlation — so that a correla-

tional feature selector would have discarded a genuine driver. Performance is reported in full: a Sharpe ratio of 0.377 with a win rate of 48.8 per cent, indicating asymmetric profit capture, and an 8.95 per cent shortfall against buy-and-hold in a rising market, which characterises the framework as a capital-preservation instrument. The 77.67 per cent peak volatility during the 2024 Union Budget window identifies stationary debiasing as a specific, addressable weakness.

Sentiment noise was quantified rather than assumed. The composite carries the expected sign against four independently published series, is significant in first differences against all four, holds the correct sign in all sixteen series-by-sub-period cells, and separates the identified market states at $p = 1.8 \times 10^{-8}$ on a variable withheld from the fitting procedure. Its forward predictive content, measured separately, is not usable. Keeping construct validity and predictive usefulness apart resolves the question Chapter 6 left open: the limitation is not one of measurement, and the value of the system must therefore be sought at the decision layer. Chapter 8 tests that directly.

Chapter 8

Decision-Theoretic Allocation and Cross-Market Validation

Research Objective 4. *To evaluate and validate the result and practical application of the proposed model across different stock markets, exploring transfer learning techniques for cross-market applications and validating the model against benchmark datasets and real-world market data.*

This chapter converts a fused signal into an executable portfolio and then takes the framework apart to determine which of its layers produces the measured outcome. The second half of that sentence is the contribution. It is straightforward to build an allocation framework that beats a benchmark on a backtest; it is considerably harder to establish *why* it beat the benchmark, and the answer obtained here is not the answer the design anticipated.

Section 8.1 specifies the framework, Section 8.2 the data and protocol, and Section 8.3 documents two implementation defects that had silently invalidated an earlier version of this study — both of which produce no error message and both of which generalise well beyond this framework. Sections 8.4 to 8.8 report the evidence, Section 8.9 the cross-market test, and Section 8.10 the limits.

The work reported in this chapter appears in publication [P5]; supporting sectoral and regime-level evidence comes from [P1] and [P2].

8.1 Design of the Decision-Theoretic Allocation Framework

8.1.1 The separation principle

The framework is built around one architectural decision from which everything else follows: the allocation problem is separated into two questions that are answered by different mechanisms.

- **Relative composition:** what to hold relative to everything else. This is answered by

a constrained quadratic program over the risky sleeve.

- **Total risk:** how much of the portfolio to expose to that sleeve at all. This is answered by a regime-dependent risk budget, with the residual held in cash.

The separation is not a presentational convenience. It is what makes the framework *falsifiable layer by layer*. Because composition and risk level are produced by distinct mechanisms, either can be disabled while the other is held fixed, and — crucially — a control can be constructed that carries the same *average* exposure as the full framework but allocates it without regard to the regime. The difference between the framework and that control isolates the *timing* of risk from its *level*. Without the separation, that experiment cannot be constructed at all, and Section 8.5 is the return on the design decision.

8.1.2 Stage 1: cross-sectional feature construction

Nine causal features are computed per asset per session and standardised *cross-sectionally* into two composites: a technical composite $T_{i,t}$ and a sentiment composite $M_{i,t}$.

Cross-sectional rather than time-series standardisation is deliberate and has two effects. A standardised value expresses how an asset ranks against its peers *on that day* rather than against its own history, which is the relevant comparison for a cross-sectional allocation. And it removes the common market factor: when every asset rises together, every raw indicator moves together, and a time-series standardisation would register a signal on every asset simultaneously — which is not a cross-sectional signal at all but a disguised market-timing bet.

8.1.3 Stage 2a: regime identification

The market state r_t is identified by a three-component Gaussian mixture over a two-dimensional feature vector of rolling realised volatility and rolling momentum, following Section 2.7.2. Two protocol requirements apply and both are load-bearing.

The mixture is **refitted at every rebalance** on an expanding window of strictly earlier sessions. Fitting once on the full sample and labelling the history backwards would let the identification of “stress” be informed by knowing when stress occurred.

The components are **anchored by ascending fitted volatility** into calm, transitional and stress states. Expectation maximisation returns component indices in arbitrary order that

permute freely between refits, so without anchoring, component 0 in one refit need not correspond to component 0 in the next. The failure mode is silent: no error is raised, and the downstream results look entirely plausible while resting on a scrambled state series.

8.1.4 Stage 2b: signal fusion and scaling

The two composites are fused with state-dependent weights and a confidence term $c_{i,t}$ that discounts assets whose composites disagree,

$$z_{i,t} = c_{i,t} (w_{r_t}^T T_{i,t} + w_{r_t}^M M_{i,t}), \quad (8.1)$$

and the fused score is placed on a return scale by information-ratio scaling [114],

$$\alpha_{i,t} = \text{IC} \cdot \sigma_{i,t} \cdot z_{i,t}. \quad (8.2)$$

The state dependence in Equation (8.1) encodes the hypothesis that the relative informativeness of the two composites varies with the market state — specifically, that sentiment becomes relatively more informative under stress. Section 8.6 tests that hypothesis directly and does not find support for it.

Equation (8.2) is not cosmetic, for reasons Section 8.3 makes concrete.

8.1.5 Stage 3a: relative composition

The risky sleeve is obtained by solving

$$\tilde{w}_t = \arg \max_w \alpha_t^\top w - \lambda w^\top \Sigma_t w - \kappa \|w - w_{t-1}\|_1 \quad \text{s.t.} \quad w \geq 0, \quad w \leq 0.25, \quad \mathbf{1}^\top w = 1, \quad (8.3)$$

where Σ_t is estimated by Ledoit–Wolf shrinkage [107], the position cap of 25 per cent limits concentration, the long-only constraint reflects the scope delimitation of Section 1.7, and the turnover penalty κ annualises a one-way transaction cost of ten basis points so that the optimiser trades only when the expected gain exceeds the cost of trading.

8.1.6 Stage 3b: total risk

The sleeve is then scaled by the regime risk budget,

$$w_t^* = k_{r_t} \tilde{w}_t, \quad k = \{1.00, 0.75, 0.50\} \text{ for calm, transitional and stress states,} \quad (8.4)$$

with the residual $1 - k_{r_t}$ held in cash. Two properties of Equation (8.4) matter.

The budget is expressed as a *fraction of the sleeve's own volatility* rather than as an absolute annualised target. Section 8.3 documents what happens when it is not.

The scaling is *one-sided*: $k \leq 1$ always, so the framework de-risks toward cash in disturbed states but never levers up. This guarantees that any measured performance difference against a fully invested benchmark is attributable to risk *reduction* rather than to leverage, which removes an entire class of alternative explanations before the evaluation begins.

8.1.7 Stage 4: evaluation and ablation

The evaluation applies a 21-session walk-forward with ten basis points of transaction costs, an 18-cell specification sweep, matched-exposure ablation, four pre-selected stress episodes, a 665-session holdout postdating every design decision, and a secondary market universe, with Newey–West [151] and stationary block bootstrap [152] inference.

8.2 Data and Experimental Protocol

8.2.1 Universes

The **primary universe** is ten liquid multi-asset exchange-traded instruments spanning United States large capitalisation, technology-weighted large capitalisation, small capitalisation, developed international equity, emerging-market equity, long-duration and intermediate-duration government bonds, investment-grade credit, gold and real estate. The **secondary universe**, used as a cross-market test, is ten mega-capitalisation United States equities over 2019–2026.

The choice of universe requires justification, because this thesis is otherwise centred on the Indian market. Two requirements bind. The allocation study needs eighteen years of clean, survivorship-free daily history including at least two severe crises, because a

claim about drawdown behaviour cannot be made on a sample containing no drawdowns. And it needs genuine dispersion in asset volatility, because a regime-conditional risk budget that de-risks into cash has nothing to rotate toward in a universe where every instrument carries similar risk. The instruments satisfying both requirements are the multi-asset exchange-traded funds used here. The study is therefore conducted as a test of the *mechanism* rather than as a claim about Indian allocation, and Section 8.9 reports what happens when the transferability condition is violated. Complementary evidence at sectoral granularity on the Indian market comes from the ten-sector evaluation of Chapter 4 and the regime-level behaviour of Chapter 7.

8.2.2 Period, protocol and benchmarks

The primary sample runs from January 2008 to August 2026, comprising 4,689 trading sessions, with rebalancing every 21 sessions — 224 rebalance dates in total. Table 8.1 records the full specification.

Table 8.1: Complete specification of the reported allocation framework configuration

Parameter	Value	Basis
Primary universe	10 multi-asset instruments	Volatility dispersion and clean history
Sample	Jan 2008 – Aug 2026, 4,689 sessions	Includes 2008 and 2020 crises
Rebalance interval	21 sessions (224 rebalances)	Monthly; swept over {10, 21, 42}
Covariance lookback	252 sessions	Swept over {126, 252}
Covariance estimator	Ledoit–Wolf shrinkage	Conditioning on 10 assets
Transaction cost	10 bp one way	Charged to the strategy only
Position cap	25%	Concentration limit
Risk aversion λ	2.0	Fixed <i>a priori</i>
Assumed information coefficient	0.05	Conservative; scaling only
Number of regimes K	3	Calm / transitional / stress
Regime multipliers k	{1.00, 0.75, 0.50}	Fixed <i>a priori</i> , one-sided
Random seed	42	Swept over {0, 7, 42}

Three benchmarks are used. **Equal weight** is the primary benchmark, following DeMiguel et al. [108]: a framework that cannot beat $1/N$ has not earned its complexity. **Inverse volatility** is a stronger risk-based competitor. **Equal weight at strategy volatility** scales the equal-weight portfolio to the same realised volatility as the framework, which is the fair comparison for a Sharpe ratio.

All three benchmarks are rebalanced daily and charged *no* transaction costs, while the

framework pays ten basis points on every trade. This is deliberately unfavourable to the framework and is stated so that the reported margins are read as conservative.

8.3 Two Silent Implementation Defects

Before the results, two defects are documented that had invalidated an earlier version of this study. Both were found only by disassembling the framework to construct the robustness analysis; both produce no error message; and both generalise well beyond this work. They are reported here rather than buried because the corrected framework cannot be understood without them, and because the class of failure they represent is under-reported in the literature.

8.3.1 Scale mismatch in signal-to-allocation conversion

In the defective version, the fused signal entered the quadratic program of Equation (8.3) in *raw indicator units*. Among the nine causal features is an on-balance volume accumulator, which on liquid instruments takes values of order 10^8 to 10^9 . The risk term $\lambda w^\top \Sigma_t w$ is of order 10^{-4} .

The consequence is that the linear term dominated the quadratic term by twelve orders of magnitude, so the covariance matrix was effectively irrelevant to the solution. The optimiser placed the entire budget on whichever asset had the largest raw signal, subject only to the position cap, on almost every rebalance date. The reported behaviour of the framework was therefore determined by the *ratio of measurement units* between the two terms of the objective, not by any information in the signal.

Nothing in the code failed. The optimiser converged, the constraints were satisfied, the backtest ran, and the resulting equity curve was plausible enough to be published. The remedy is Equation (8.2): information-ratio scaling places the signal on a return metric so that the two terms of the objective are commensurable, and the risk model resumes doing work.

The general lesson is worth stating independently of this framework. Whenever a signal enters a mean-variance objective, the units of the linear and quadratic terms must be reconciled explicitly. A concentrated, nearly single-asset solution on almost every date is the diagnostic symptom.

8.3.2 An inert risk layer

In the same defective version, the regime risk budget was specified as *absolute* annualised volatility targets of 20, 15 and 10 per cent for the three states. The risky sleeve of a diversified ten-instrument multi-asset portfolio carries a realised volatility of about 6.4 per cent.

An absolute target binds only when the portfolio's own volatility exceeds it. At 6.4 per cent against a floor of 10 per cent, the target bound in **one rebalance out of 224**. The regime layer — the central mechanism of the framework — was present in the specification, consumed computation at every rebalance, and did nothing on 223 of 224 occasions.

Again nothing failed. The regime model fitted, the states were identified, the multipliers were computed, and the results were reported as though the layer had been active. The remedy is Equation (8.4): expressing the budget as a *fraction of the sleeve's own volatility* makes it scale-free, so it remains active in any universe.

The general lesson is that an absolute volatility target is not scale-free, and moving a strategy between universes can therefore disable it silently. Any risk layer should be instrumented with a count of how often it actually binds — a one-line diagnostic that would have caught this immediately.

8.4 Main Results

Table 8.2 consolidates the evaluation across the primary universe, the holdout, the specification sweep and the secondary market.

8.4.1 The primary universe

On the primary universe the framework raises the Sharpe ratio from 0.628 to 0.878, a gain of 40 per cent, and reduces maximum drawdown from -36.08 to -18.98 per cent, a reduction of 47 per cent, while earning a lower compound return of 5.65 against 8.45 per cent per annum.

The correct characterisation is therefore a *risk-reduction instrument whose return sacrifice is smaller than its risk reduction*, and it is presented as such rather than as a return-enhancing one. An investor unwilling to give up return should not deploy it.

Table 8.2: Portfolio-level evaluation. Benchmarks are rebalanced daily and charged no transaction costs, which is deliberately unfavourable to the proposed framework.

Configuration	CAGR	Vol.	Sharpe	Max DD	Calmar
<i>Primary universe: ten multi-asset instruments, Jan 2008 – Aug 2026, 4,689 sessions</i>					
Proposed regime-conditional framework	5.65%	6.44%	0.878	–18.98%	0.298
Equal-weight benchmark	8.45%	13.45%	0.628	–36.08%	0.234
Inverse-volatility benchmark	7.46%	8.78%	0.849	–24.09%	0.310
Equal-weight at strategy volatility	4.19%	6.44%	0.651	–18.69%	0.224
<i>Out-of-sample holdout: 665 sessions, Jan 2024 – Aug 2026</i>					
Proposed framework	12.32%	7.63%	1.614	–5.48%	2.246
Equal-weight benchmark	15.44%	11.10%	1.391	–10.13%	1.524
Inverse-volatility benchmark	11.89%	8.51%	1.397	–6.70%	1.775
Equal-weight at strategy volatility	7.28%	5.32%	1.370	–7.04%	1.473
<i>Specification sweep: 3 seeds $\times$ 2 covariance windows $\times$ 3 rebalance intervals</i>					
Range across all 18 cells	—	—	0.813 – 0.956	–21.1% – –19.0%	—
Cells beating the benchmark on both measures	18 of 18				
<i>Secondary universe: ten mega-capitalisation equities, 2019 – 2026 (cross-market transfer)</i>					
Proposed framework	17.02%	14.87%	1.145	–20.45%	0.832
Equal-weight benchmark	33.52%	24.45%	1.371	–33.63%	0.997

Against the volatility-matched benchmark the reading is more precise and less flattering. The framework retains its Sharpe advantage — 0.878 against 0.651 — but loses marginally on drawdown, –18.98 against –18.69 per cent. This bounds the drawdown claim exactly: the framework reduces drawdown relative to an unlevered diversified benchmark, *not* relative to every portfolio held at the same volatility. Stating the bound is what makes the unbounded version of the claim unnecessary.

8.4.2 The holdout

The holdout block is the single strongest piece of evidence in this thesis, and the reason is procedural rather than statistical. Every specification decision — universe, feature set, number of regimes, risk budget multipliers, rebalancing interval, cost assumption — was fixed before any session in the holdout was examined. The 665 sessions from January 2024 postdate the entire design.

On those sessions the framework records a higher Sharpe ratio and a smaller drawdown than all three benchmarks, and two of those comparisons *reverse* the in-sample ordering: the inverse-volatility benchmark beats the framework on Calmar in sample and does not in the holdout. Figure 8.1 shows the comparison.

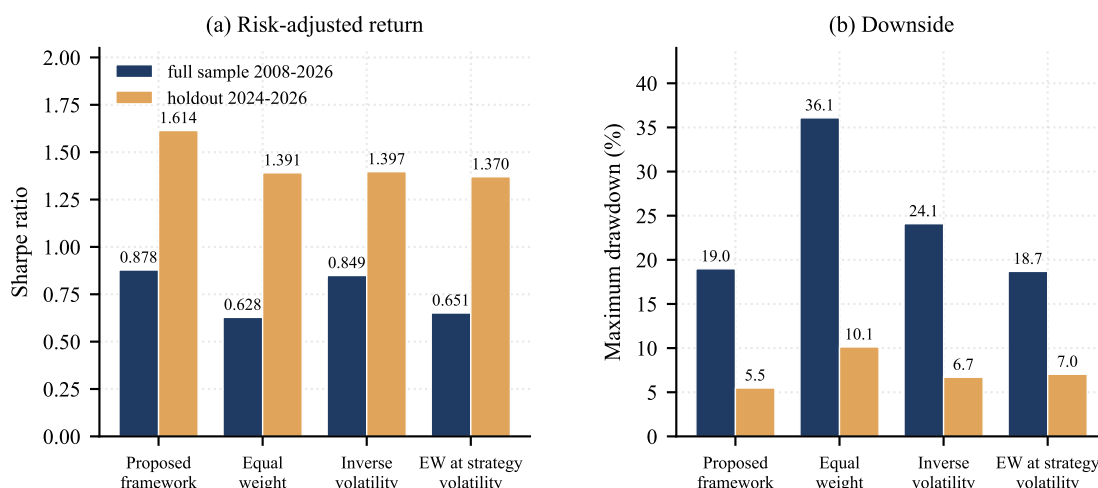


Figure 8.1: Full-sample and holdout performance against the three benchmarks. Every specification decision was fixed before any holdout session was examined.

Three qualifications are stated alongside, and they are not perfunctory. Six hundred and sixty-five sessions cannot settle the question on their own. The period contains no episode comparable to 2008 or 2020, so the holdout does not test the drawdown mechanism where it matters most. And the framework again earns less in absolute terms than the benchmark it beats on risk-adjusted measures.

8.4.3 The specification sweep

Reporting the best cell of a grid and omitting the grid is a misstatement of evidence, for the reasons given in Section 2.11.3. The entire sweep is therefore reported: three seeds $\times$ two covariance lookback windows $\times$ three rebalance intervals, eighteen cells. Figure 8.2 plots all of them.

Sharpe ratios range from 0.813 to 0.956 and maximum drawdowns from -21.1 to -19.0 per cent. All eighteen cells beat the equal-weight benchmark on *both* measures. The reported specification, at Sharpe 0.878, sits near the middle of that range rather than at its top — which is the honest position to report from and is itself informative, since a headline drawn from the best cell would have overstated the framework by roughly nine per cent of Sharpe ratio.

8.5 Component Ablation at Matched Exposure

This section contains the central result of the chapter, and the instrument that produces it is the matched-exposure control described in Section 2.10.4.

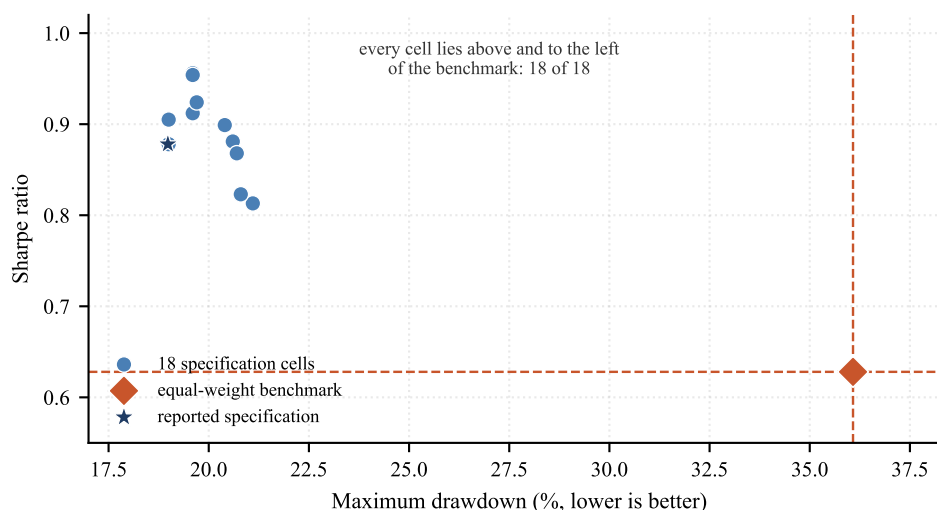


Figure 8.2: All eighteen cells of the specification sweep on the primary universe, plotted against the equal-weight benchmark. Every cell lies above and to the left of the benchmark, that is, higher Sharpe ratio and smaller drawdown.

8.5.1 The control

Comparing a de-risked framework against a fully invested benchmark confounds two effects: the framework holds *less* risk on average, and it holds risk at *different times*. The first alone will improve a drawdown statistic without any skill in timing whatever. The control therefore carries the same average invested exposure as the full framework — measured at 92.1 per cent — but allocates it *uniformly* rather than by regime. Any difference between the framework and this control is attributable to the timing of risk alone.

8.5.2 Regime-conditional risk timing works

Holding average exposure fixed at 92.1 per cent, allocating that exposure according to the identified regime rather than uniformly raises the Sharpe ratio from 0.795 to 0.878 and improves maximum drawdown from -22.26 to -18.98 per cent, with *no significant difference in mean return* ($p = 0.695$).

That combination is the signature of a genuine risk-timing effect. Both risk-adjusted measures improve while mean return is statistically unchanged, which is what one expects if the framework is rearranging *when* it carries risk rather than simply carrying less of it. Had the mechanism been merely holding less, mean return would have fallen measurably.

The cost of that timing is also measured, because a mechanism that improves risk-adjusted return is not free. Against a fully invested variant, the framework gives up

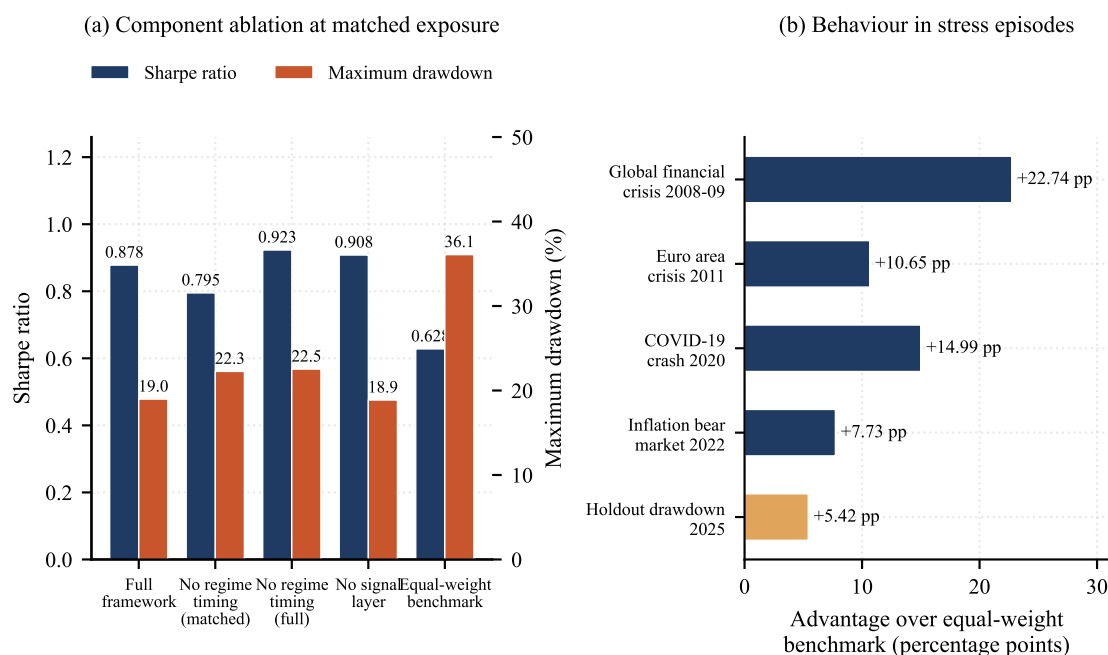


Figure 8.3: (a) Component ablation. The matched-exposure control carries the same average invested exposure as the full framework but applies it without regard to the regime, so that only the timing of risk differs. (b) Cumulative advantage over the benchmark in four stress episodes chosen on documented market history, plus the deepest benchmark drawdown in the holdout.

approximately 1.3 percentage points of annual return, significantly so ($p = 0.016$), in exchange for 3.55 percentage points of maximum drawdown. That is the actual trade a practitioner is being offered, and it is stated in those terms.

8.5.3 The signal fusion layer returns a null

The same ablation removes the signal fusion layer entirely, so that the objective of Equation (8.3) reduces to a constrained minimum-variance problem. The Sharpe ratio moves from 0.878 to 0.908 and maximum drawdown from -18.98 to -18.87 per cent. *Both movements favour removal*, and neither is significant ($p = 0.650$).

The signal fusion layer — the component that a reader of Chapters 4 to 7 would expect to be central — contributes nothing detectable to cross-sectional selection in this framework on this universe.

This null is retained in full rather than removed, for two reasons. The first is that it is what the evidence shows. The second is methodological: component-level negative results are rarely published, which is one reason architectures in this literature accumulate layers whose contribution has never been isolated. Reporting which layer works is more useful to a subsequent researcher than an aggregate number that conceals the distinction, and a

framework whose useless component has been identified is a better engineering artefact than one whose aggregate performance is slightly higher for unknown reasons.

8.6 Why the Signal Layer Fails: Information Coefficients

The ablation establishes *that* the signal layer adds nothing. The information coefficients establish *why*. Table 8.3 and Figure 8.4 report them.

Table 8.3: Cross-sectional information coefficients of each composite against next-session returns, with Newey–West t statistics

Universe	Technical		Sentiment		Fused	
	IC	t	IC	t	IC	t
Multi-asset, 2008–2026	−0.030	−2.14	−0.026	−2.26	−0.033	−2.42
Sector rotation, 2008–2026	−0.007	−0.49	−0.008	−0.62	−0.005	−0.37
Mega-capitalisation, 2019–2026	+0.035	+1.52	+0.029	+1.51	+0.044	+2.02

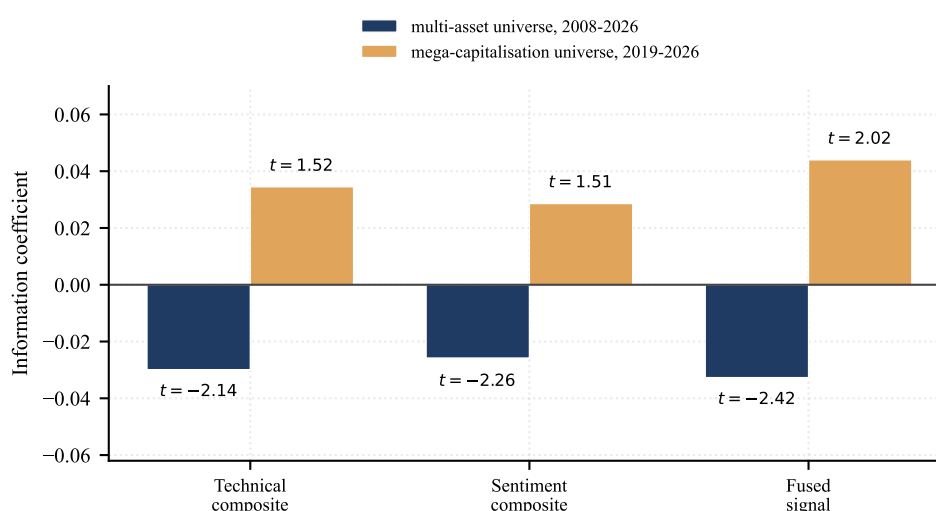


Figure 8.4: Cross-sectional information coefficients of the technical, sentiment and fused composites against next-session returns, on the primary multi-asset universe and on the secondary mega-capitalisation universe, with Newey–West t statistics.

Three findings follow.

The composites are significantly negatively related to next-session returns on the primary universe. All three information coefficients are negative and all three carry Newey–West t statistics beyond -2 . A signal that is significantly negatively related to the target is information, in principle; a long-only optimiser that treats it as positive will allocate away from the assets it should hold.

The state-dependent weighting hypothesis is not supported. Equation (8.1) encodes

the hypothesis that sentiment becomes relatively more informative under stress. Measured directly, the point estimates run the *other way*, and the calm-to-stress contrast is insignificant ($t = -1.46$, $p = 0.145$). The mechanism was built on a hypothesis the data do not support, and saying so is more useful than retaining the layer on the grounds that it does no visible harm.

The relationship is not stable enough to exploit by inversion. The obvious response to a significantly negative information coefficient is to invert the signal. The sub-period breakdown forecloses that: the coefficient is -0.039 before 2024 and $+0.009$ afterwards. A sign that flips between sub-periods is not a tradable relationship; it is an in-sample artefact that inversion would have converted into an out-of-sample loss.

The sector-rotation row is included because it is the closest available analogue to a sectoral Indian deployment, and there the coefficients are indistinguishable from zero in either direction. The mega-capitalisation row is positive and marginally significant on the fused signal, which is consistent with a genuine but small cross-sectional signal existing in a universe of heavily-researched single names and not in a universe of broad index instruments — exactly the structural argument of Section 3.2.3 about where instrument-level information exists.

8.7 Behaviour Under Stress

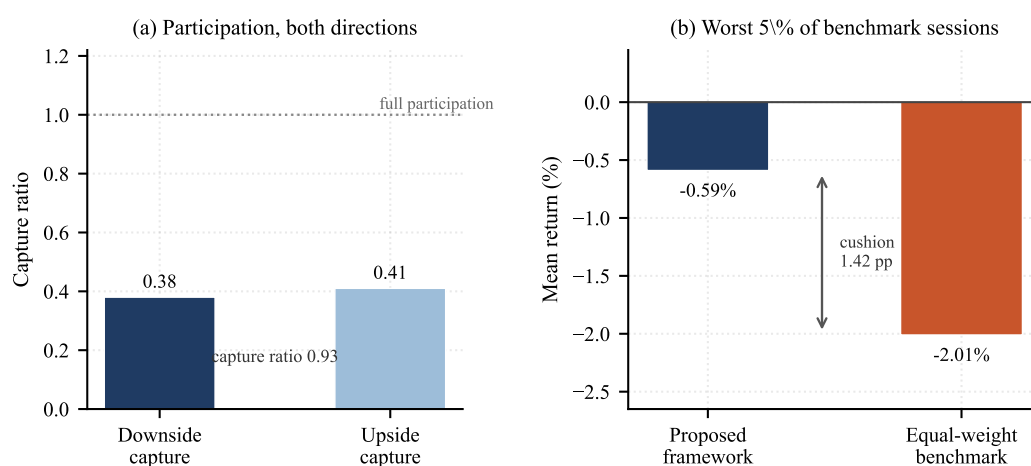
Four stress episodes were selected *before* the analysis on documented market history rather than by inspection of results: the global financial crisis of 2008–09, the euro area crisis of 2011, the COVID-19 crash of 2020 and the inflation-driven bear market of 2022. The deepest benchmark drawdown within the holdout period is reported alongside as a fifth, out-of-sample episode.

The framework outperforms in *all four* pre-selected episodes, by margins between 7.73 and 22.74 percentage points, and by a further 5.42 points in the holdout drawdown (Figure 8.3(b)). Table 8.4 reports two complementary tail diagnostics.

The capture figures are the honest interpretation of the stress results and deserve emphasis because they are easy to overstate. Downside capture is 0.38 and upside capture 0.41, giving a capture ratio of 0.93 — slightly *below* one. The framework participates less in both directions in roughly equal proportion. It has no directional skill; what it has is a lower average exposure in adverse states, and the cushion of 1.42 percentage points on

Table 8.4: Tail behaviour on the primary universe: mean return on the worst five per cent of benchmark sessions, and directional capture ratios

Measure	Framework	Benchmark	Reading
Mean return on worst 5% of benchmark sessions	−0.59%	−2.01%	Cushion of 1.42 percentage points per session
Downside capture	0.38	1.00	Participates in 38% of benchmark declines
Upside capture	0.41	1.00	Participates in 41% of benchmark advances
Capture ratio (up/down)	0.93	1.00	Below one: no directional skill

**Figure 8.5:** (a) Downside and upside capture against the benchmark. (b) Mean return on the worst five per cent of benchmark sessions. The framework participates less in both directions in roughly equal proportion, so its advantage comes from carrying less risk in adverse states rather than from directional skill.

the worst sessions is the accumulated benefit of that.

This is consistent with the ablation of Section 8.5 and inconsistent with an interpretation in which the framework anticipates declines. It does not anticipate them. It carries less risk when the identified state suggests risk is expensive, and the identified state is correlated enough with subsequent stress for that to be worth doing.

8.8 Statistical Significance, Stated Directly

The Sharpe difference against the equal-weight benchmark is +0.250. Its 95 per cent stationary bootstrap interval is $[-0.102, 0.574]$, which **includes zero**. The shortfall in mean return is likewise not significant ($p = 0.171$).

The reason is structural rather than a deficiency of the analysis. Eighteen years of daily data contain 4,689 observations but only four independent observations of the events

that matter for a drawdown claim. No amount of daily sampling converts four crises into statistical power, because the daily observations within a crisis are not independent draws from the distribution of crises.

What *is* robust is consistency, and the distinction between robustness and significance made in Section 2.11.4 applies here directly:

- the Sharpe advantage appears in 18 of 18 specifications;
- the drawdown advantage appears in 18 of 18 specifications and in 18 of 19 calendar years;
- the outperformance appears in all four pre-selected stress episodes and in the holdout drawdown;
- the holdout, which postdates every design decision, beats all three benchmarks on both measures.

Robustness and significance are different properties and neither implies the other. This thesis claims the former and explicitly does not claim the latter.

8.9 Cross-Market Transfer

8.9.1 Direct transfer, not transfer learning

One point of scope is stated plainly rather than glossed, because the objective's wording invites a stronger reading than the evidence supports. Research Objective 4 refers to *transfer learning* for cross-market application. What has been carried out is *direct transfer*: the framework, its feature definitions and its fitted procedure were applied to the second universe without modification and without any adaptation step. No parameter was fine-tuned on the target market, no representation was pre-trained and re-used, and no domain-adaptation objective was optimised.

Direct transfer is the stricter test of whether the *mechanism* generalises, and the result is therefore a lower bound on what an adapted framework would achieve. Adaptation-based transfer is identified as open work in Chapter 9 rather than claimed here.

8.9.2 Result: defensive behaviour survives, risk-adjusted return does not

On ten mega-capitalisation United States equities over 2019–2026, the outcome is mixed and is reported as such.

The framework **loses** on risk-adjusted return: Sharpe 1.145 against 1.371 for equal weight, giving up 15.09 percentage points of annual return, significantly so ($t = -2.791$, $p = 0.005$). Against the volatility-matched benchmark at Sharpe 1.350 it also loses.

The defensive behaviour nevertheless survives intact. Maximum drawdown improves from -33.63 to -20.45 per cent; the framework wins on drawdown in eight of eight calendar years and in *all 36* specification cells examined on that universe; and it outperforms in both stress episodes the window contains, by 12.50 percentage points in the 2020 crash and 12.60 points in the 2022 bear market. Its mean return on the worst five per cent of benchmark sessions is -2.35 per cent against -3.99 per cent, a cushion of 1.64 points.

Most importantly, the *regime layer itself continues to behave as designed*. Against the matched-exposure control on this universe — average exposure 91.6 per cent — the framework improves both the Sharpe ratio (1.145 against 1.119) and the drawdown (-20.45 against -30.36 per cent). The risk-timing effect reproduces in a universe where the framework loses overall, which is strong evidence that the mechanism is real and that the overall loss has a different cause.

8.9.3 The cause is a property of the universe, and it is testable in advance

Ten mega-capitalisation equities are close to a single factor bet. Correlations among them are high, there is no low-volatility asset to rotate *toward*, and the risky sleeve carries roughly 15 per cent annualised volatility against 6 per cent in the multi-asset case. De-risking in such a universe can only move capital to cash, so the framework systematically forgoes return during an exceptional advance — and 2019–2026 was an exceptional advance for that universe, with the benchmark compounding at 33.52 per cent per annum.

The transferability condition is therefore explicit and can be checked *before* deployment rather than discovered after it: **the framework requires a universe with genuine dispersion in asset volatility**. Where that dispersion exists, de-risking is a rotation and the framework adds risk-adjusted value; where it does not, de-risking is a cash drag and the framework delivers drawdown protection at the cost of return.

A bounded claim of this kind is more useful to a practitioner than an unbounded one, because it tells them where not to use the tool.

8.10 Limitations

Five limitations bound this chapter.

The Sharpe improvement is not statistically significant. Stated in Section 8.8 and not softened.

Few independent stress events. Four crises in eighteen years is the effective sample for the claim that matters most.

Lower absolute return. The framework earns less than the benchmark it outperforms on risk-adjusted measures, in every universe tested.

The universe is not Indian. The mechanism is tested where the data requirements can be met, and Section 8.2 states why. Applying the framework to an Indian multi-asset universe — Nifty indices, government securities, gold and real-estate instruments — is feasible and is identified as future work.

The signal layer null is a null about these composites on these universes. It is not a general result about signal-driven allocation, and as Section 7.8 argued, the sentiment composite involved is market-derived rather than textual. A test on a universe with instrument-level news coverage remains outstanding.

8.11 Summary

This chapter converted a fused signal into an executable portfolio and then disassembled the result to find out which of its layers produced the outcome.

Over eighteen years and 4,689 sessions of ten liquid multi-asset instruments, with transaction costs charged to the framework and not to its benchmarks, the framework attains a Sharpe ratio of 0.878 against 0.628 for equal weighting and reduces maximum drawdown from -36.08 to -18.98 per cent, at the cost of a lower compound return. It beats the equal-weight benchmark on both measures in all eighteen cells of a fully disclosed specification sweep, wins on drawdown in 18 of 19 calendar years, outperforms in all four pre-selected stress episodes, and beats all three benchmarks on both measures over

a 665-session holdout that postdates every design decision.

The matched-exposure ablation locates the source of the gain precisely. Holding average exposure fixed at 92.1 per cent, timing risk by identified market state improves both the Sharpe ratio and the drawdown with no significant change in mean return — the signature of a genuine risk-timing effect — at a measured cost of 1.3 percentage points of annual return. The signal fusion layer, by contrast, contributes nothing detectable: removing it improves both measures insignificantly, and the information coefficients explain why, being significantly negative on the primary universe, unstable in sign across sub-periods, and inconsistent with the state-dependent weighting hypothesis the layer was built on.

Two silent implementation defects were documented because both generalise: a signal entering a mean-variance objective in raw units, so that the ratio of measurement units rather than the information in the signal determined the solution; and an absolute volatility target that bound in one rebalance out of 224, leaving the central mechanism of the framework present in the specification and inert in operation.

Cross-market transfer reproduces the defensive behaviour and the regime effect in a concentrated mega-capitalisation universe while losing on risk-adjusted return, for a reason that is a property of that universe rather than of the framework. The resulting transferability condition — genuine dispersion in asset volatility — is stated in a form that can be checked before deployment.

Chapter 9

Conclusion and Future Scope

This chapter consolidates what the thesis has established across its four research objectives, states the limits of the evidence explicitly, explains why two accuracy figures reported in different chapters are not in conflict, sets out the research significance, and identifies the work that remains.

9.1 Consolidated Outcomes Against the Objectives

Table 9.1 summarises what each objective produced and the status of the evidence behind it. The final column is stated deliberately, because the strength of a result and the confidence that can be placed in it are separate properties and have been treated separately throughout this work.

Read as a ledger, the pattern is consistent. Adaptive fusion beats fixed-weight combination, but index-level accuracy does not transfer to single-name daily direction. Language-model sentiment is orthogonal and highest-importance, but does not overturn market efficiency at a daily horizon. The sentiment construct is validated against four independent external series, but a validated construct does not thereby forecast returns. Regime risk timing improves both the Sharpe ratio and the drawdown at matched exposure, but the Sharpe gain is not statistically significant on eighteen years of data. The defensive behaviour transfers to a second universe, but the framework does not add risk-adjusted value in a concentrated single-factor one.

These bounding statements are not shortcomings appended to the results. They are the boundary of the contribution, and stating them is what makes the positive claims credible.

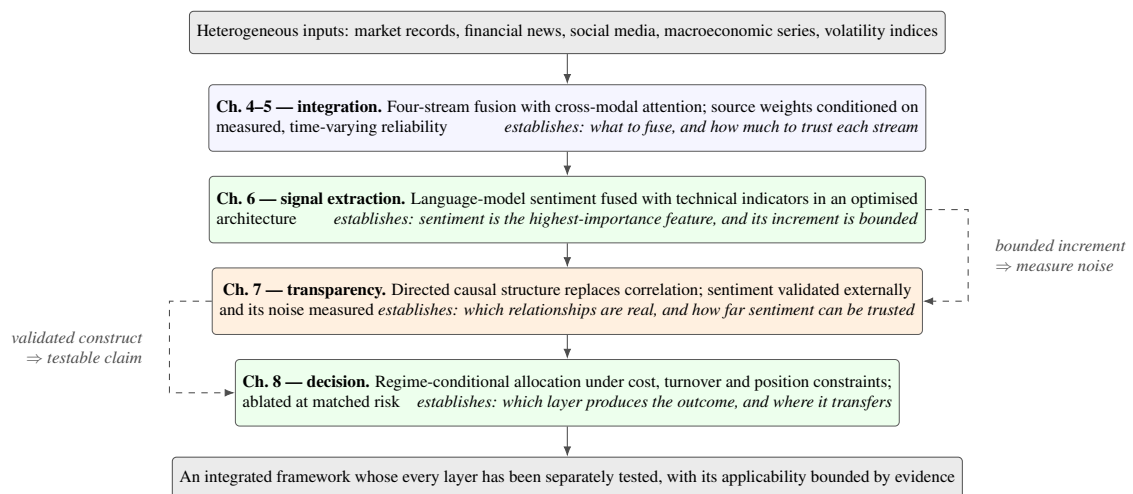
9.2 How the Contributions Compose

The four contributions are not four independent studies with a shared subject. Each was produced in response to a limitation exposed by the one before it, and Figure 9.1 shows the composed system together with the two places where a finding redirected the design

Table 9.1: Objective-wise consolidated outcomes and the status of the supporting evidence

Obj.	Principal outcome	Key quantitative evidence	Status and limits
RO1	Four-stream fusion model with adaptive, then uncertainty-conditioned, source weighting (Ch. 4–5)	MAE 0.018, R^2 0.988, DA 84.7% against 76.3% for a fixed-weight ensemble on identical inputs; both textual sources contribute 10.9% of R^2 superadditively; dynamic weighting adds 22.7% Sharpe and removes 18.3% of drawdown	Established. Index-level, one-day horizon; macro interpolation understates announcement-day effects; dynamic result reported relative to a baseline
RO2	Hybrid LLM sentiment and indicator framework in a reproducible environment (Ch. 6)	Sentiment is the highest-gain feature; correlation with price 0.04, with momentum 0.74; accuracy 47.72% against a 46.43% baseline mean; highest prediction confidence; buy-signal success 72% and 65%	Established but bounded. All models near the random classifier at a single-name daily horizon
RO3	Causal, multi-granularity transparency and a measured account of sentiment noise (Ch. 7)	Causal graph density 0.850 with an identifiable hub; correct external sign in 16 of 16 series-by-period tests; market states separated at $p = 1.8 \times 10^{-8}$ on a withheld variable	Established for construct validity. Predictive content separately measured and not usable; five instruments only
RO4	Regime-conditional decision framework, ablated at matched risk and tested across markets (Ch. 8)	Sharpe 0.878 against 0.628; drawdown -18.98% against -36.08% ; wins in 18/18 specifications, 18/19 years, 4/4 stress episodes; holdout Sharpe 1.614	Established as consistent, not as statistically significant. Bootstrap interval $[-0.102, 0.574]$ includes zero; signal layer returns a null

of the next stage.

**Figure 9.1:** Integration of the four contributions. Each layer is a response to a limitation exposed by the previous one, and the dashed paths show the two places where a finding in one study redirected the design of the next.

The logic of the progression is as follows. Chapter 4 established that fusing four sources improves prediction and that the value of each source is state-dependent, which raised the question of how to weight a source whose reliability changes — answered in Chapter 5 by deriving the weight from measured predictive uncertainty rather than from learned average relevance. Chapter 6 isolated the language-model sentiment channel and found it to be the most important single feature yet to deliver only a bounded increment in raw accuracy, which raised the question of whether the limitation lay in the measurement or in the horizon. Chapter 7 answered by separating the two: the sentiment construct was validated externally and found sound, while its predictive content was measured separately and found unusable, so the limitation is *not* one of measurement. Chapter 8 followed directly. If the signal channel is bounded and the boundedness does not stem from poor measurement, the framework must earn its value at the decision layer — and the matched-exposure ablation confirms exactly this: risk timing by market state works, and the signal layer does not.

The final system is therefore not the one originally envisaged, and it is better supported for that reason. An architecture whose components have each been tested, one of which has been shown not to work, is a more reliable engineering artefact than one whose aggregate performance conceals which of its parts is doing the work.

9.3 Reconciling the Divergent Accuracy Figures

A reader comparing Chapters 4 and 6 will notice an apparent contradiction: the first reports 84.7 per cent directional accuracy, the second 47.72 per cent. The two numbers are not in conflict, and understanding why is central to the significance of this research.

They answer different questions about different objects. Chapter 4 predicts the next-day level of a *diversified index* from four fused sources. An index aggregates hundreds of constituents, so idiosyncratic noise largely cancels and the remaining signal is dominated by persistent market-wide state — exactly the component that macroeconomic conditions, aggregate sentiment and index-level technical structure describe well. Chapter 6 predicts the next-day *direction of an individual equity*, which carries its full idiosyncratic component, has close to balanced classes, and offers no aggregation to average the noise away. Predictability is therefore expected to be far lower, and this is exactly what the empirical asset pricing literature reports [137, 131].

Three implications follow, and together they define the trajectory of the research programme.

An accuracy figure is meaningless without its object and horizon. Comparing 84.7 per cent to 47.72 per cent as though they measured the same thing is a category error, and the fact that both numbers appear in this thesis makes the point more sharply than any argument could.

The level at which a system is applied should match where the signal lives. Aggregate market state is predictable; single-name daily direction is largely not. A system that attempts to extract single-name daily alpha from the same information that describes aggregate state is applying a tool at the wrong granularity.

If the signal channel is weak where decisions are taken, value must be created at the decision layer. This is the most consequential implication and it is what Chapter 8 tests directly. The matched-exposure ablation confirms it: risk timing works where signal selection does not.

The apparent contradiction between the two figures is therefore not an inconsistency to be explained away. It is the finding that organises the thesis.

9.4 Research Significance

9.4.1 Methodological significance

The work advances multi-source fusion from a question of *what* to combine to a question of *how* to combine under changing reliability. Expressing fusion weights as a function of measured, time-varying predictive uncertainty gives a source-weighting rule that self-corrects within days rather than at the next retraining cycle, and does so in response to causes that need not be represented in the training inputs. Distribution-aware normalisation, fitted on an expanding window, prevents a heavy-tailed input from dominating the fused representation without leaking future information into the scaler.

Separating the allocation decision into relative composition and total risk is a second methodological contribution. It makes each layer independently falsifiable, and without it the central ablation of this thesis could not have been constructed at all — the matched-exposure control requires that the level of risk be settable independently of its composition.

9.4.2 Evaluation significance

Three evaluation practices used here are uncommon in this literature and are, in the author's judgement, the most transferable part of the work.

The first is **ablation at matched risk**, which separates the timing of risk from its level. Comparing a de-risked strategy against a fully invested benchmark confounds the two and credits a framework for merely holding less. The literature contains many risk-managed strategies whose reported advantage has never been decomposed in this way.

The second is **disclosure of the entire specification sweep** rather than its best cell — the minimum required once multiple testing is taken seriously [146, 147, 150]. In this thesis the reported specification sits near the middle of its sweep rather than at the top, and reporting the range rather than the maximum costs roughly nine per cent of headline Sharpe ratio. That cost is the price of a credible number.

The third is **separation of construct validity from predictive usefulness**. The sentiment composite is shown to measure sentiment — sixteen of sixteen correct signs against independent external series, market states separated at $p = 1.8 \times 10^{-8}$ on a withheld variable — and, separately, shown not to forecast returns. Conflating those questions is a recurring error, and keeping them apart changes what a null result means: it converts “the measure is bad” into “the predictability is absent”, which points research in a completely different direction.

9.4.3 Engineering significance

Two failure modes were diagnosed that generalise beyond this framework, and both are silent.

Scale mismatch in signal-to-allocation conversion. Passing an unscaled indicator into a mean-variance objective lets the ratio of units between the linear and quadratic terms, rather than the information in the signal, determine the solution, producing a degenerate nearly single-asset portfolio. Information-ratio scaling [114] removes it. The diagnostic symptom is a concentrated solution on almost every rebalance date.

An inert risk layer. An absolute volatility target calibrated for a concentrated equity portfolio binds in one rebalance out of 224 on a diversified universe, so the layer is present, consumes computation and does nothing. A relative budget is scale-free and

remains active. The diagnostic is a one-line counter of how often the constraint binds.

Mixture-model label anchoring has the same character: omitting it produces no error message, only a silently scrambled regime series. The common property of all three is that the software runs correctly and the results are plausible. This class of defect is under-reported precisely because it leaves no trace, and documenting instances of it is a contribution in itself.

9.4.4 Practical significance

For a practitioner the results support a specific and limited prescription.

- **Fuse sources, but weight them by measured reliability** rather than by historical average importance. The weighting rule, not the expert networks, produced the measured gain in Chapter 5.
- **Use language-model sentiment for what it demonstrably provides** — orthogonal information and better-calibrated confidence — rather than for a directional edge it does not provide at a daily horizon. Its buy-side signal quality is materially better than its sell-side quality.
- **Prefer causally selected features where the structure is recoverable**, because they survive regime change better and, as Chapter 7 showed, retain relationships that a correlational filter discards silently.
- **Implement a regime risk budget in relative rather than absolute form**, evaluate it against a volatility-matched benchmark, and expect it to help only where the universe contains genuine dispersion in asset volatility.
- **Size positions conservatively**, in the fractional form the growth-optimal criterion recommends once estimation error is admitted [105].
- **Instrument every risk layer with a binding counter**, so that an inert mechanism announces itself.

For the Indian market specifically, the measured weights matter. Macroeconomic indicators carry 25 per cent of average attention, reflecting the influence of policy rates, inflation and industrial output on valuations. Sentiment sources jointly carry 40 per cent, with the social media share rising from 18 to 28 per cent at market extremes — corroborating independently published findings on the asymmetric influence of social sentiment on Indian sectoral returns [35]. Sectoral predictability varies systematically

with information availability and factor exposure, from R^2 above 0.99 in financials to below 0.96 in real estate. These are measured properties of that information environment rather than assumptions imported from elsewhere.

9.5 Conclusion

This research set out to integrate multi-source data with sentiment analysis and language models in order to improve stock market decision making, and to determine which parts of such an integration actually contribute to the outcome. Both aims are met, though not entirely in the direction originally anticipated, and the divergence is itself the most useful result.

A four-stream fusion model integrating financial metrics, news sentiment, social media sentiment and macroeconomic indicators was developed and evaluated on ten years of Indian market data across ten sectoral indices. It attains a normalised mean absolute error of 0.018, a coefficient of determination of 0.988 and a directional accuracy of 84.7 per cent against 76.3 per cent for a fixed-weight ensemble over identical inputs, with stability confirmed by walk-forward evaluation at 0.0016 ± 0.0004 normalised error across five sequential folds. Removing both textual sources costs 10.9 per cent of explained variance, exceeding the sum of their individual removals of 5.3 and 3.6 per cent — the quantitative signature of complementary rather than redundant information. The fusion rule was then advanced from learned average relevance to measured instantaneous reliability through a Bayesian formulation in which weights are obtained by a softmax over negative predictive uncertainties, improving the Sharpe ratio by 22.7 per cent and reducing maximum drawdown by 18.3 per cent against the strongest static-fusion baseline while the same experts fused statically add nothing.

A hybrid framework coupling a transformer language model with fifteen conventional technical indicators was implemented in a reproducible computational environment and evaluated under a strict temporal protocol. Language-model sentiment emerged as the highest-gain feature in the fused space, correlating 0.74 with momentum yet only 0.04 with price, establishing that it supplies orthogonal information; it produced the highest prediction confidence of the models examined, the most stable accuracy across sentiment regimes, and buy-signal success rates near 72 and 65 per cent. The gain in raw accuracy is modest — 47.72 per cent against a baseline mean of 46.43 per cent — and all models operate close to the random classifier at a single-name daily horizon. That limit is

reported rather than concealed, and it redirected the research toward the decision layer.

Transparency was established structurally rather than by post-hoc attribution. A directed causal adjacency matrix recovered by conditional independence testing yields a graph of density 0.850 with an identifiable causal hub whose outgoing influence systematically exceeds its incoming influence, and demonstrates that strong directed influence can coexist with weak linear correlation — the precise justification for causal rather than correlational feature selection, since a correlational selector would have discarded a genuine driver. Sentiment noise was quantified rather than assumed: the composite carries the expected sign against four independently published series, is significant against all four in first differences, holds the correct sign in all sixteen series-by-sub-period combinations, and separates the identified market states at $p = 1.8 \times 10^{-8}$ on a variable withheld from the fitting procedure, while its predictive content, measured separately, is not usable. Keeping construct validity and forecasting power apart is a contribution in its own right.

The decision framework was evaluated over eighteen years on ten liquid multi-asset instruments under a walk-forward protocol with transaction costs charged to the framework and not to its benchmarks. It attains a Sharpe ratio of 0.878 against 0.628 for an equal-weighted benchmark and reduces maximum drawdown from -36.1 to -19.0 per cent, outperforming in all four pre-selected stress episodes, in 18 of 19 calendar years on drawdown and in all 18 cells of a fully disclosed specification sweep; on a 665-session holdout that postdates every design decision it records a higher Sharpe ratio and a smaller drawdown than all three benchmarks. Ablation at matched average exposure locates the source precisely: timing risk by identified market state improves both the Sharpe ratio and the drawdown with no significant change in mean return, whereas the signal fusion layer contributes nothing detectable, for reasons the information coefficients make plain. Cross-market transfer reproduces the defensive behaviour and the regime effect in a concentrated mega-capitalisation universe while losing on risk-adjusted return, bounding applicability to universes with genuine dispersion in asset volatility.

The overall conclusion is precise rather than expansive. Multi-source integration and language-model sentiment demonstrably improve the *description* of the market, and causal structure demonstrably improves the *transparency* of that description. At the horizon and granularity at which decisions are actually taken, however, the measurable value of the system arises at the allocation layer, through regime-conditional risk timing, rather than at the signal layer. A framework whose every layer has been separately tested, and

whose applicability has been bounded by evidence, is a more useful engineering artefact than one whose aggregate performance conceals which of its components is doing the work.

9.6 Limitations of the Thesis

The limitations specific to each study are stated in the chapter that reports it. Six apply to the thesis as a whole and are consolidated here.

Horizon. Every result is at a daily decision horizon. Nothing in this thesis speaks to intraday microstructure or to execution.

Direction and leverage. Every portfolio is long-only and unlevered, and risk budgeting is one-sided. The framework cannot express a short view, and its defensive behaviour is achieved entirely by moving toward cash.

Language coverage. Sentiment extraction is English-only. A substantial share of Indian retail participants consume financial news in regional languages, so the measured sentiment channel observes part of the information reaching the market and its measured influence is correspondingly a lower bound.

Direct rather than adaptive transfer. The cross-market evidence comes from applying the framework without modification. No claim about transfer *learning* is made.

No attribution experiments. Transparency rests on structure; Shapley additive explanations and local surrogate models are reviewed and cited but not run, so no comparative claim between structural and attributional transparency is supported.

The allocation universe is not Indian. The allocation study is conducted on instruments that satisfy its data requirements, and it is presented as a test of the mechanism rather than as a claim about Indian allocation.

9.7 Future Scope

Eight directions follow from the evidence, ordered by how directly they address a limitation identified above.

1. Textual sentiment in a universe with instrument-level news coverage. The null reported in Chapter 8 is a null about a market-derived proxy, not about textual sen-

timent. Applying the fusion hypothesis where a time-stamped, instrument-level news corpus exists would convert a proxy null into a genuine test, and would connect the language-model pipeline of Chapter 6 directly to the allocation layer of Chapter 8. The information-coefficient evidence of Section 8.6 — positive and marginally significant on single names, indistinguishable from zero on broad instruments — suggests this is where the signal, if any, will be found [60, 14].

2. Bayesian neural experts for epistemic uncertainty. The uncertainty estimate of Chapter 5 is a rolling error variance, which captures aleatoric variation. Replacing the point-estimating experts with Bayesian neural models would additionally expose epistemic uncertainty, producing a confidence measure that distinguishes a noisy environment from an unfamiliar one [75, 76, 16]. The operational difference is that the system could report low confidence *before* accumulating errors rather than after, which is exactly the behaviour required at the onset of an unprecedented event.

3. Non-stationary debiasing. The stationarity assumption in the multi-scale debiasing layer fails at exactly the moments that matter, as the 77.67 per cent peak volatility during the 2024 Union Budget window shows. A formulation whose noise model is itself state-dependent — estimated separately within identified market states, or allowed to follow the volatility process — is the direct remedy, and it connects naturally to the regime identification already present in Chapter 8.

4. Scalable causal discovery. Extending conditional independence testing from five instruments to a full fifty-constituent index requires an algorithm whose cost does not grow combinatorially. This is a prerequisite for causal transparency at production scale, and without it the structural-transparency argument of Chapter 7 remains a demonstration rather than a deployable method [88, 98].

5. Adaptation-based cross-market transfer. The natural extension of the direct-transfer result is genuine transfer learning: pre-training the regime model and the source-reliability estimates on a data-rich market and fine-tuning them on a thinner one, so that an emerging-market deployment inherits structure learned elsewhere instead of starting from its own limited history. The transferability condition identified in Section 8.9 — dispersion in asset volatility — gives a testable criterion for when this should be expected to work.

6. An Indian multi-asset allocation study. The allocation framework requires a uni-

verse with volatility dispersion and long clean history. Nifty sectoral indices, government securities, gold and real-estate instruments together satisfy the first requirement, and the history available is now long enough to include the 2008 and 2020 episodes. This would close the one structural gap between the prediction chapters and the decision chapter of this thesis.

7. Learned risk budgets and explicit state persistence. The regime multipliers are fixed at 1.00, 0.75 and 0.50. Learning them — reinforcement learning being the natural candidate given the sequential structure [124] — would adapt the budget to the universe, subject to the overfitting controls this thesis argues are mandatory [146]. Replacing the Gaussian mixture with a hidden Markov or jump formulation would represent regime persistence directly rather than recovering it incidentally [119, 116].

8. Regional-language and higher-frequency sentiment, and deployment engineering. Sentiment extraction is currently English-only and daily. A large share of Indian retail participants read financial news in regional languages, and intraday resolution would expose incorporation dynamics that daily aggregation averages away — particularly the two distinct incorporation regimes identified in Section 7.9. On deployment, inference cost is negligible; the binding constraints are the latency of retrieving and scoring text at scale, feed rate limits, coordinated manipulation of social signals at exactly the market extremes where they carry most weight, transaction costs, and compliance with algorithmic trading regulation. These are treated here as first-class engineering problems rather than as afterthoughts, because a framework that cannot be operated is not a decision-support system.

Publications on PhD Work

The research reported in this thesis has been published or communicated in the following works. The labels [P1]–[P6] are used throughout the thesis to refer to them, and publication status is as of the date of submission.

International Journals

- [P3] A. Pardeshi and S. Deshmukh, “Fusing LLM-Extracted News Sentiment and Market Data for Enhanced Stock Movement Prediction,” *Liberte Journal*, ISSN 0024-2020, vol. 14, no. 6, pp. 424–439, 2026. **(Published)**

Indexing: [indexing, e.g. Scopus / UGC-CARE Group II / not indexed]

[article URL or DOI]

Supports Research Objective 2; reported in Chapter 6 and, for the sentiment-noise diagnostics, Chapter 7.

- [P4] A. Pardeshi and S. Deshmukh, “Dynamic Uncertainty-Weighted Fusion for Multi-Source Financial Data: A Bayesian Framework for Time-Varying Source Reliability,” *International Journal of Computer Information Systems and Industrial Management Applications* (IJCISIM). **(Communicated, [submission month and year])**

[submission acknowledgement URL or manuscript ID, if any]

Supports Research Objective 1; reported in Chapter 5.

- [P5] A. Pardeshi and S. Deshmukh, “From Signal Fusion to Asset Allocation: A Decision-Theoretic Model for Portfolio Construction Under Regime-Based Sentiment and Volatility,” *International Journal of Applied Decision Sciences* (Inderscience), manuscript IJADS-312370. **(Under review — revised manuscript submitted after a major-revision decision)**

[Inderscience submission URL, if you wish to include it]

Supports Research Objective 4; reported in Chapter 8 and, for the sentiment construct validation, Chapter 7.

International Conferences

- [P1] A. Pardeshi and S. Deshmukh, “A Multi-Source Data Fusion Framework for En-

hanced Stock Market Prediction,” *2026 International Conference on Sustainability, Innovation and Technology (ICSIT)*, [**host institute, city, and conference dates**], IEEE, 2026, [**pages, or “in press”**].

[**IEEE Xplore DOI or proceedings URL**]

Supports Research Objective 1; reported in Chapter 4.

[P2] A. Pardeshi and S. Deshmukh, “A Dynamic-Causal Hybrid Framework for NIFTY50 Stock Decision Making: A Multi-Granularity Temporal Learning Approach,” [**full conference name, host institute, city and dates**]. (Communicated)

[**submission acknowledgement URL or paper ID, if any**]

Supports Research Objective 3; reported in Chapter 7.

[P6] A. Pardeshi and S. Deshmukh, “Deep Learning in Stock Market Forecasting: Comparative Insights and Future Directions,” *2025 International Conference on Emerging Trends in Industry 4.0 Technologies (ICETI4T)*, Navi Mumbai, India, IEEE, June 2025, [**pages**].

[**IEEE Xplore DOI or proceedings URL**]

Comparative groundwork that established the research gap; cited in Chapter 3.

Table P.1 states the relationship between each publication and each research objective. A *primary* mapping means the publication was designed to answer the objective and supplies its principal evidence; a *partial* mapping means it contributes a specific, identifiable component without answering the objective as a whole. Blank cells are deliberate and should be read as an accurate statement of scope rather than as an omission.

Table P.1: Mapping of published work to the research objectives and to the chapters in which each contribution is developed

Ref.	Publication	RO1	RO2	RO3	RO4	Chapter	Nature of the contribution
[P1]	Multi-source data fusion framework	P	p	—	p	4, 6	Four-stream architecture, cross-modal attention, source ablation, ten-sector evaluation
[P2]	Dynamic-causal hybrid framework	p	—	P	p	7	Directed causal feature selection, multi-granularity memory, credibilistic allocation
[P3]	LLM news sentiment and market data	p	P	p	—	6, 7	LLM sentiment pipeline, feature-level fusion, sentiment-regime analysis
[P4]	Bayesian dynamic uncertainty-weighted fusion	P	p	p	—	5	Time-varying source reliability, softmax weighting over negative uncertainties
[P5]	Signal fusion to asset allocation	—	—	p	P	7, 8	Regime detection, constrained allocation, matched-exposure ablation, holdout, cross-market test
[P6]	Deep learning in stock forecasting	p	p	—	—	3	Comparative groundwork that established the research gap

P = primary contribution; **p** = partial contribution; — = no contribution claimed.

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